WSTECH

APS-Series

Additional Technical Data

WSTECH is a Wind&Sun Technologies and Siemens Joint Venture.

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1 Introduction

This document is presenting several additional technical information for the APS-Series inverters.

2 Power capabilities

2.1 Active power capability P_{AC} (Udc)

The function P_{AC}(Udc) depends on the grid voltage and DC voltage. The diagrams in Figure 1 and Figure 2 show the functions with different nominal voltage levels at 50 Hz. The upper limit is caused by the maximum DC current limit.

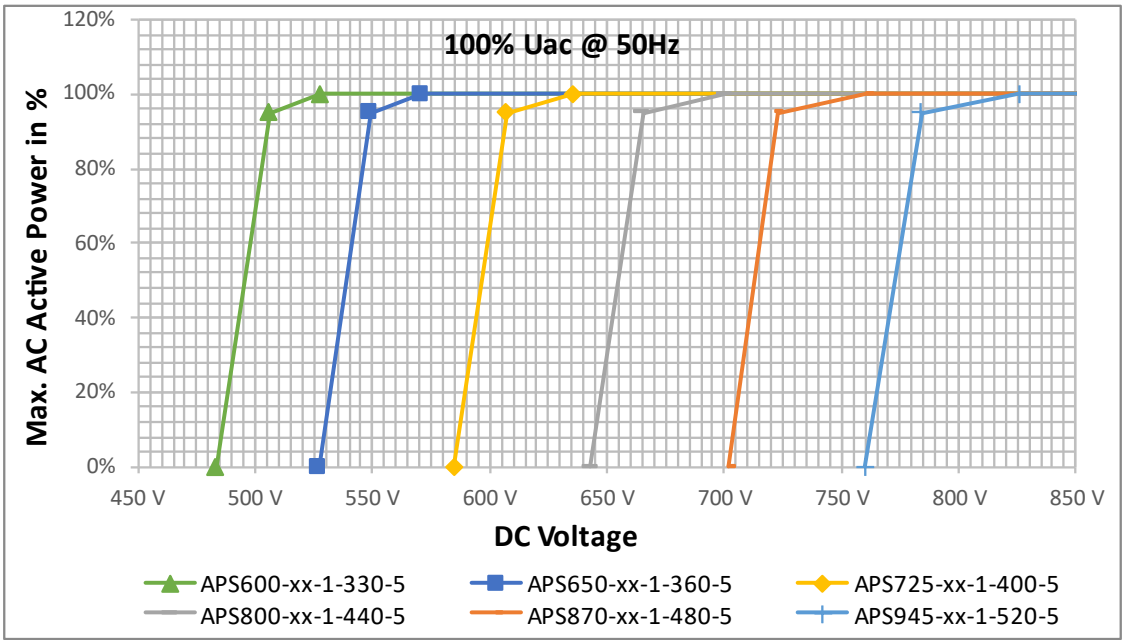


Figure 1: Pac(Udc) at 330 Vac … 520 Vac @ 50 Hz

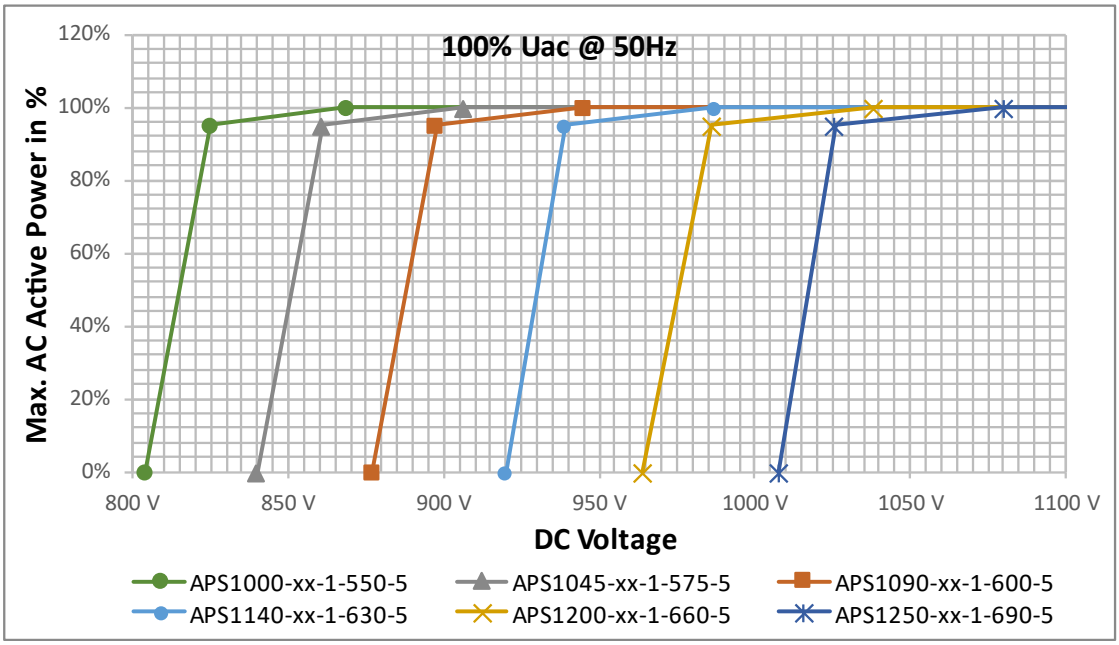


Figure 2: Pac(Udc) at 550 Vac …690 Vac @ 50 Hz

The diagrams in Figure 3 and Figure 4 show the functions with different nominal voltage levels at 60 Hz. The upper limit is caused by the maximum DC current limit.

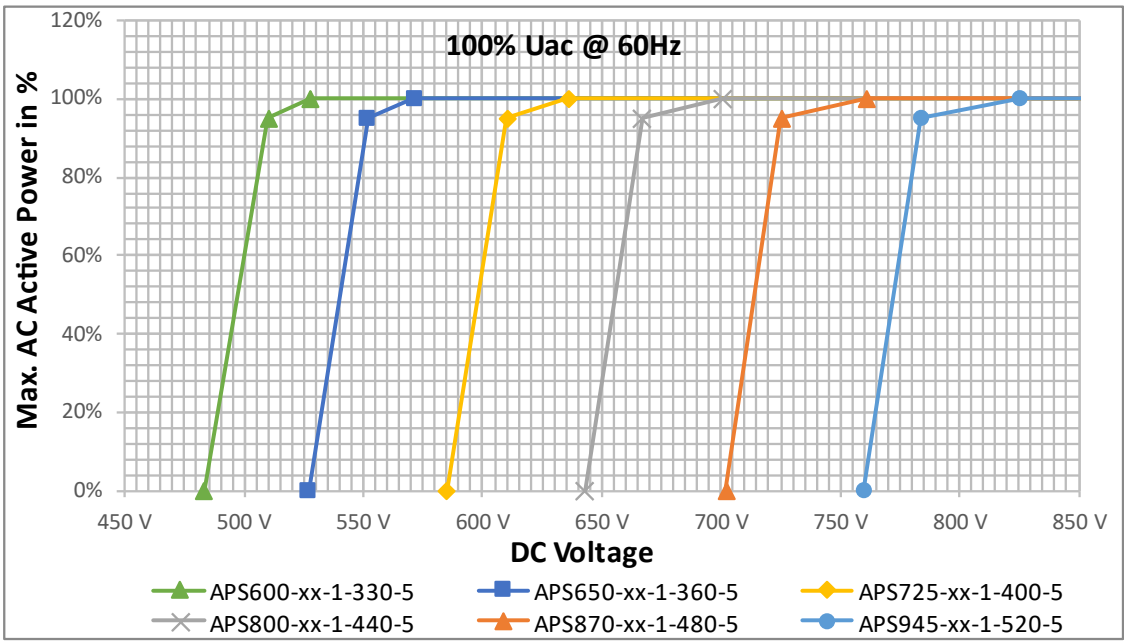


Figure 3: Pac(Udc) at 330 Vac … 520 Vac @ 60 Hz

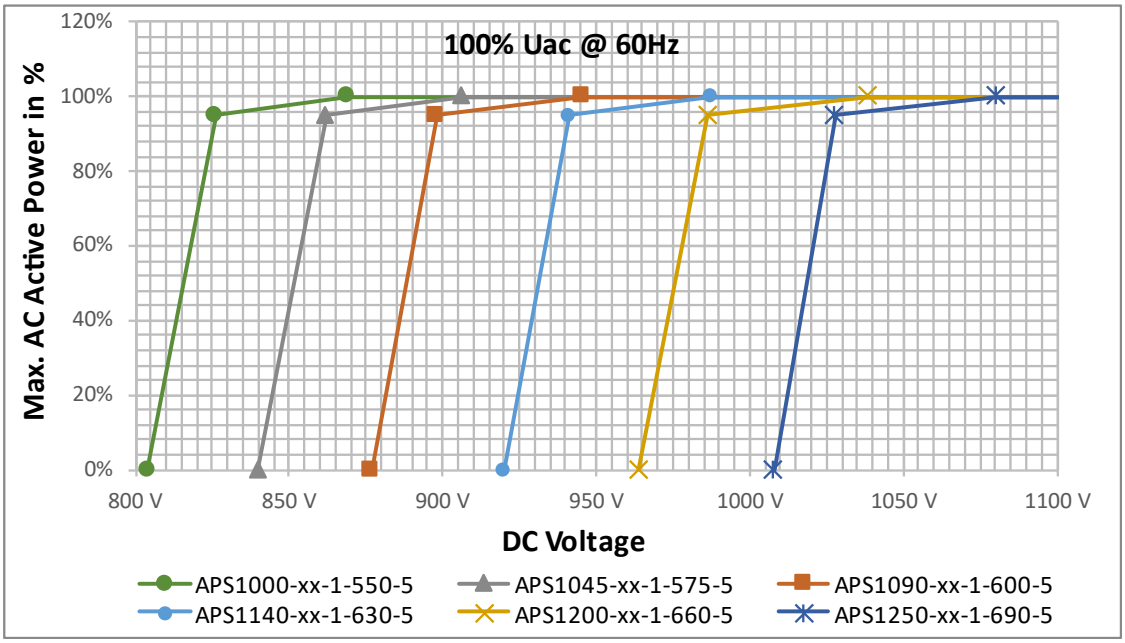


Figure 4: Pac(Udc) at 550 Vac … 690 Vac @ 60 Hz

The diagram in Figure 5 shows the Pac limitation at high DC voltages. The limitation at 50 Hz is equal like the limitation at 60 Hz. For all different inverter types the limitation is the same. The different graphs of the inverter types overlap in the figure.

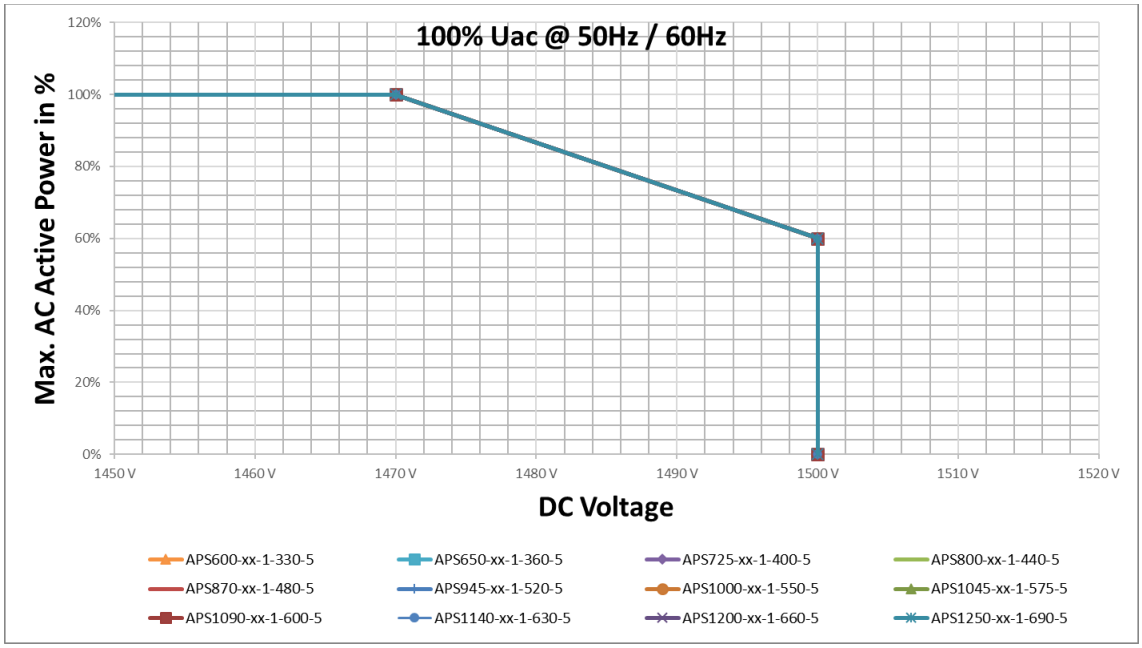


Figure 5: High Udc Pac limitation at 330 Vac … 690 Vac @ 50/60 Hz

2.2 Reactive power capability Udc(S, Uac, f, cosPhi)

The following figures show the minimum DC voltage required for operation with reactive load. The minimum DC-voltage is related to the following parameters:

- Apparent power load
- Power factor
- Grid voltage
- Grid frequency

IMPORTANT NOTE

There is an additional limitation on the DC-link current. This limitation is not concern in the following figures. In some special cases the dc current in the following figures could be higher than possible. This is not allowed. In any case the dc-current has to be considered in the plant design in before defining the operating points. These limits are marked in the previous chapter. This limitation could cause a higher required dc-link voltage in some operating points.

In case of an inductive power factor the minimum dc-link voltage is lower. These values are not shown in any of these diagrams. When selecting the inverter, at least the voltage at power factor one must be considered. Inductive reactive power can always provide up to the maximum apparent power. An example is shown in Figure 6.

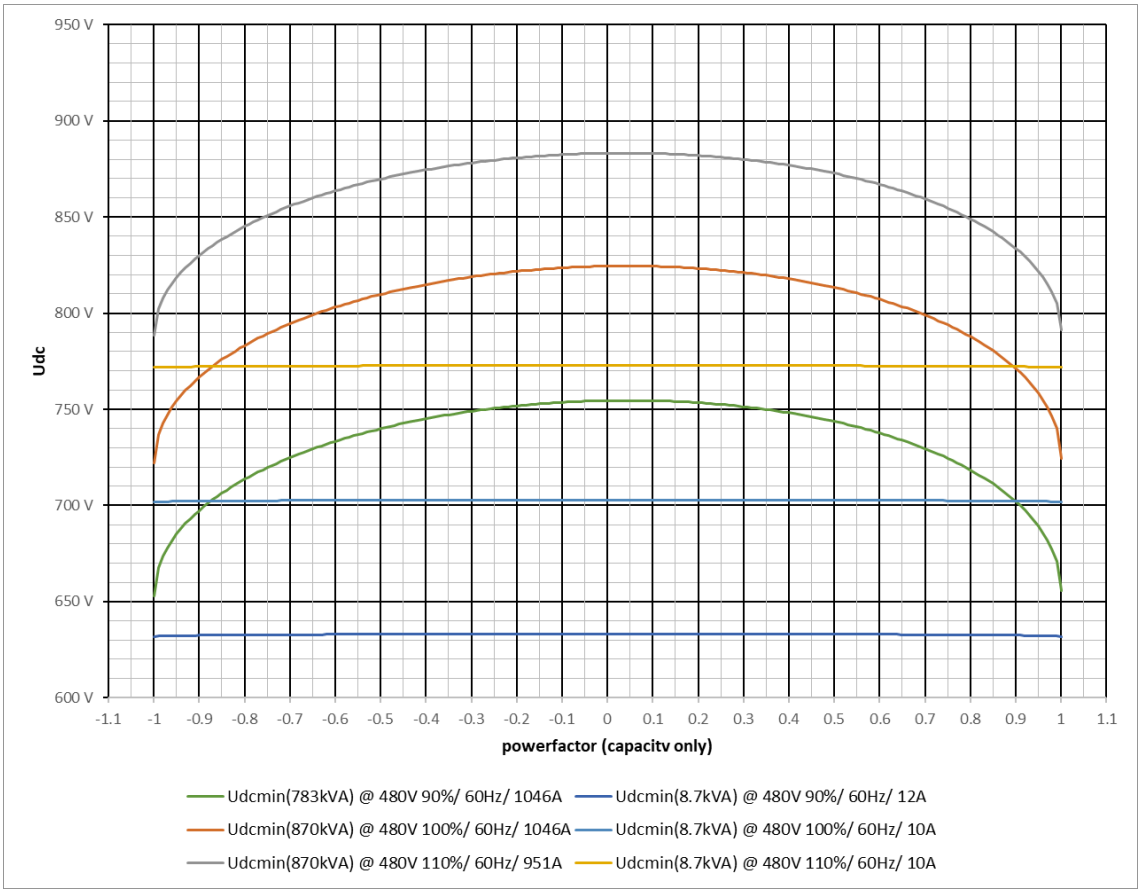


Figure 6: Example power factor diagram

2.2.1 How to read the diagrams

The following power factor curves are to be used as follows:

2.2.1.1 Method 1: What is the required min dc-link voltage of the inverter?

1. Define your operating points:

Table 1: Reactive power capability method 1 operating points

Operating Points	f	Uac PU	P	Q	S	Power factor
OP01	50 Hz	100.0%	1 kW	0 kvar	1 kVA	1.00
OP02	50 Hz	100.0%	870 kW	0 kvar	870 kVA	1.00
OP03	50 Hz	100.0%	0 kW	870 kvar	870 kVA	0.00
OP04	50 Hz	100.0%	-870 kW	0 kvar	870 kVA	-1.00
OP05	50 Hz	100.0%	0 kW	435 kvar	435 kVA	0.00
OP06	50 Hz	100.0%	500 kW	250 kvar	559 kVA	0.89
OP07	50 Hz	100.0%	0 kW	1 kvar	1 kVA	0.00

2. Mark the operating points in the diagram (see Figure 7)

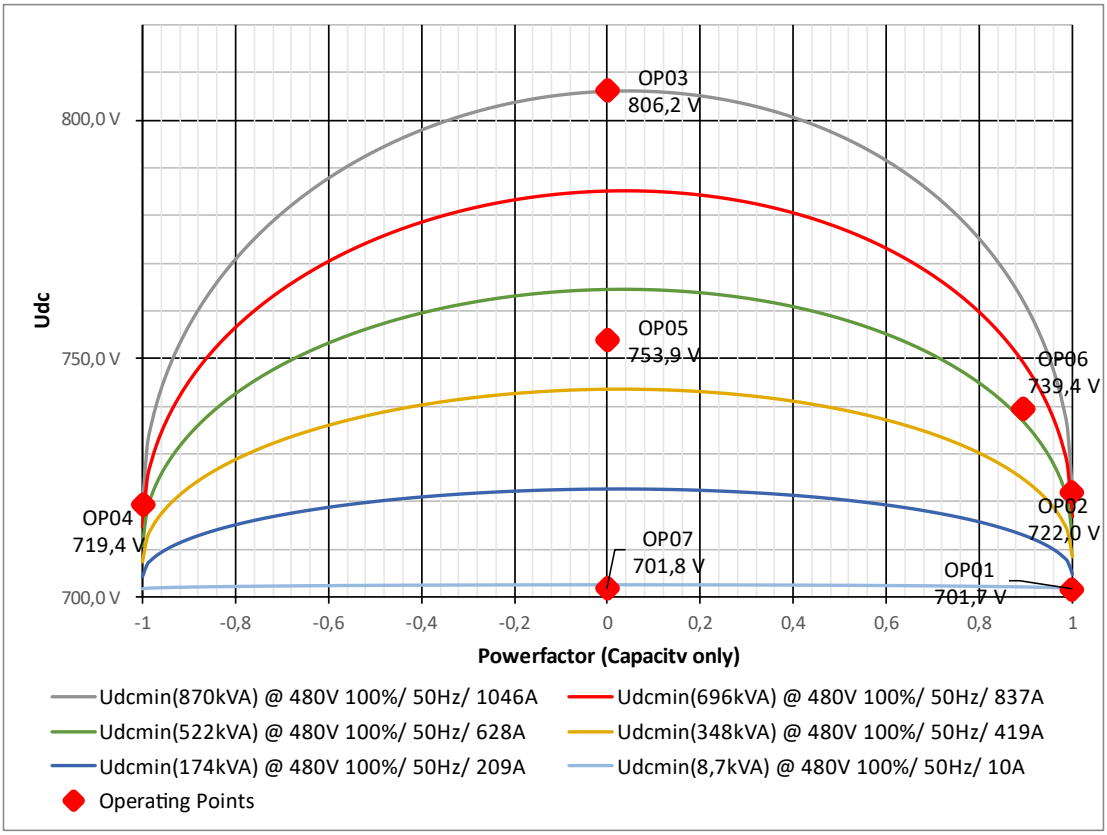


Figure 7: Power factor diagram method 1

3. Read out the required DC-link voltage

Table 2 Reactive power capability method 1 results

Operating Points	f	Uac PU	P	Q	S	Udc_Min
OP01	50 Hz	100.0%	1 kW	0 kvar	1 kVA	702 V
OP02	50 Hz	100.0%	870 kW	0 kvar	870 kVA	722 V
OP03	50 Hz	100.0%	0 kW	870 kvar	870 kVA	806 V
OP04	50 Hz	100.0%	-870 kW	0 kvar	870 kVA	719 V
OP05	50 Hz	100.0%	0 kW	435 kvar	435 kVA	754 V
OP06	50 Hz	100.0%	500 kW	250 kvar	559 kVA	739 V
OP07	50 Hz	100.0%	0 kW	1 kvar	1 kVA	702 V

2.2.1.2 Method 2: What power can be provided?

1. Define your minimum DC-link voltage from your source.
2. Define your requested power factor or load.

Table 3: Reactive power capability method 2 operating points

Operating Points	Udc min (DC-Source)	Power factor	S
OP01	760 Vdc	0.90	Read out of the diagram
OP02	760 Vdc	Read out of the diagram	522 kVA

3. Mark the minimum DC-link voltage in the diagram like in Figure 8.

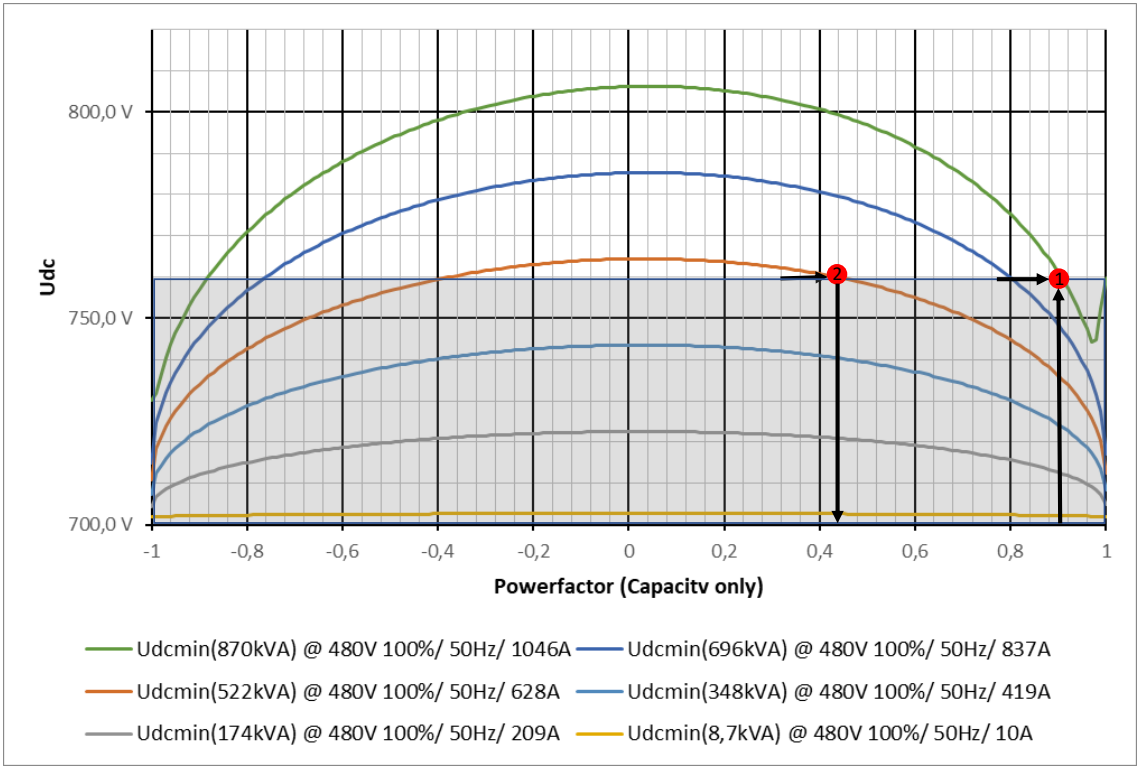


Figure 8: Power factor diagram method 2

4. Read out the possible inverter power on different power factors.

Table 4: Reactive power capability method 2 results

Operating Points	Udc min (DC-Source)	Power factor	S
OP01	760 Vdc	0.90	826,5 kVA
OP02	760 Vdc	0.43	522 kVA

2.2.2 Diagrams

2.2.2.1 APS600-xx-1-330-5

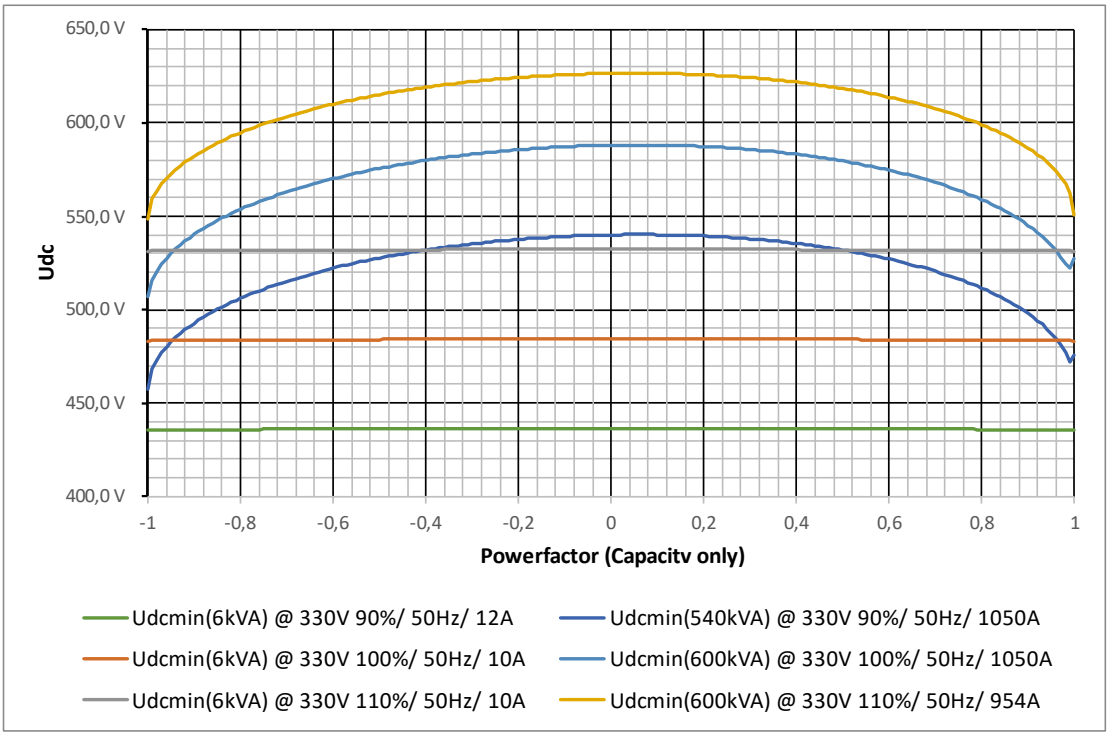


Figure 9: APS600-xx-1-330-5 Udcmin(Cos(Phi)) 50 Hz

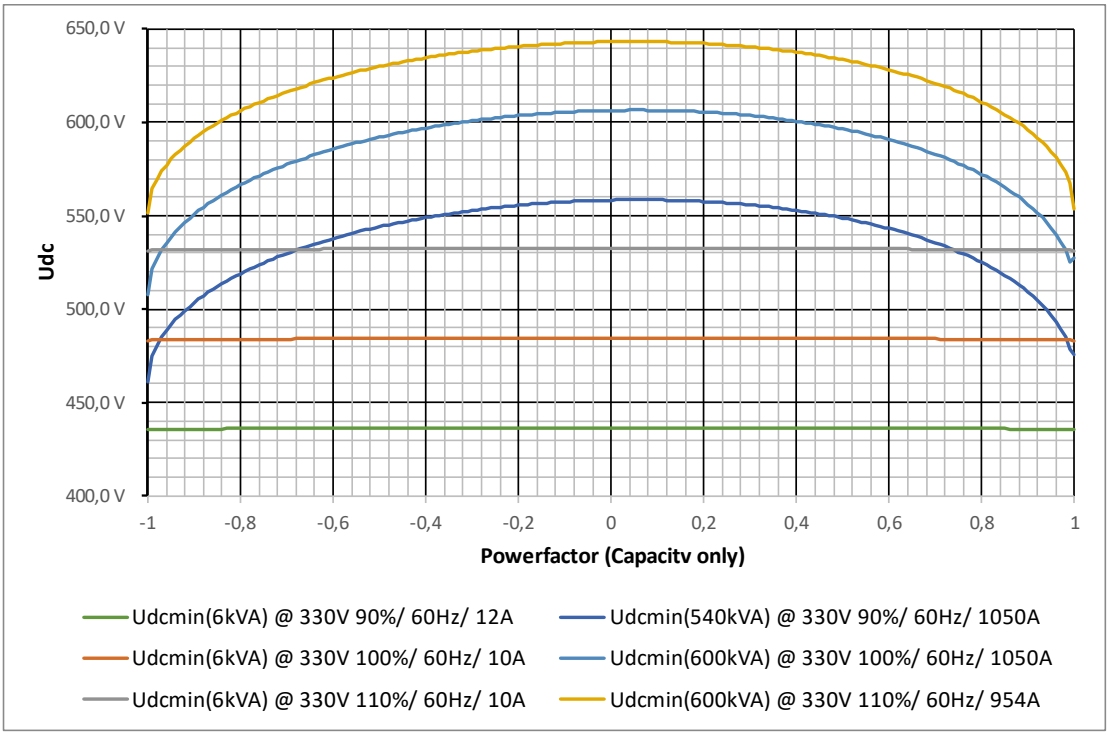


Figure 10: APS600-xx-1-330-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.2 APS650-xx-1-360-5

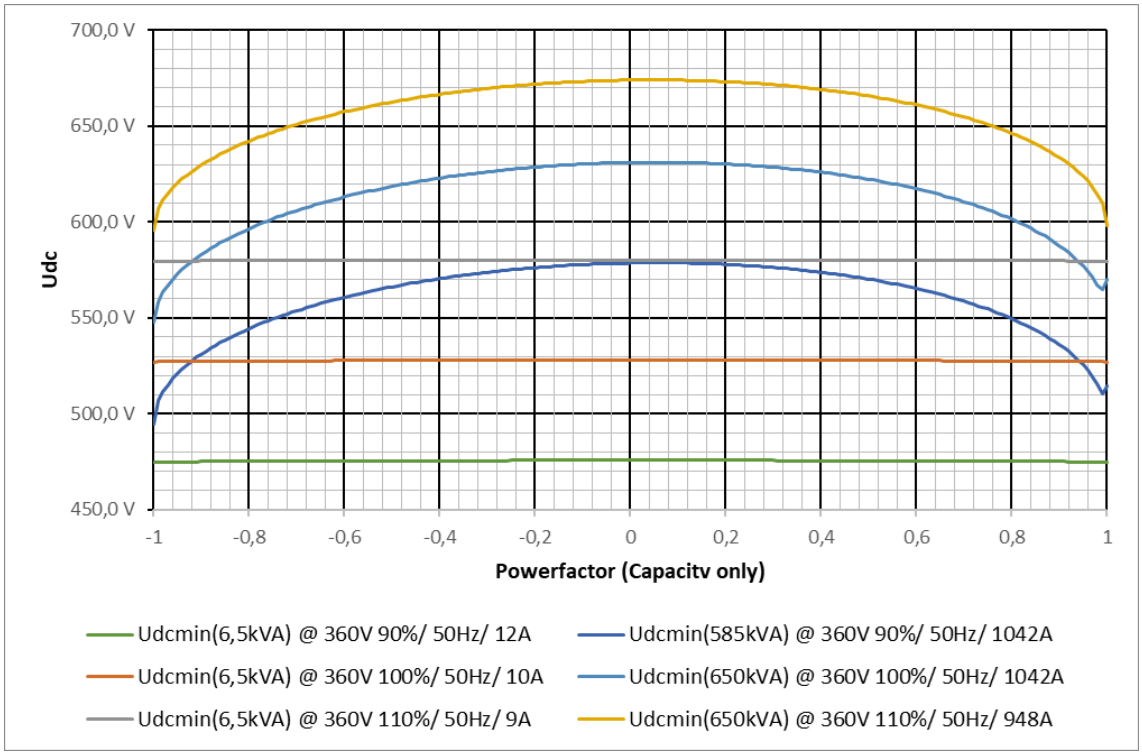


Figure 11: APS650-xx-1-360-5 Udcmin(Cos(Phi)) 50 Hz

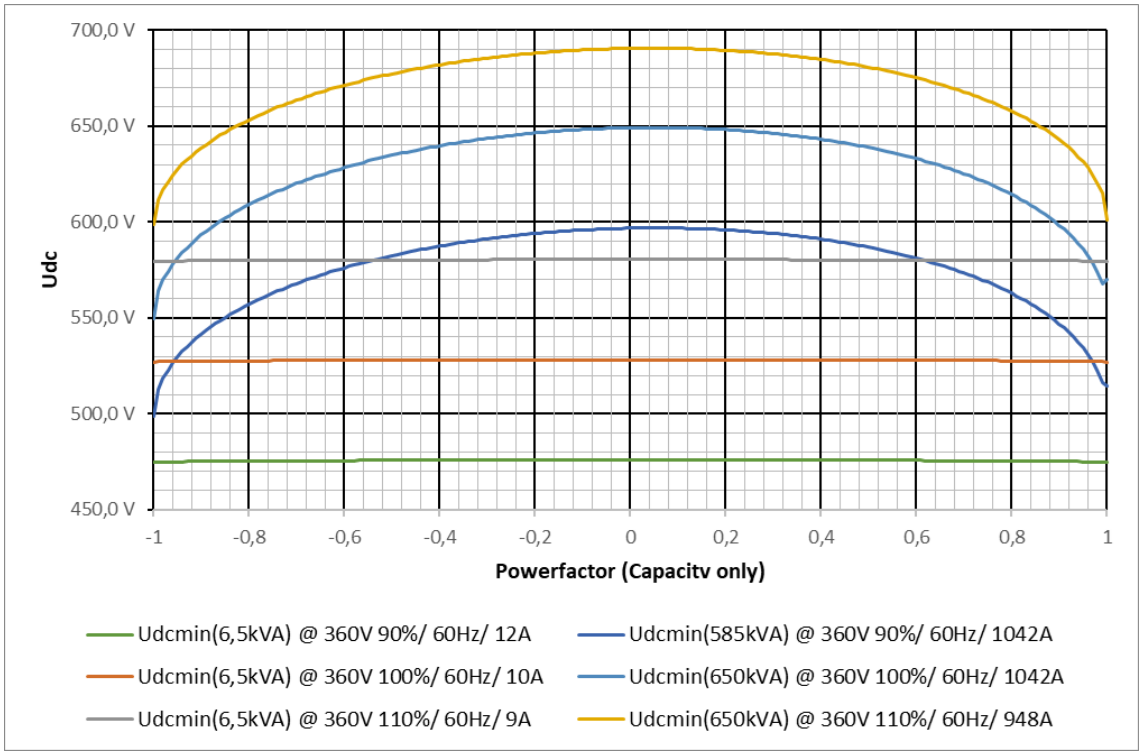


Figure 12: APS650-xx-1-360-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.3 APS725-xx-1-400-5

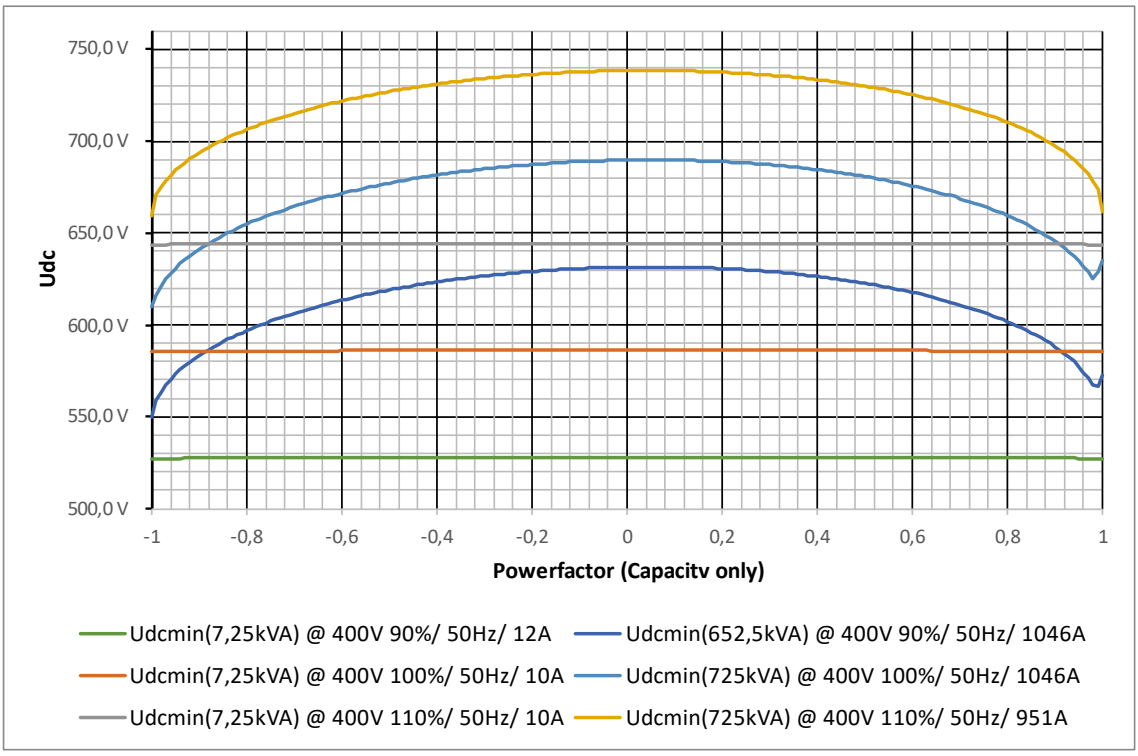


Figure 13: APS725-xx-1-400-5 Udcmin(Cos(Phi)) 50 Hz

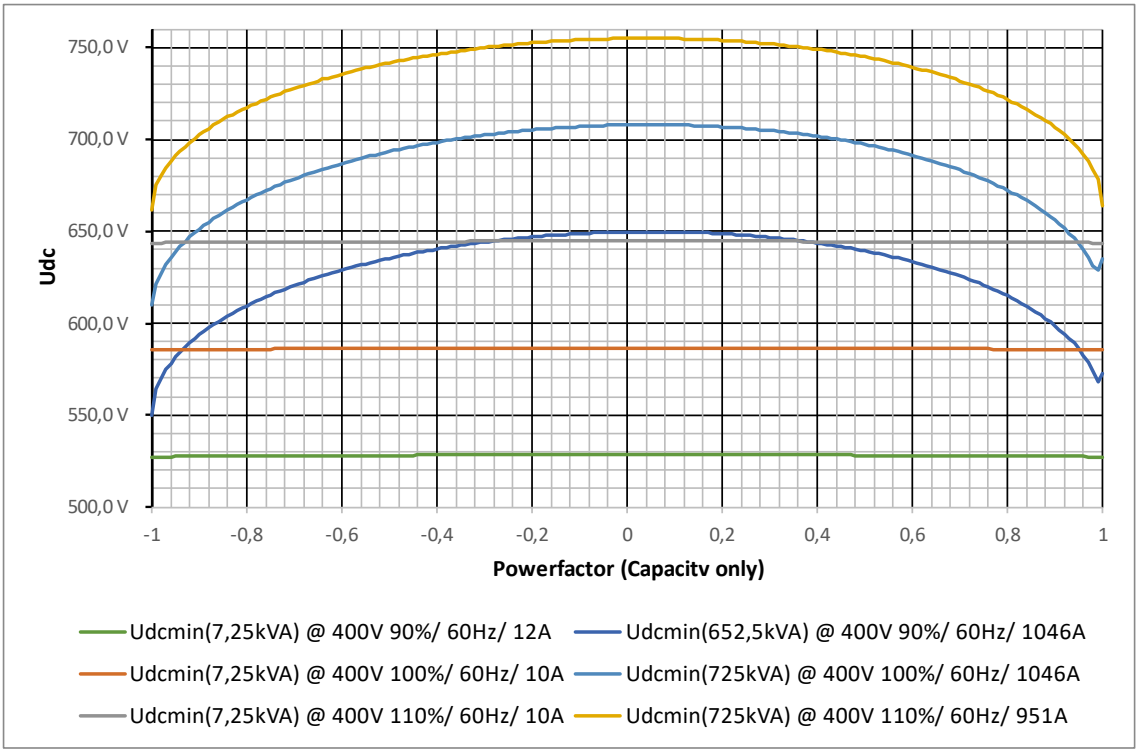


Figure 14: APS725-xx-1-400-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.4 APS800-xx-1-440-5

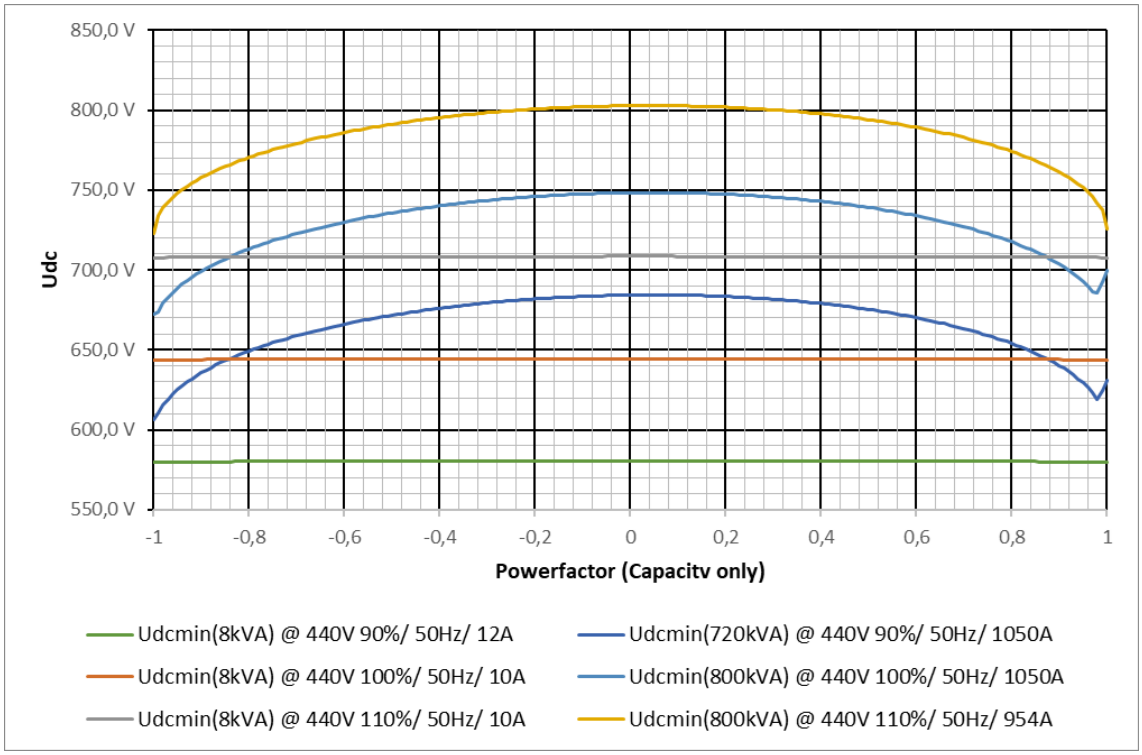


Figure 15: APS800-xx-1-440-5 Udcmin(Cos(Phi)) 50 Hz

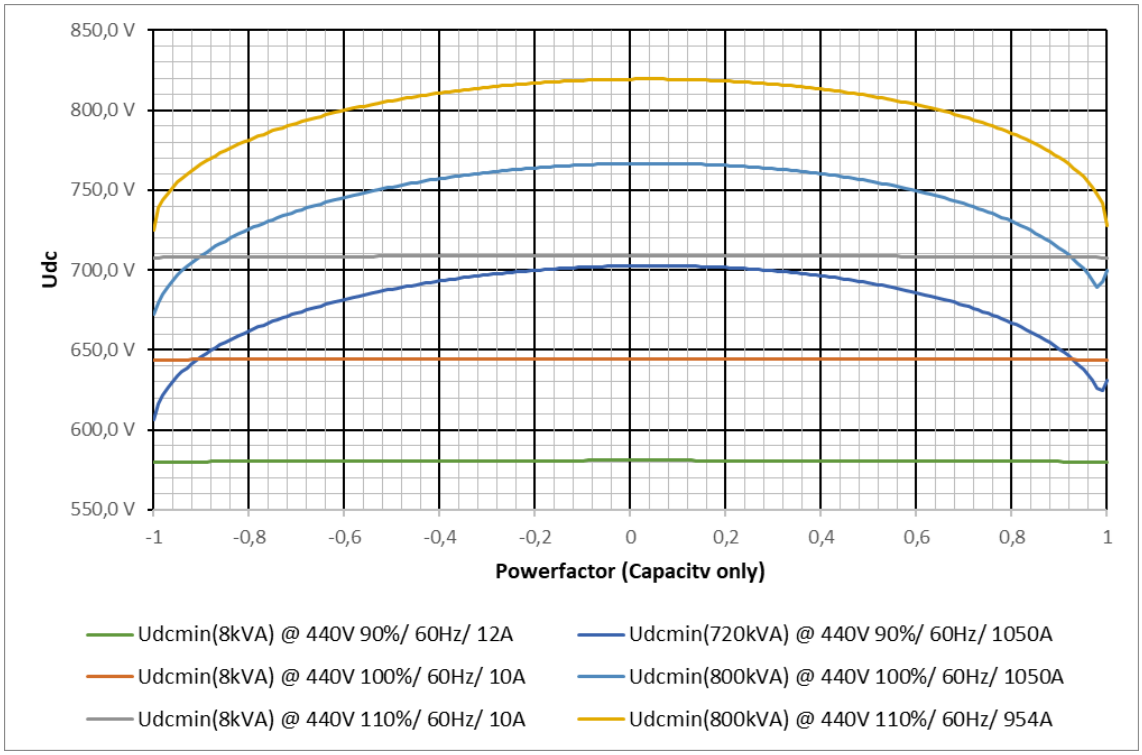


Figure 16: APS800-xx-1-440-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.5 APS870-xx-1-480-5

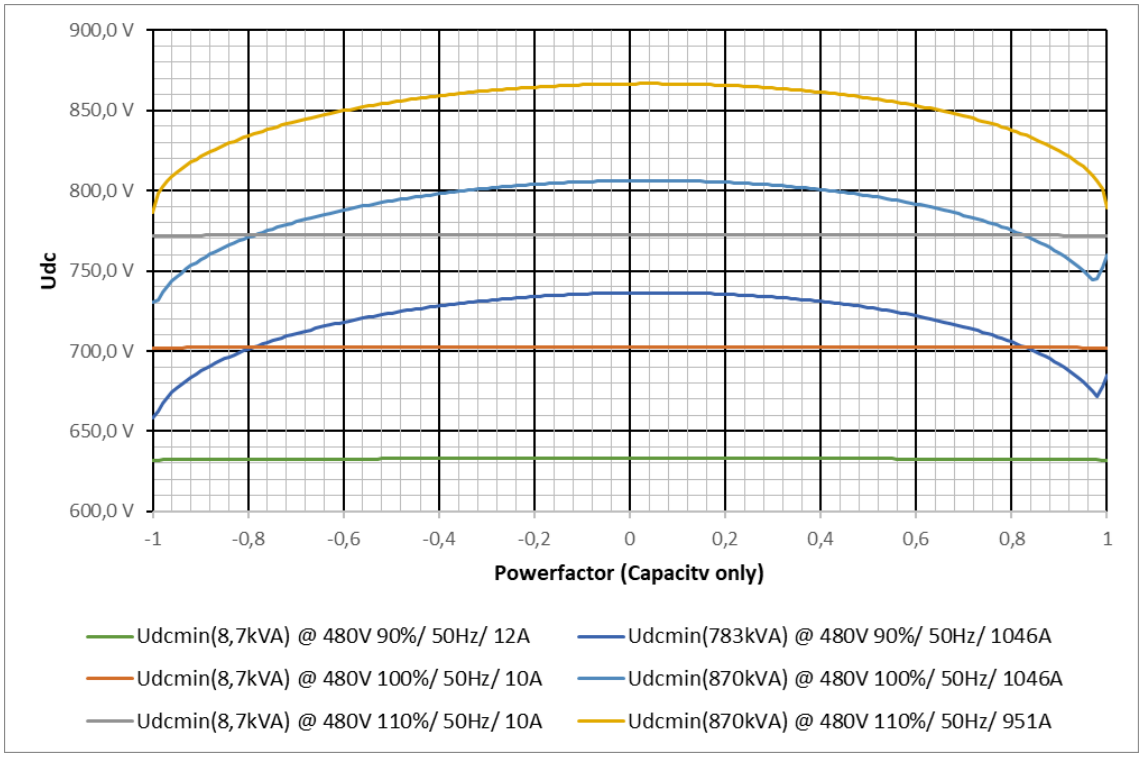


Figure 17: APS870-xx-1-480-5 Udcmin(Cos(Phi)) 50 Hz

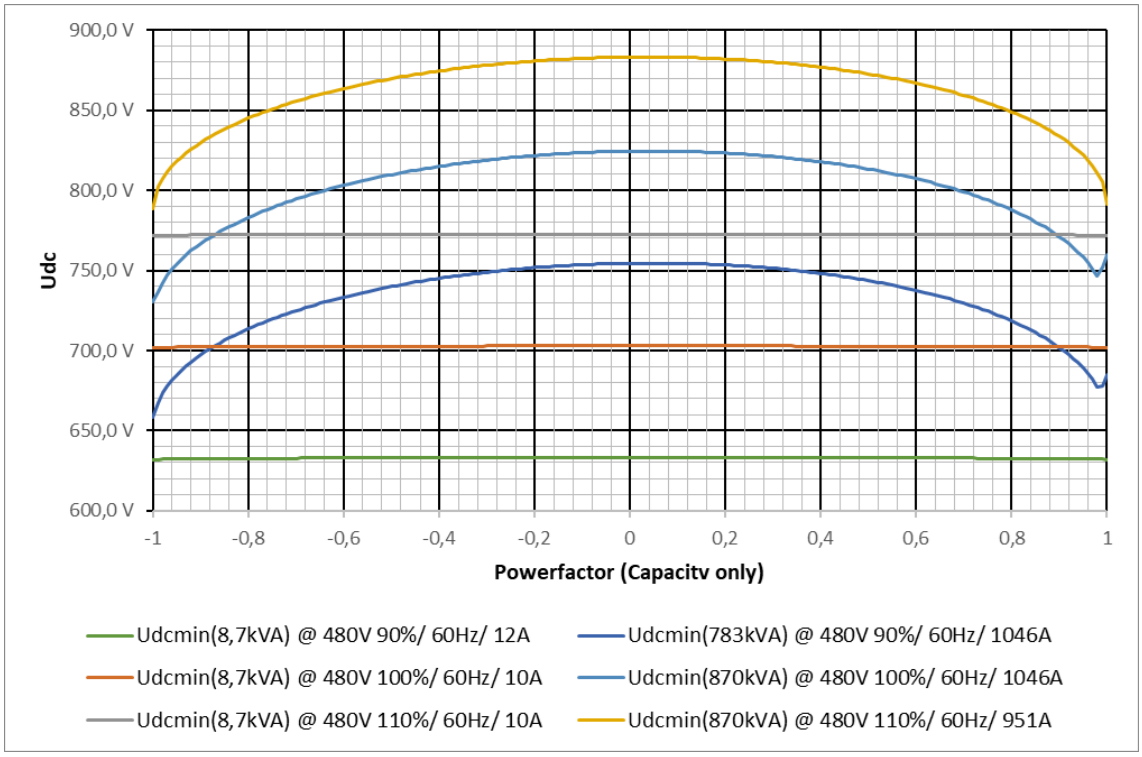


Figure 18: APS870-xx-1-480-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.6 APS945-xx-1-520-5

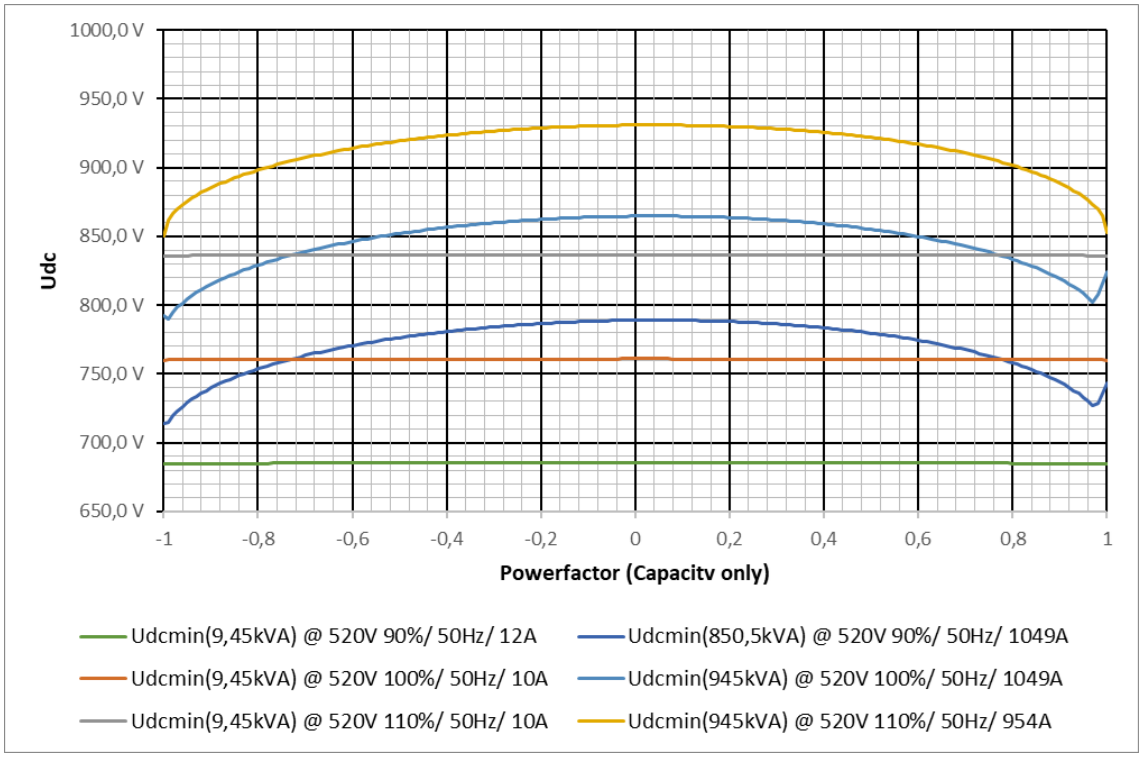


Figure 19: APS945-xx-1-520-5 Udcmin(Cos(Phi)) 50 Hz

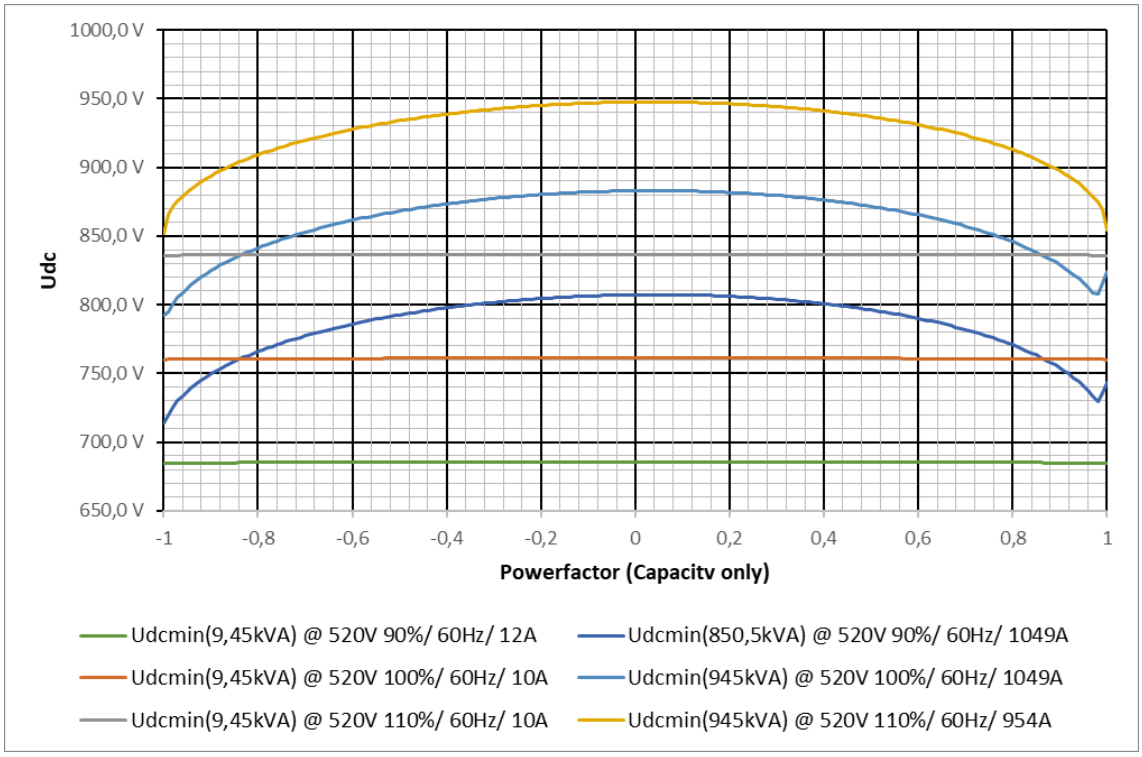


Figure 20: APS945-xx-1-520-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.7 APS1000-xx-1-550-5

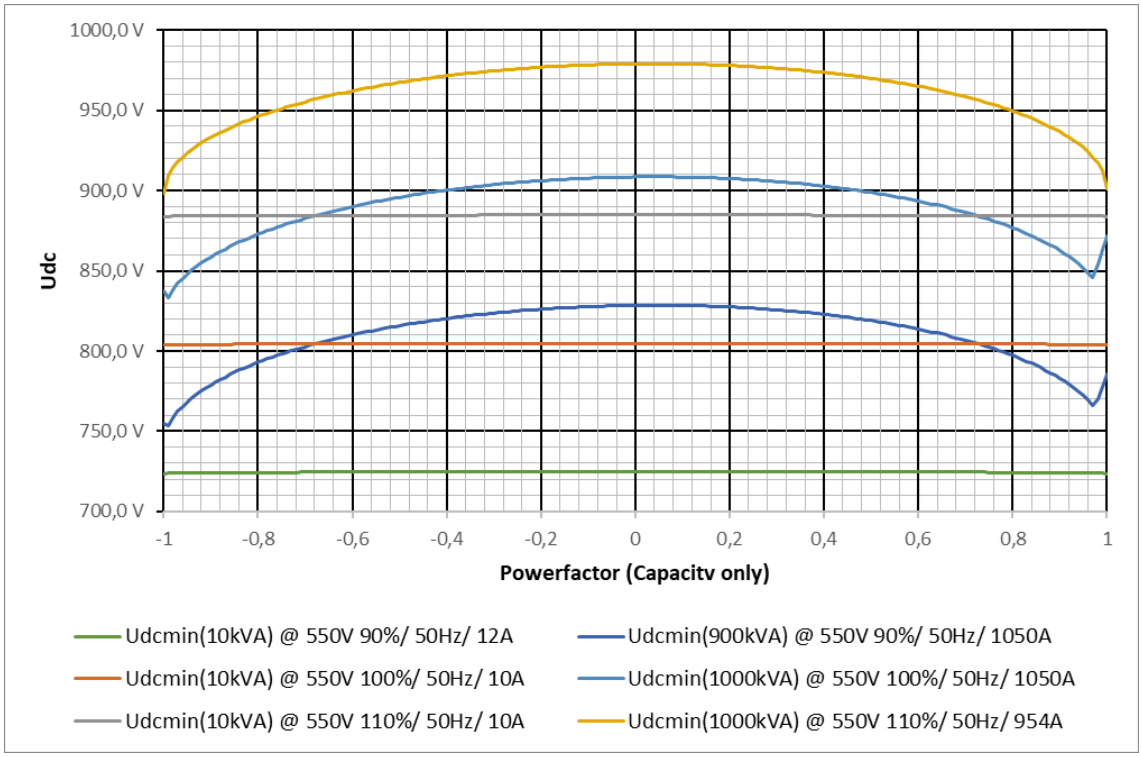


Figure 21: APS1000-xx-1-550-5 Udcmin(Cos(Phi)) 50 Hz

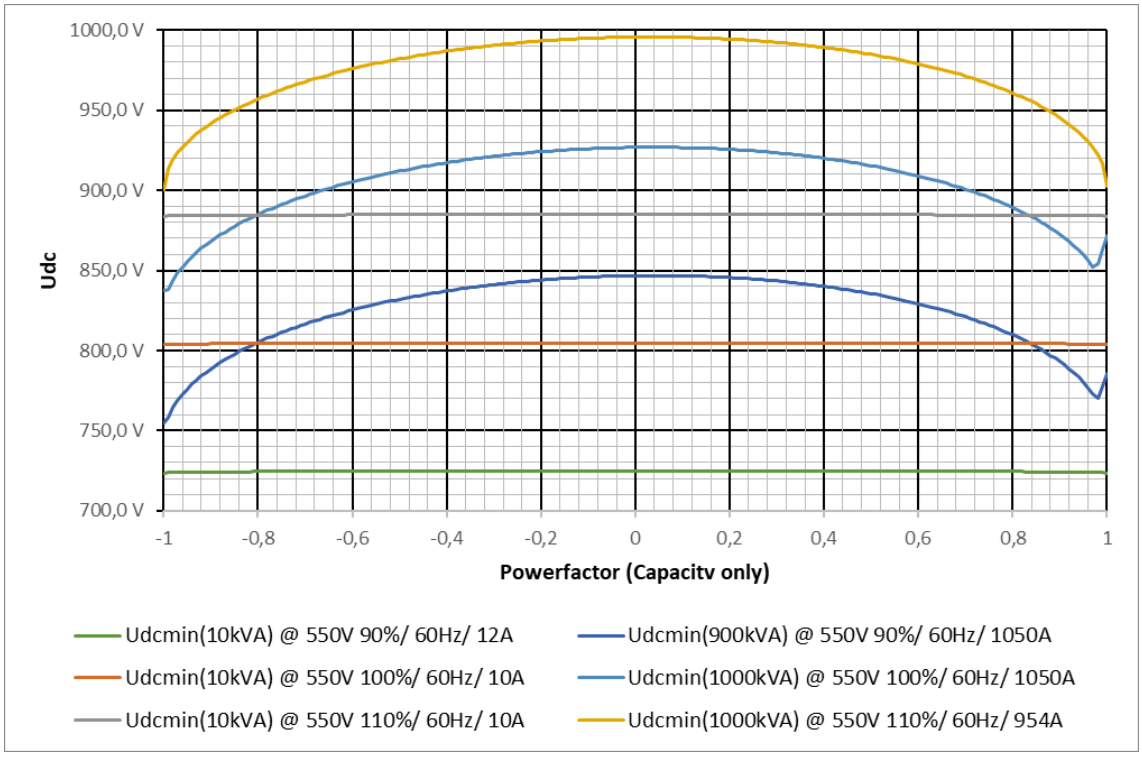


Figure 22: APS1000-xx-1-550-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.8 APS1045-xx-1-575-5

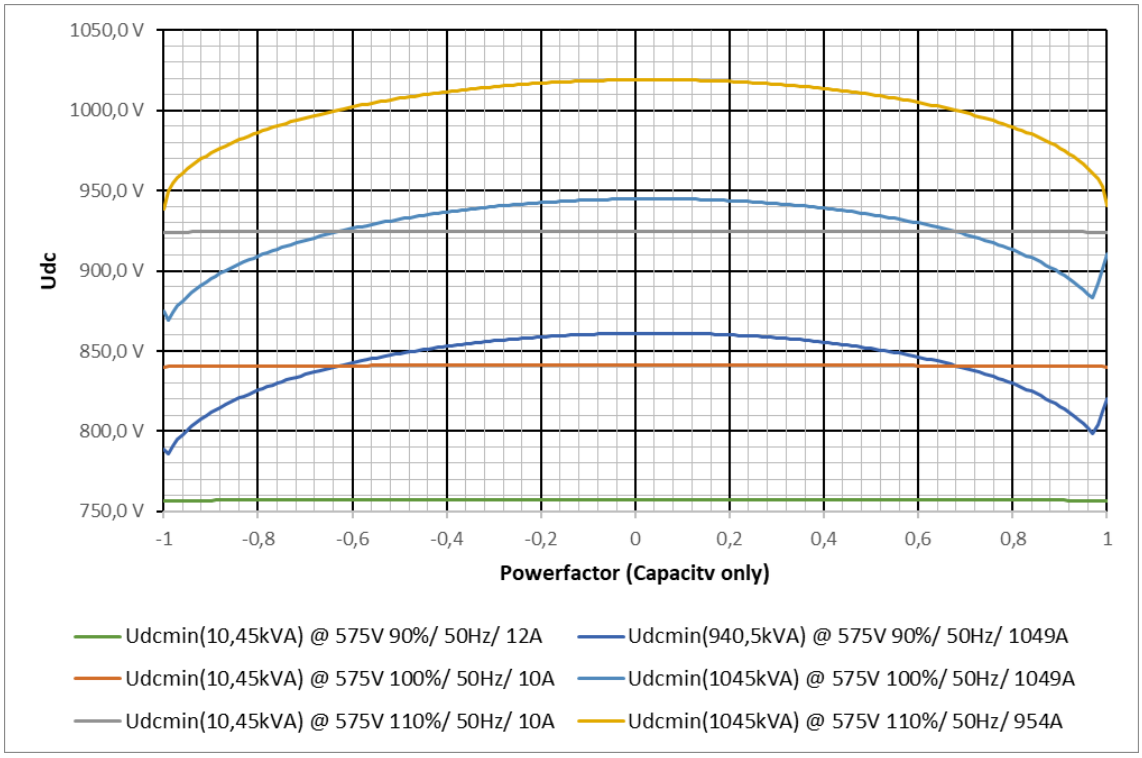


Figure 23: APS1045-xx-1-575-5 Udcmin(Cos(Phi)) 50 Hz

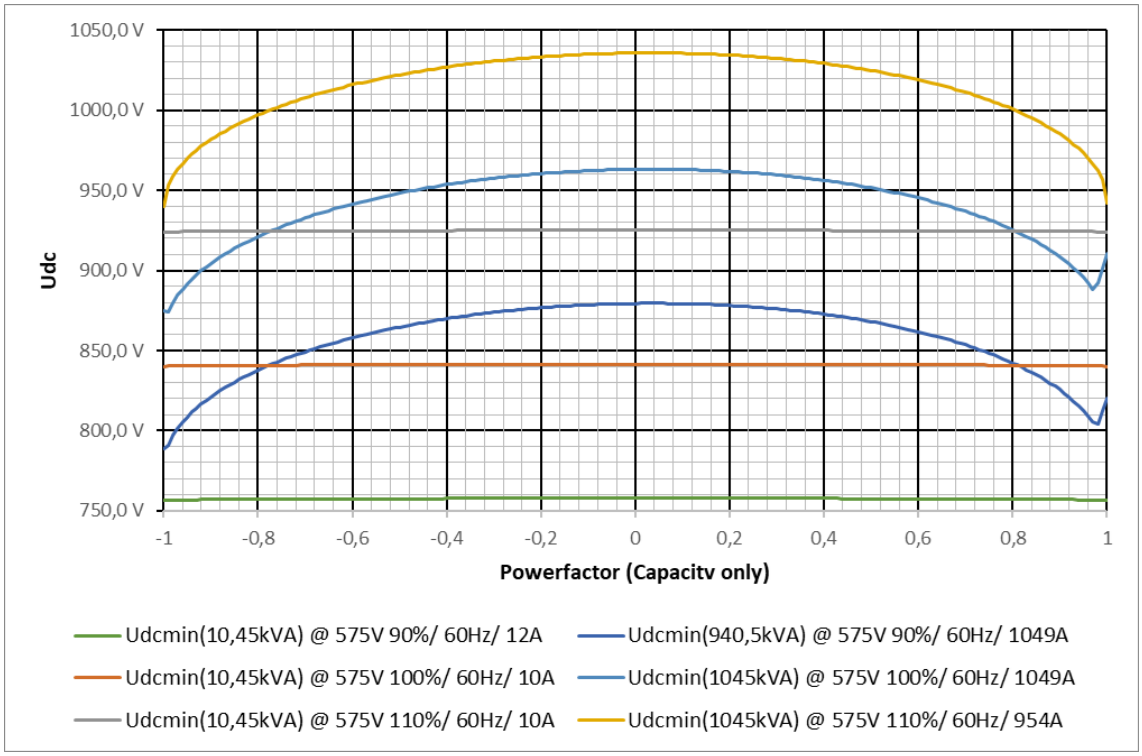


Figure 24: APS1045-xx-1-575-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.9 APS1090-xx-1-600-5

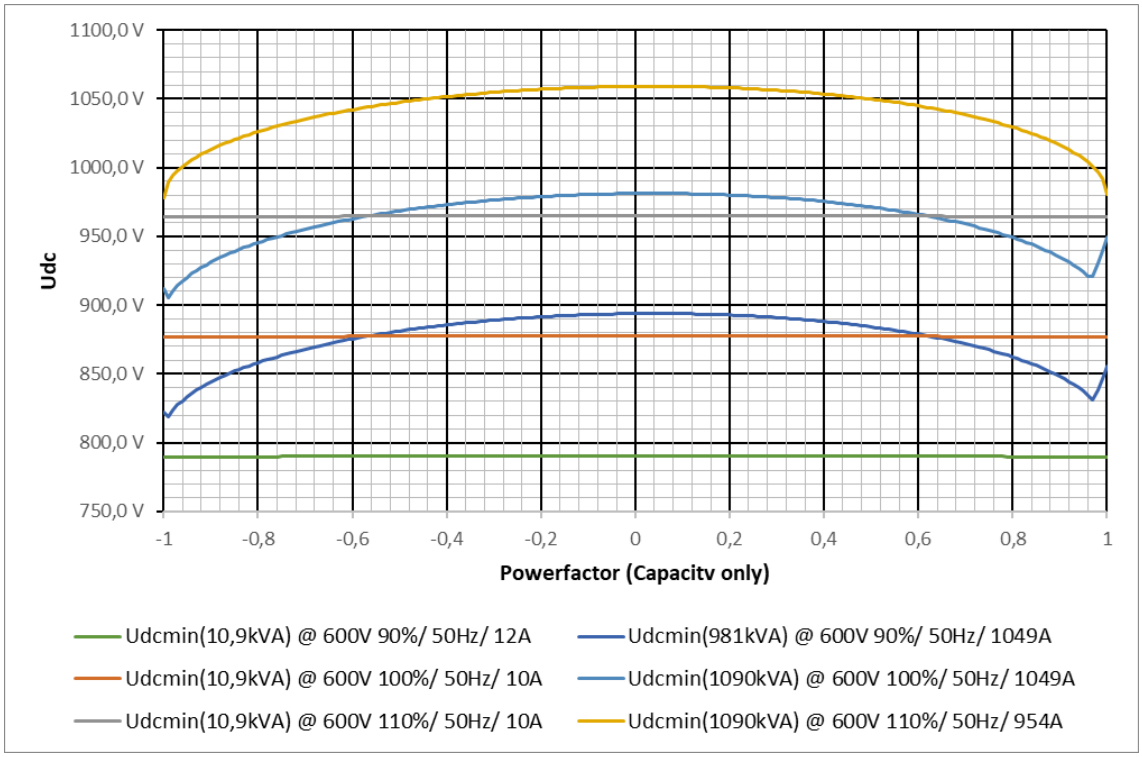


Figure 25: APS1090-xx-1-600-5 Udcmin(Cos(Phi)) 50 Hz

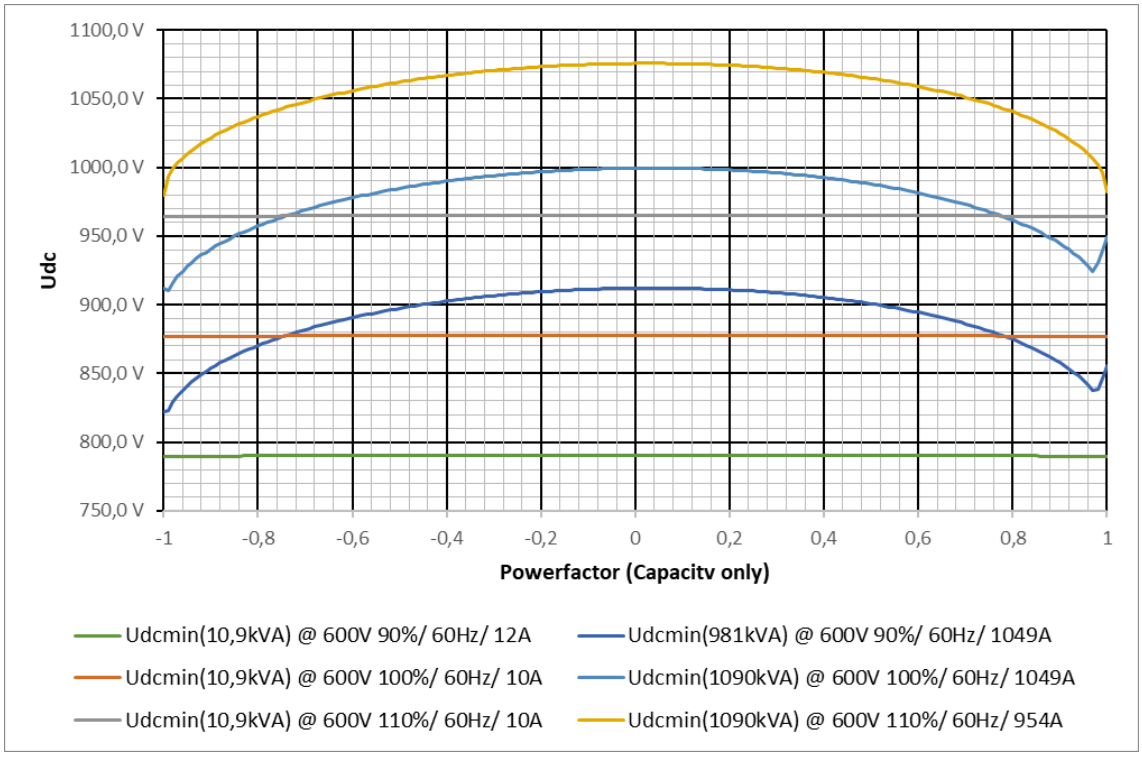


Figure 26: APS1090-xx-1-600-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.10 APS1140-xx-1-630-5

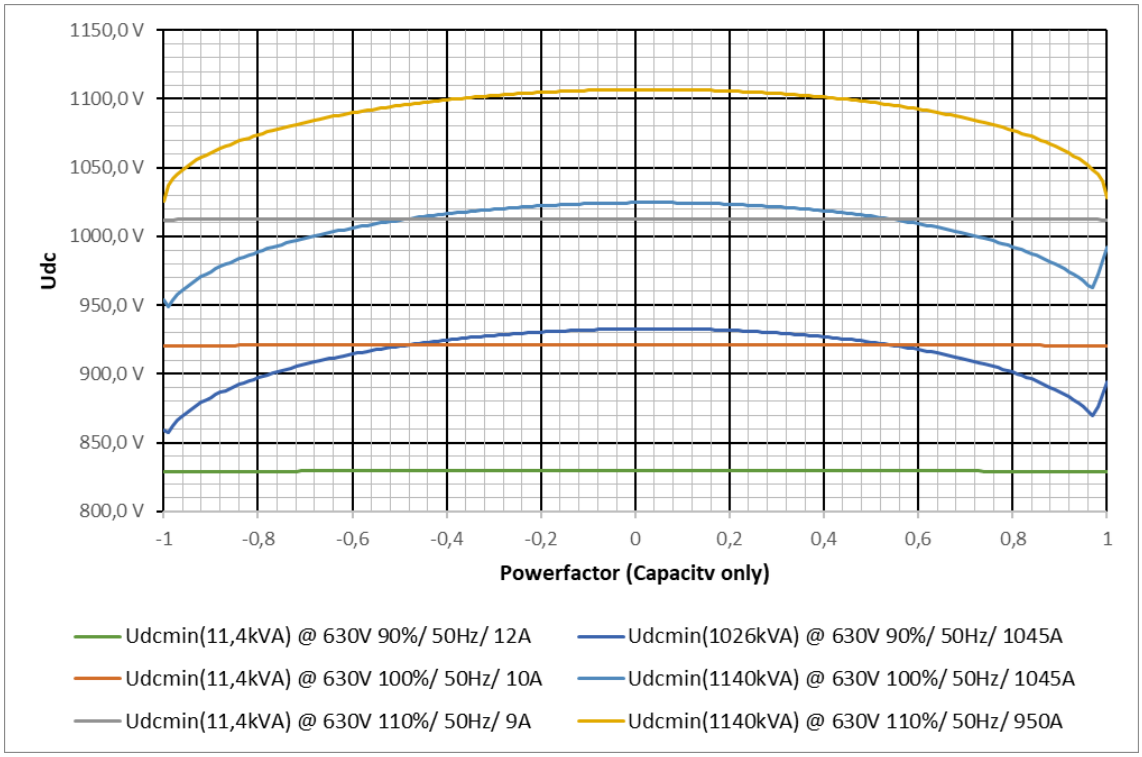


Figure 27: APS1140-xx-1-630-5 Udcmin(Cos(Phi)) 50 Hz

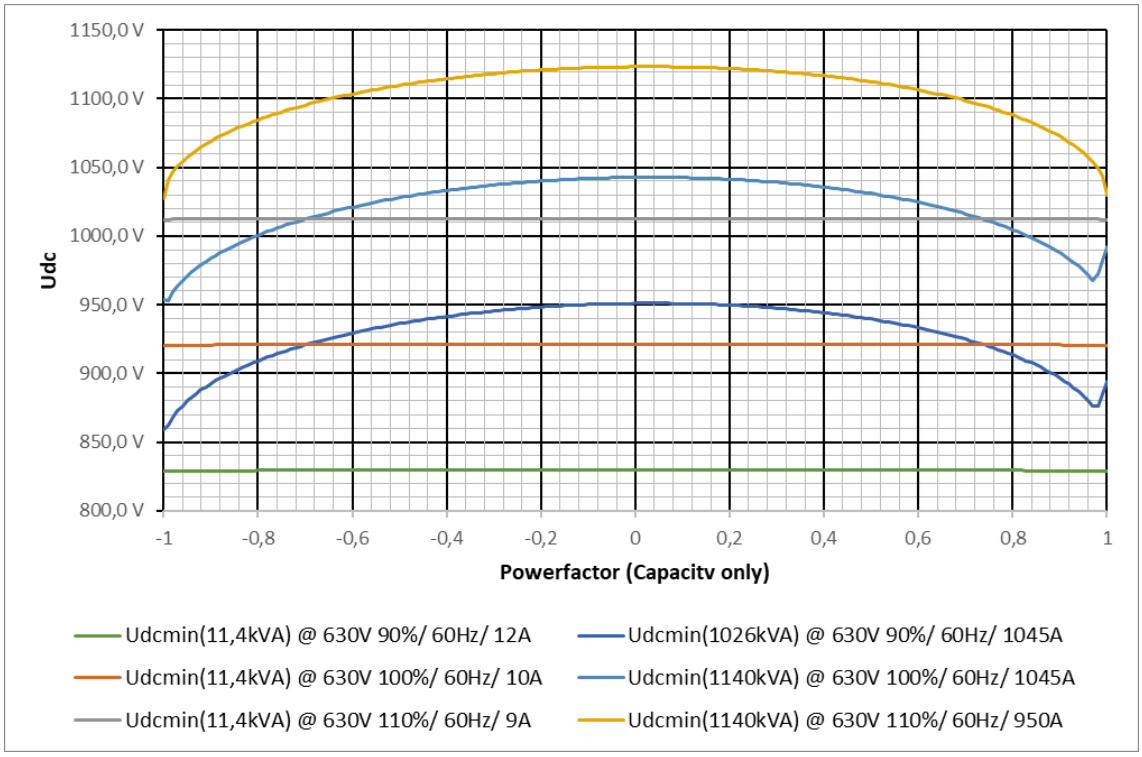


Figure 28: APS1140-xx-1-630-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.11 APS1200-xx-1-660-5

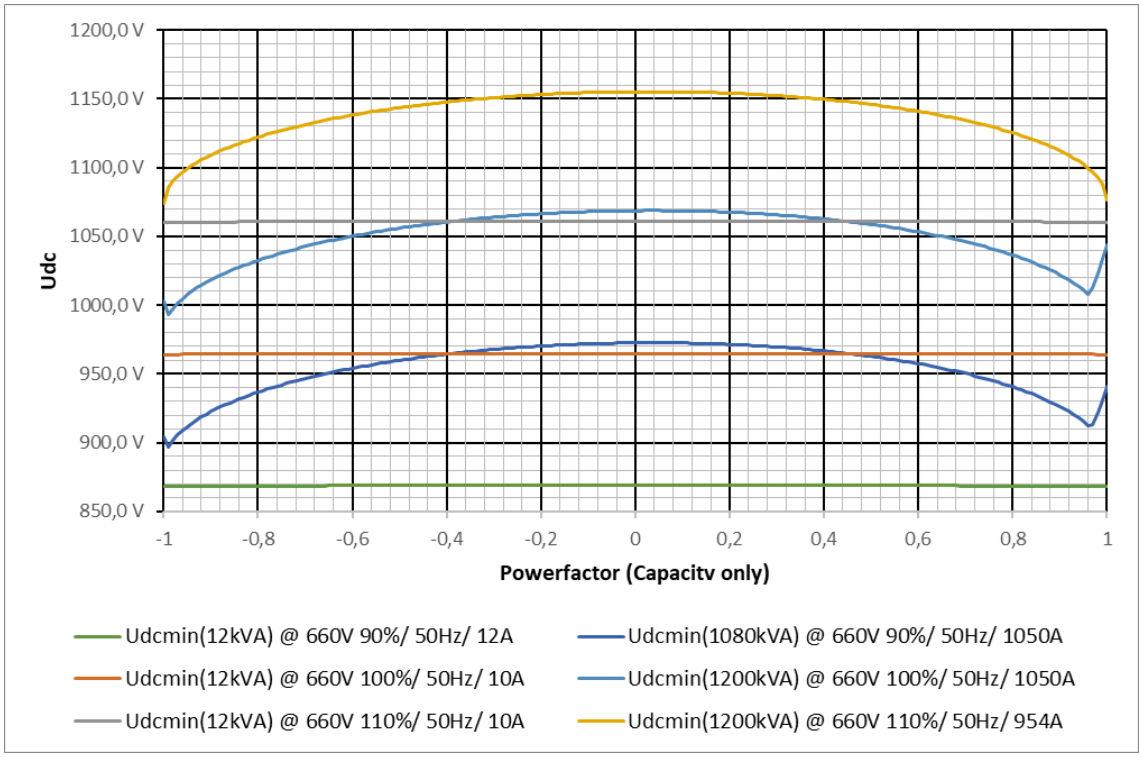


Figure 29: APS1200-xx-1-660-5 Udcmin(Cos(Phi)) 50 Hz

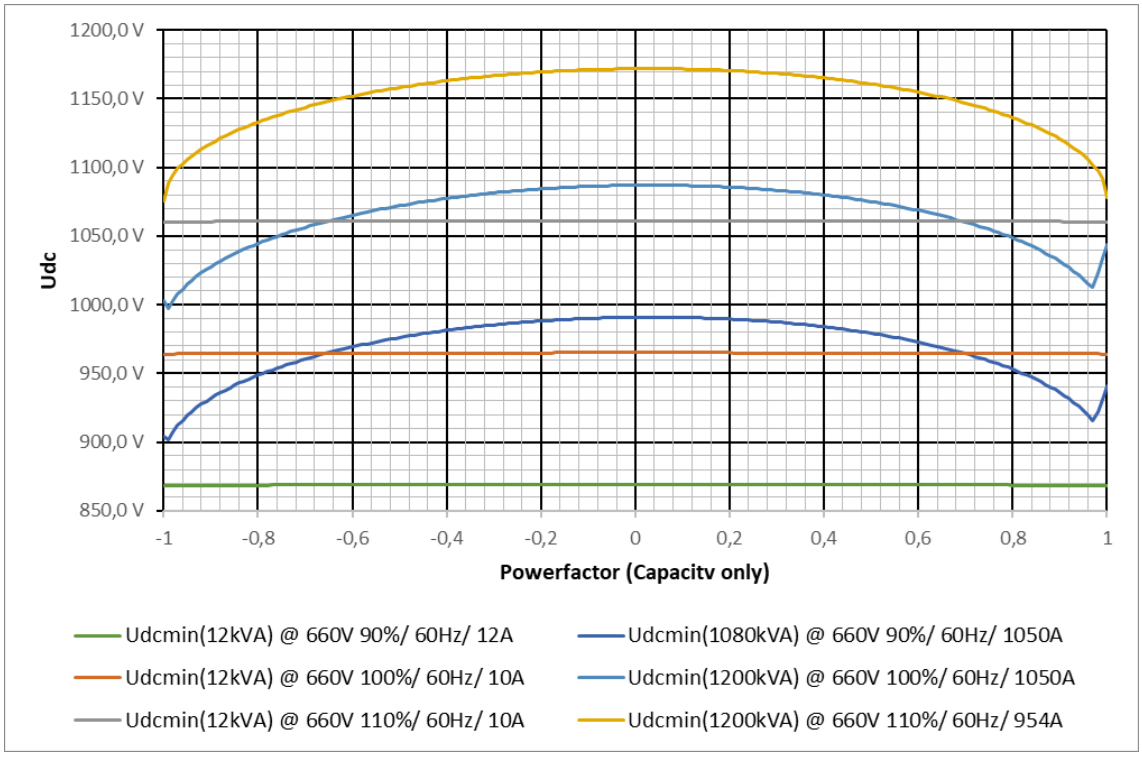


Figure 30: APS1200-xx-1-660-5 Udcmin(Cos(Phi)) 60 Hz

2.2.2.12 APS1250-xx-1-690-5

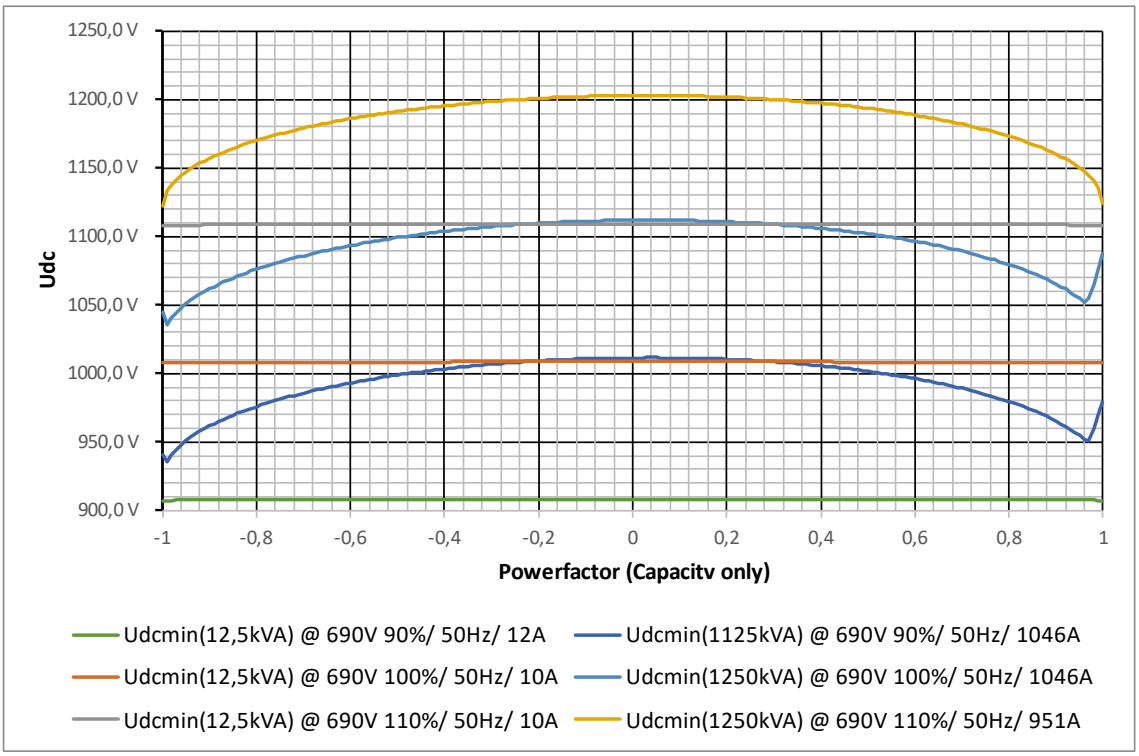


Figure 31: APS1250-xx-1-690-5 Udcmin(Cos(Phi)) 50 Hz

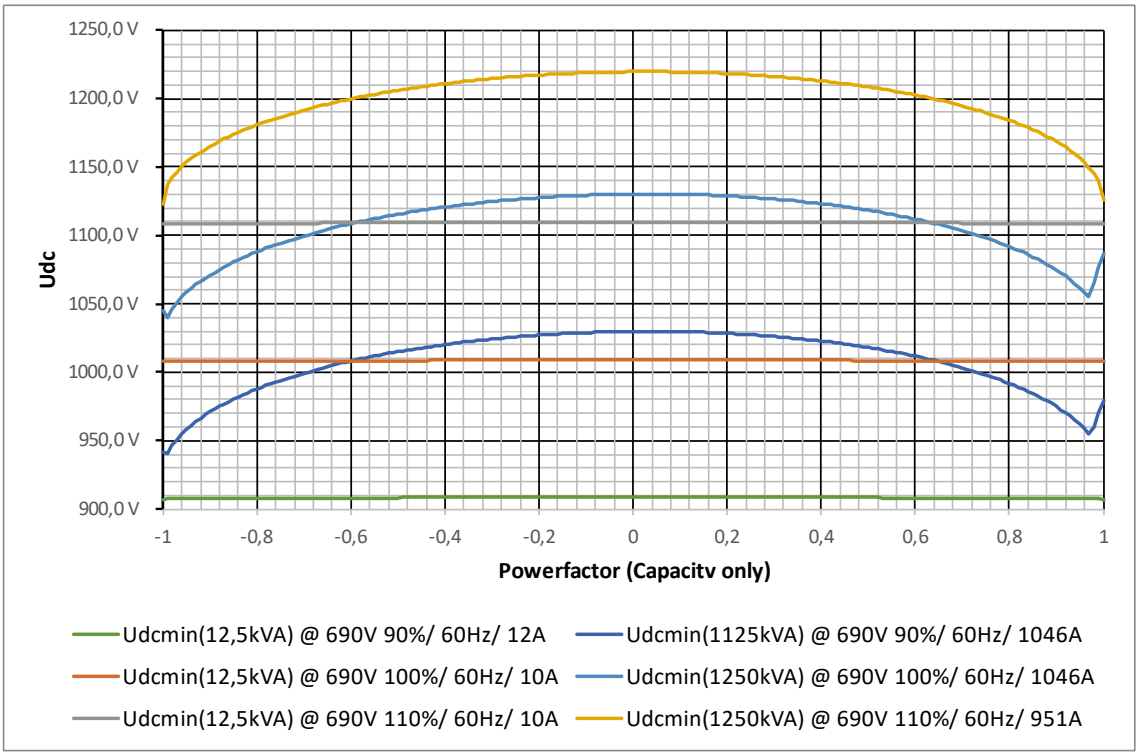


Figure 32: APS1250-xx-1-690-5 Udcmin(Cos(Phi)) 60 Hz

2.3 Rate of Change of Frequency Change (RoCoF)

The APS can operate during a frequency change of up to 4 Hz/s. There are no significant variations in active power or current.

A measurement of a RoCoF-event is shown in the following figures.

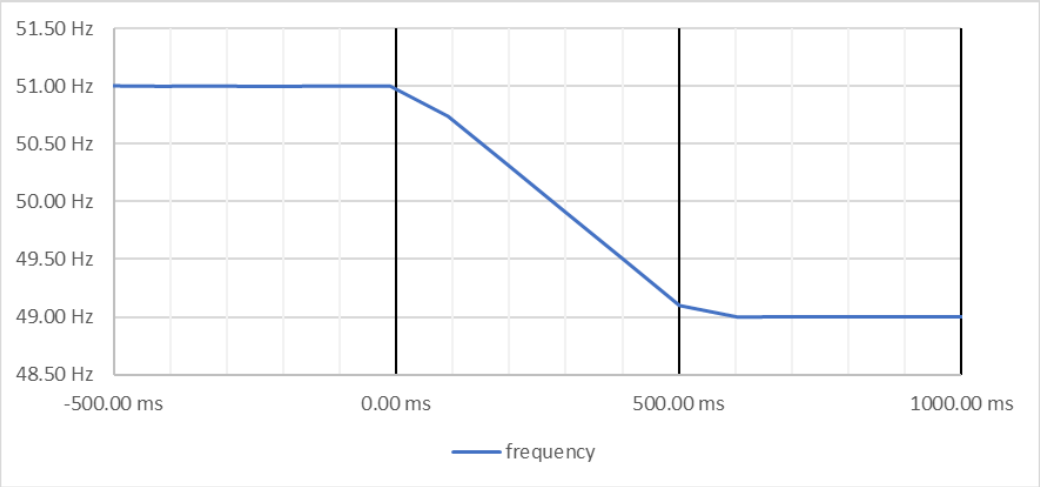


Figure 33: RoCoF-event = 4 Hz/s – frequency drop

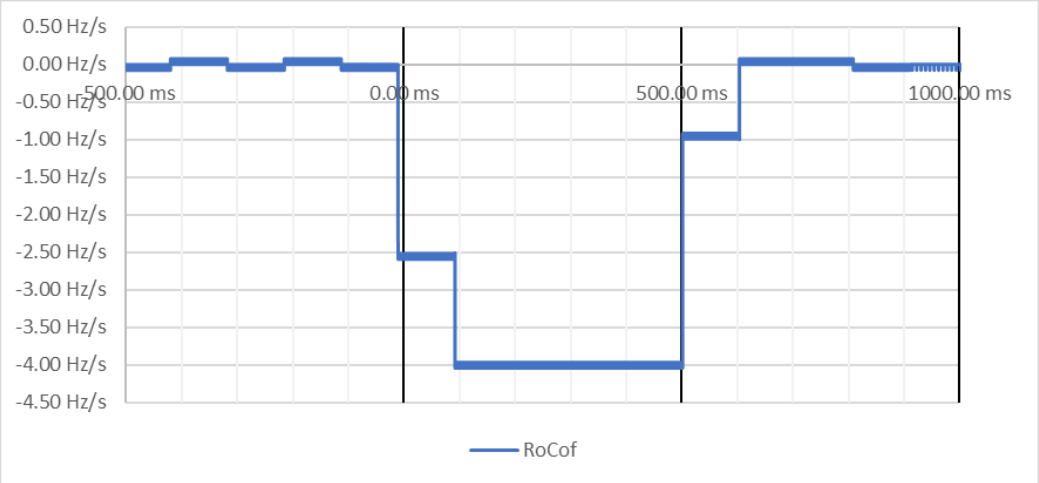


Figure 34: RoCoF-event = 4 Hz/s

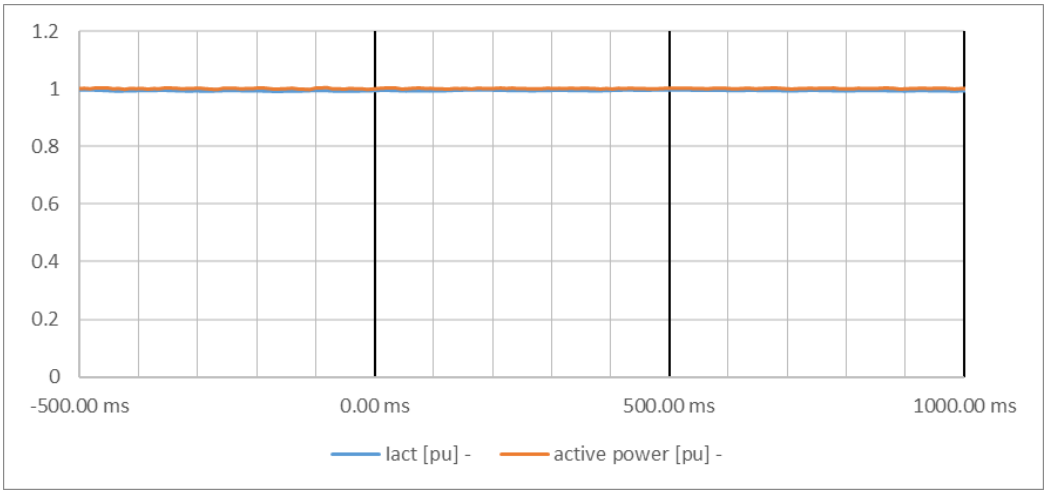


Figure 35: RoCoF-event = 4 Hz/s –Active power and current response

2.4 Performance Reduction

The APS can be operated with reduced parameters in several applications. In the next chapters, different derating scenarios are described.

2.4.1 Altitude

The altitude of the APS-installation can be increased with a limitation of several system parameters. These limitations are marked in the next chapters.

2.4.1.1 DC Voltage

The maximum voltage in the DC-link must be limited by the customer if the APS is used at higher altitudes. The correlation between the maximum DC voltage and the altitude is shown in Figure 36.

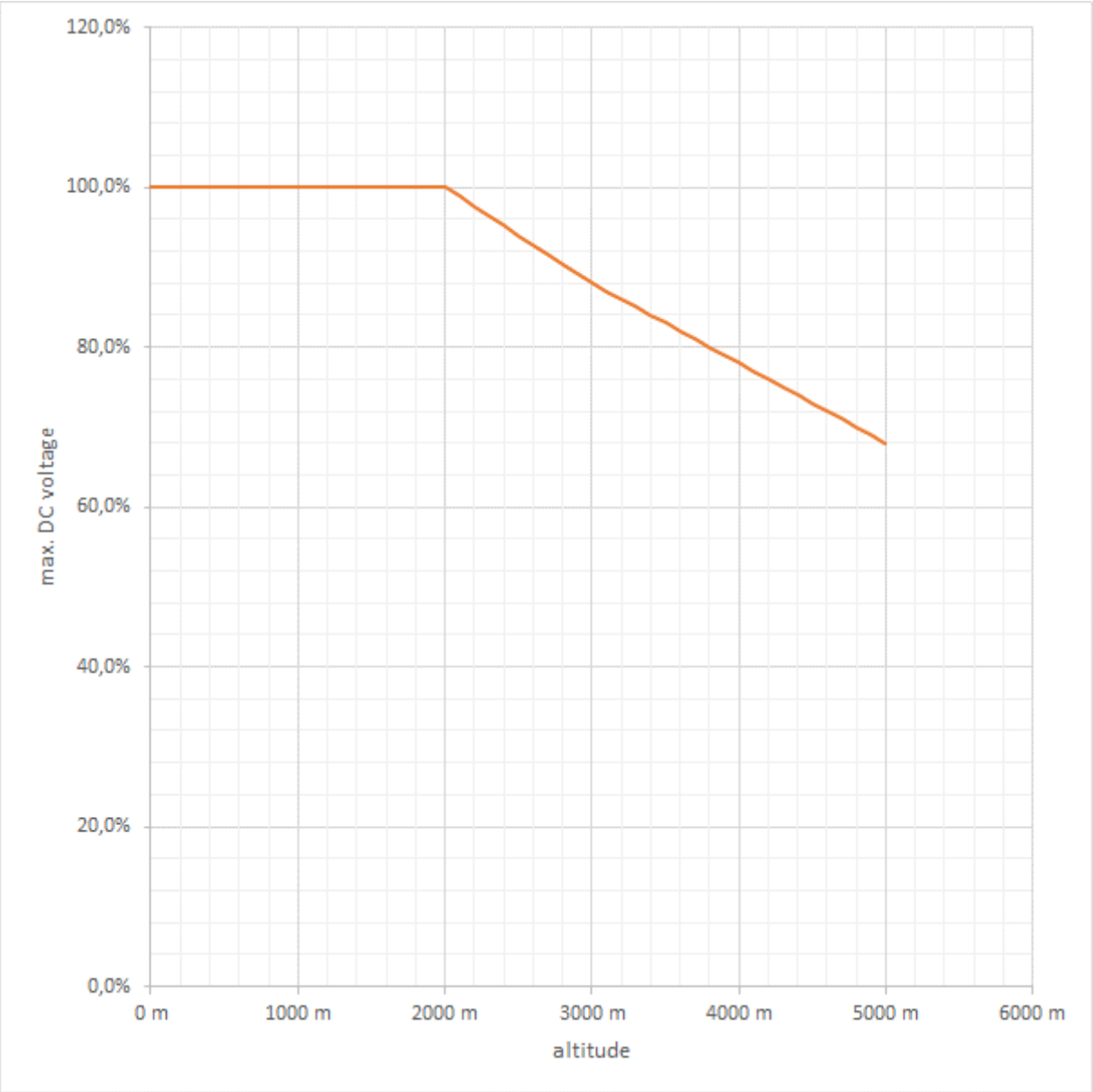


Figure 36: Maximum DC voltage vs. altitude

2.4.1.2 Power

This chapter presents the power reduction depending on the parameters of altitude and ambient temperatures. Figure 37 shows the power as a function of altitude and Figure 38 shows the power as a function of the ambient temperature.

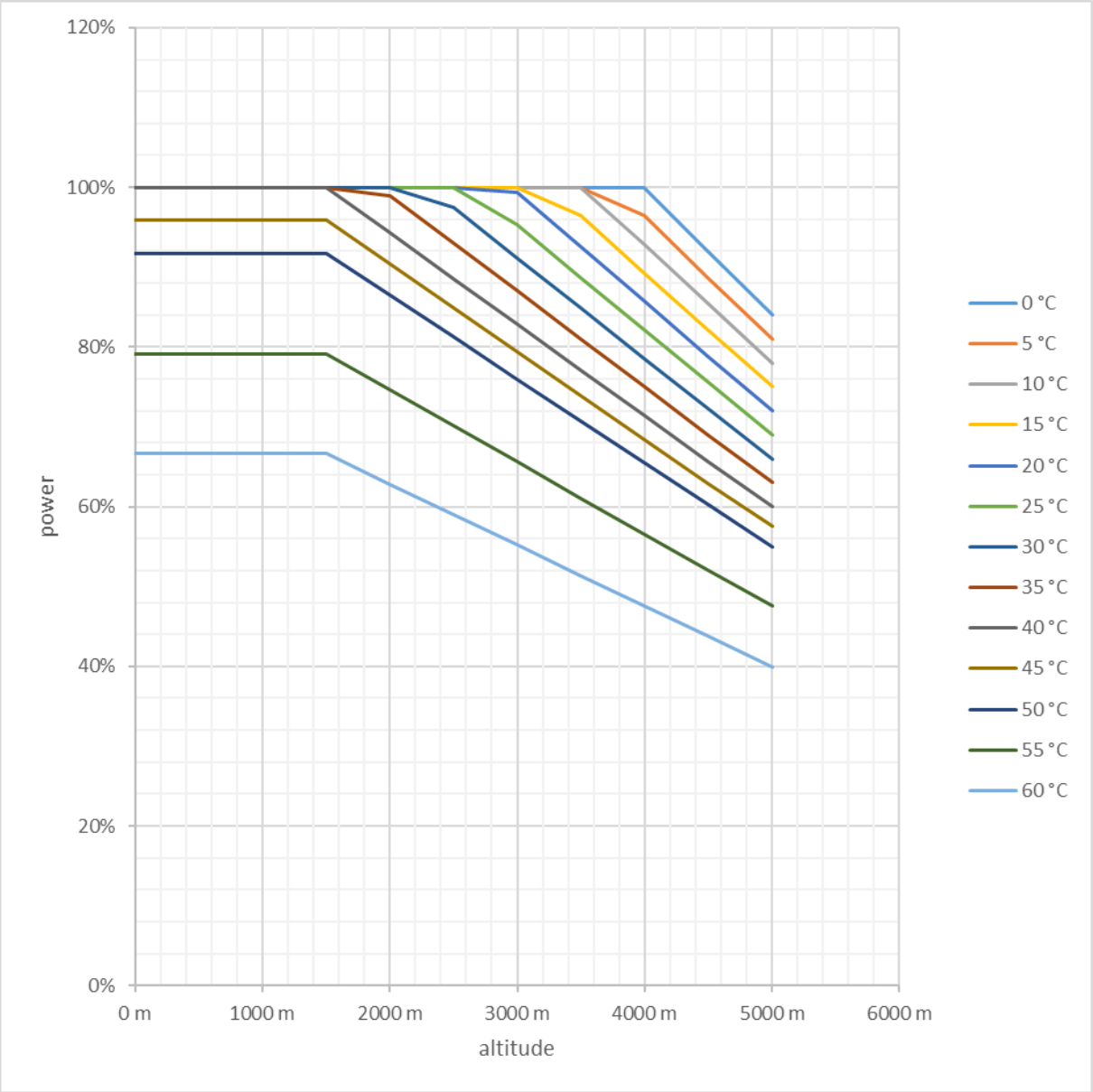


Figure 37: Maximum power vs. (altitude and ambient temperature)

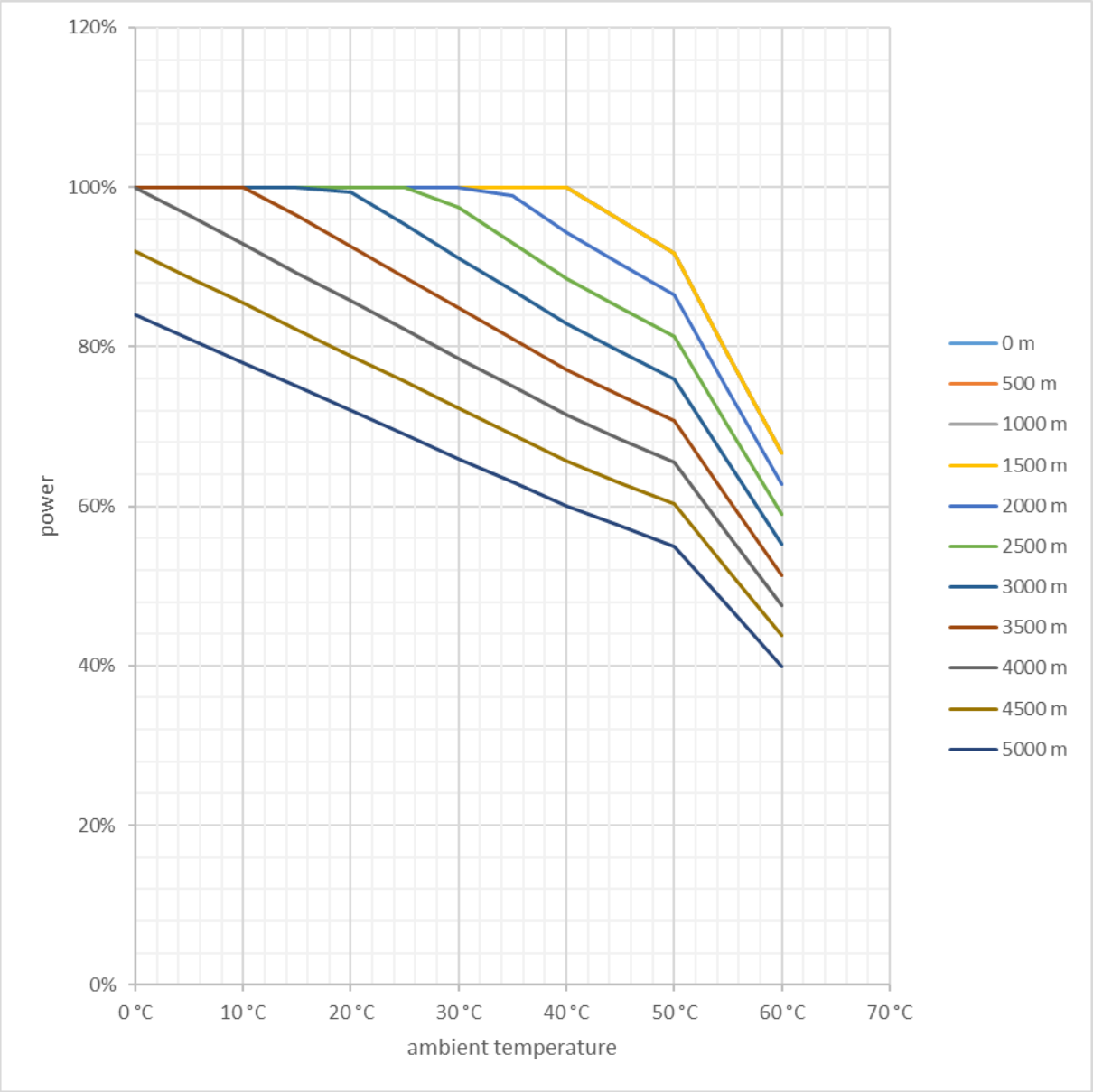


Figure 38: Maximum power vs. (ambient temperature and altitude)

2.4.2 Ambient temperature

This chapter presents the power reduction depending on the ambient temperature. The correlation is shown in Figure 39.

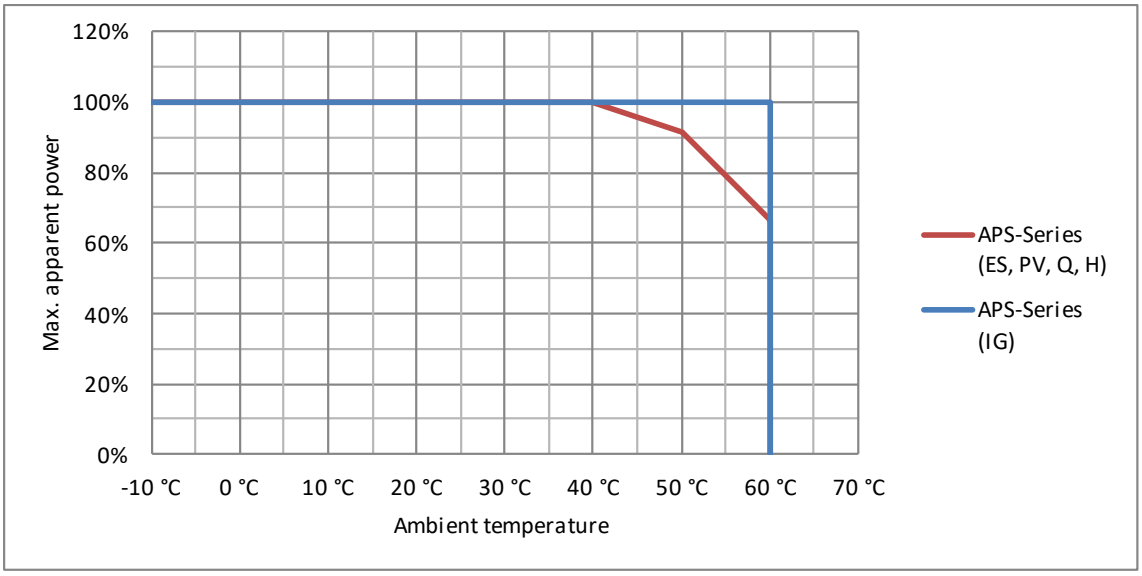


Figure 39: Power reduction vs. Ambient temperature

2.4.3 Transformer temperature

The transformer temperature can be measured with the APS. A power reduction of the inverter can be configured. Please look to the “function and interface description” for further information.

2.5 Power priority PQ(Uac, prio)

Following evaluations show the PQ capability depending on 85 – 115% nominal grid voltage and P/Q prisonization. The P/Q prisonization is configurable (see function and Interface description).

They are valid for the following models:

Table 5: List of inverter models

APS models	APS models
APS650-XX-1-360-5	APS1090-XX-1-600-5
APS1300-XX-2-360-5	APS2180-XX-2-600-5
APS1950-XX-3-360-5	APS3270-XX-3-600-5
APS2600-XX-4-360-5	APS4360-XX-4-600-5
APS725-XX-1-400-5	APS1140-XX-1-630-5
APS1450-XX-2-400-5	APS2280-XX-2-630-5
APS2175-XX-3-400-5	APS3420-XX-3-630-5
APS2900-XX-4-400-5	APS4560-XX-4-630-5
APS800-XX-1-440-5	APS1200-XX-1-660-5
APS1600-XX-2-440-5	APS2400-XX-2-660-5
APS2400-XX-3-440-5	APS3600-XX-3-660-5
APS3200-XX-4-440-5	APS4800-XX-4-660-5
APS870-XX-1-480-5	APS1250-XX-1-690-5
APS1740-XX-2-480-5	APS2500-XX-2-690-5
APS2610-XX-3-480-5	APS3750-XX-3-690-5
APS3480-XX-4-480-5	APS5000-XX-4-690-5
APS945-XX-1-520-5	
APS1890-XX-2-520-5	
APS2835-XX-3-520-5	
APS3780-XX-4-520-5	
APS1000-XX-1-550-5	
APS2000-XX-2-550-5	
APS3000-XX-3-550-5	
APS4000-XX-4-550-5	
APS1045-XX-1-575-5	
APS2090-XX-2-575-5	
APS3135-XX-3-575-5	
APS4180-XX-4-575-5	

Please note, that the grid voltage operating range of the different inverters may reduce. If the voltage level is out of the operating range these chapters have to be ignored and the inverter is not allowed to be operated.

The voltage range is defined in the datasheet. If other ranges are required, please contact our sales department.

2.5.1 PQ-Capability at 85% of nominal grid voltage

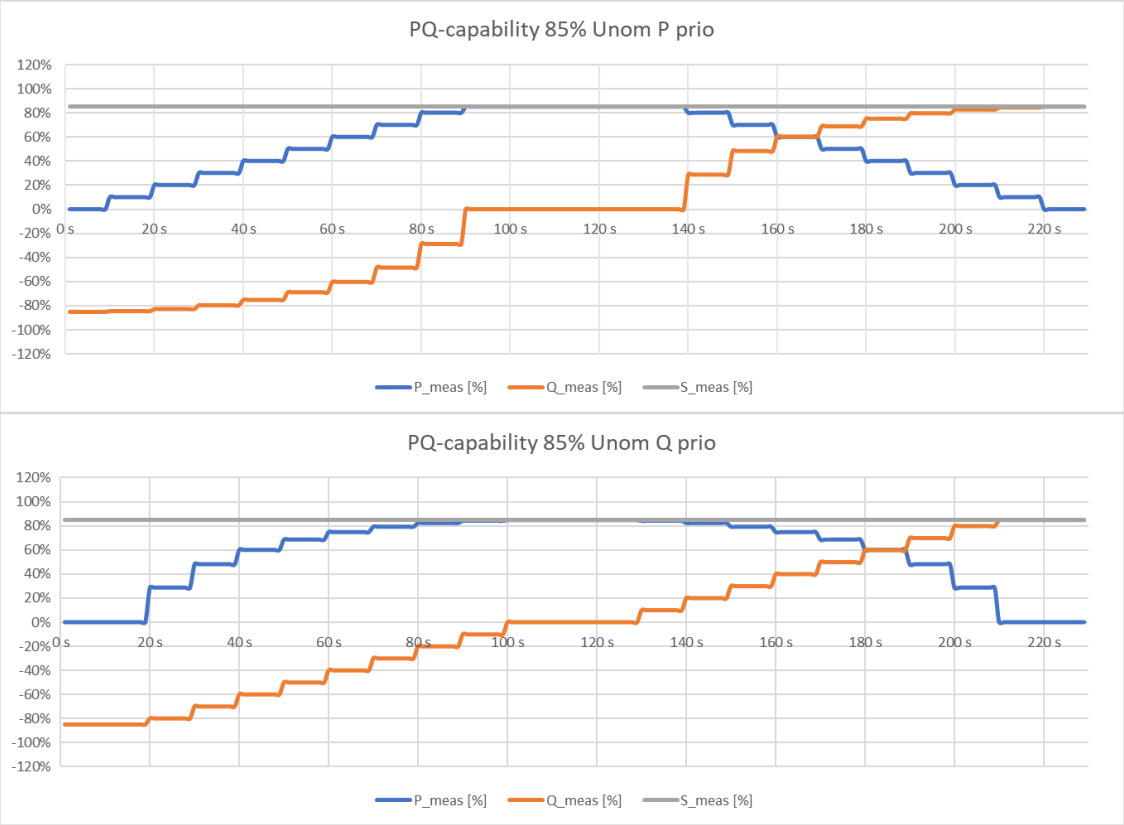


Figure 40: PQ-capability 85% Unom (P- and Q-prioritisation)

Table 6: PQ-capability at 85% Unom (P- and Q-prioritisation)

P Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-85%	85%
10%	-84%	85%
20%	-83%	85%
30%	-80%	85%
40%	-75%	85%
50%	-69%	85%
60%	-60%	85%
70%	-48%	85%
80%	-29%	85%
85%	0%	85%
85%	0%	85%
85%	0%	85%
80%	-29%	85%
70%	-48%	85%
60%	-60%	85%
50%	-69%	85%
40%	-75%	85%
30%	-80%	85%
20%	-83%	85%
10%	-84%	85%
0%	-85%	85%

Q Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-85%	85%
0%	-85%	85%
29%	-80%	85%
48%	-70%	85%
60%	-60%	85%
69%	-50%	85%
75%	-40%	85%
80%	-30%	85%
83%	-20%	85%
84%	-10%	85%
85%	0%	85%
84%	10%	85%
83%	20%	85%
80%	30%	85%
75%	40%	85%
69%	50%	85%
60%	60%	85%
48%	70%	85%
29%	80%	85%
0%	85%	85%
0%	85%	85%

2.5.2 PQ-Capability at 90% of nominal grid voltage

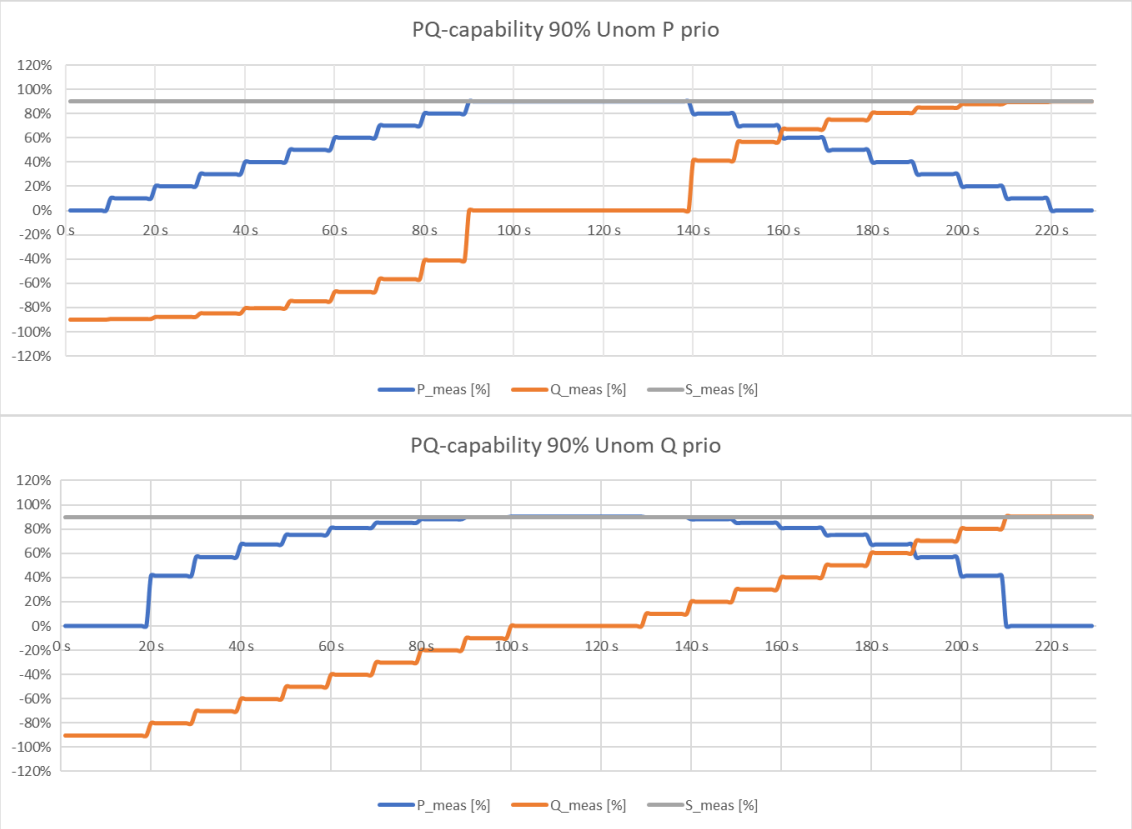


Figure 41: PQ-capability 90% Unom (P- and Q-prioritisation)

Table 7: PQ-capability at 90% Unom (P- and Q-prioritisation)

P Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-90%	90%
10%	-89%	90%
20%	-88%	90%
30%	-85%	90%
40%	-81%	90%
50%	-75%	90%
60%	-67%	90%
70%	-57%	90%
80%	-41%	90%
90%	0%	90%
90%	0%	90%
90%	0%	90%
80%	-41%	90%
70%	-57%	90%
60%	-67%	90%
50%	-75%	90%
40%	-81%	90%
30%	-85%	90%
20%	-88%	90%
10%	-89%	90%
0%	-90%	90%

Q Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-90%	90%
0%	-90%	90%
41%	-80%	90%
57%	-70%	90%
67%	-60%	90%
75%	-50%	90%
81%	-40%	90%
85%	-30%	90%
88%	-20%	90%
89%	-10%	90%
90%	0%	90%
89%	10%	90%
88%	20%	90%
85%	30%	90%
81%	40%	90%
75%	50%	90%
67%	60%	90%
57%	70%	90%
41%	80%	90%
0%	90%	90%
0%	90%	90%

2.5.3 PQ-Capability at 95% of nominal grid voltage

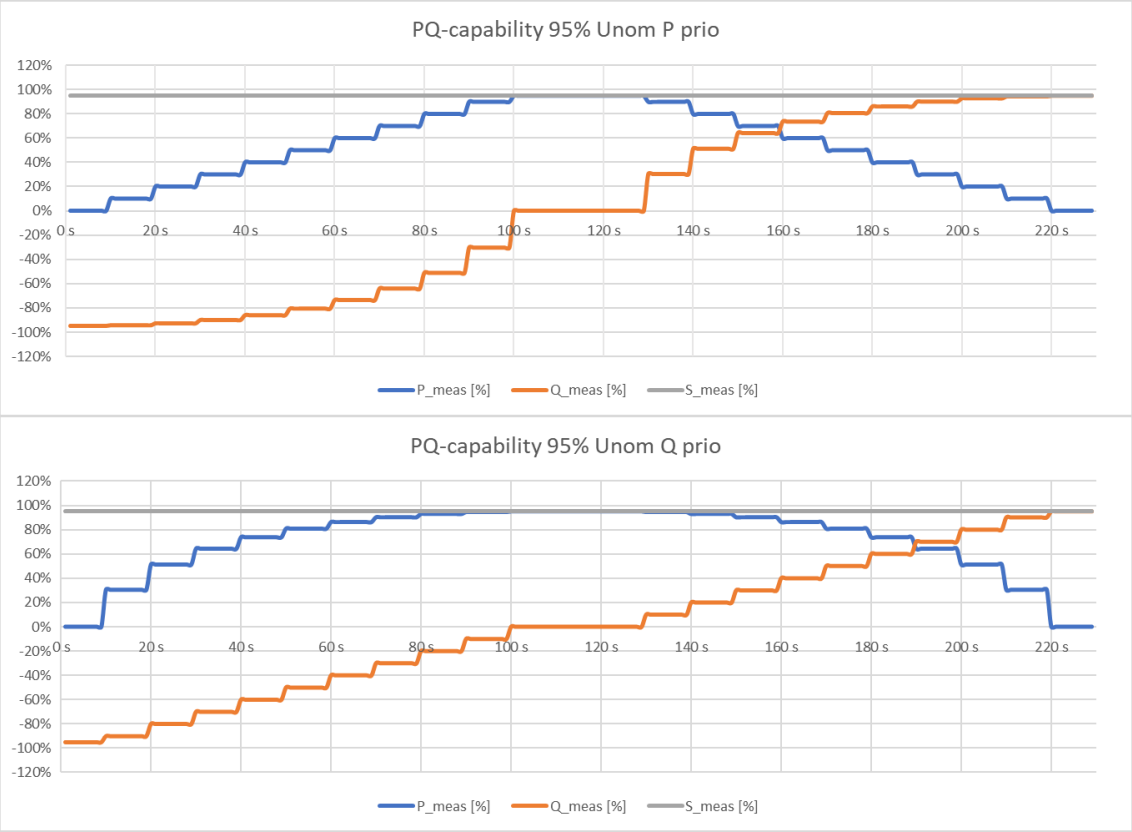


Figure 42: PQ-capability 95% Unom (P- and Q-prioritisation)

Table 8: PQ-capability at 95% Unom (P- and Q-prioritisation)

P Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-95%	95%
10%	-94%	95%
20%	-93%	95%
30%	-90%	95%
40%	-86%	95%
50%	-81%	95%
60%	-74%	95%
70%	-64%	95%
80%	-51%	95%
90%	-30%	95%
95%	0%	95%
90%	-30%	95%
80%	-51%	95%
70%	-64%	95%
60%	-74%	95%
50%	-81%	95%
40%	-86%	95%
30%	-90%	95%
20%	-93%	95%
10%	-94%	95%
0%	-95%	95%

Q Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-95%	95%
30%	-90%	95%
51%	-80%	95%
64%	-70%	95%
74%	-60%	95%
81%	-50%	95%
86%	-40%	95%
90%	-30%	95%
93%	-20%	95%
94%	-10%	95%
95%	0%	95%
94%	10%	95%
93%	20%	95%
90%	30%	95%
86%	40%	95%
81%	50%	95%
74%	60%	95%
64%	70%	95%
51%	80%	95%
30%	90%	95%
0%	95%	95%

2.5.4 PQ-Capability at 100% of nominal grid voltage

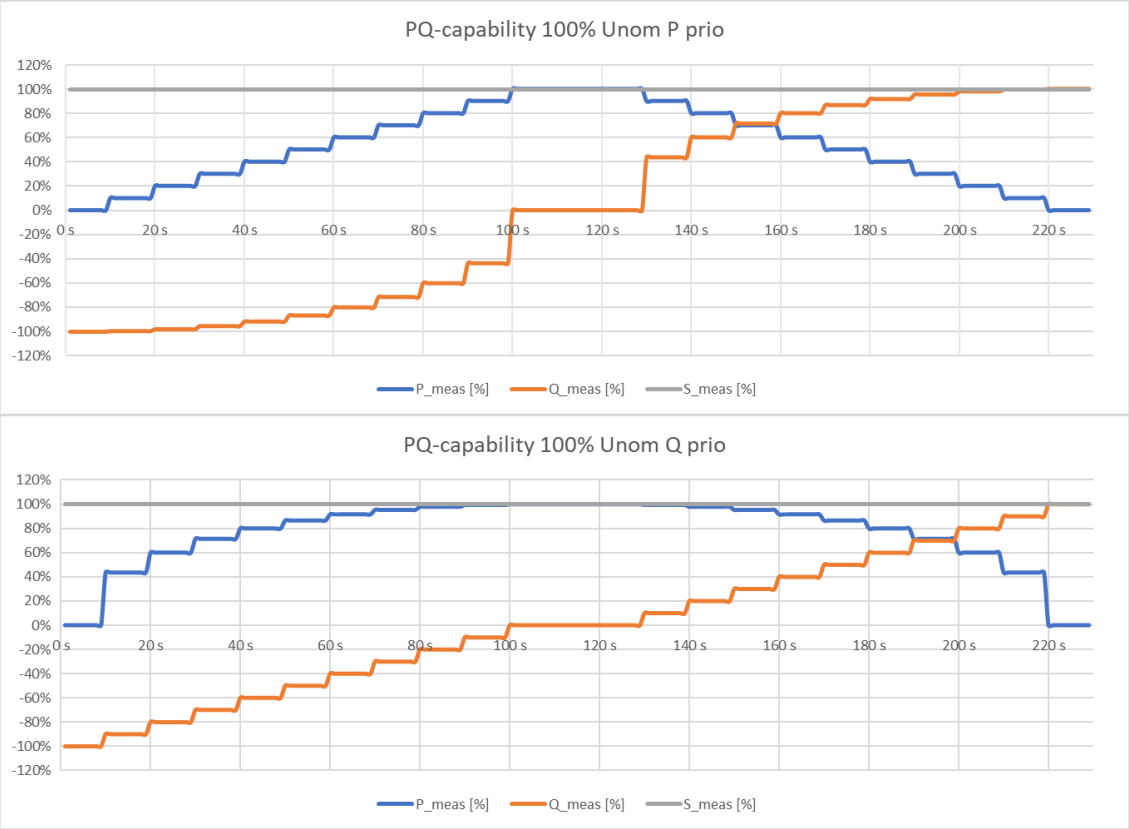


Figure 43: PQ-capability 100% Unom (P- and Q-prioritisation)

Table 9: PQ-capability at 100% Unom (P- and Q-prioritisation)

P Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-100%	100%
10%	-99%	100%
20%	-98%	100%
30%	-95%	100%
40%	-92%	100%
50%	-87%	100%
60%	-80%	100%
70%	-71%	100%
80%	-60%	100%
90%	-44%	100%
100%	0%	100%
90%	-44%	100%
80%	-60%	100%
70%	-71%	100%
60%	-80%	100%
50%	-87%	100%
40%	-92%	100%
30%	-95%	100%
20%	-98%	100%
10%	-99%	100%
0%	-100%	100%

Q Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-100%	100%
44%	-90%	100%
60%	-80%	100%
71%	-70%	100%
80%	-60%	100%
87%	-50%	100%
92%	-40%	100%
95%	-30%	100%
98%	-20%	100%
99%	-10%	100%
100%	0%	100%
99%	10%	100%
98%	20%	100%
95%	30%	100%
92%	40%	100%
87%	50%	100%
80%	60%	100%
71%	70%	100%
60%	80%	100%
44%	90%	100%
0%	100%	100%

2.5.5 PQ-Capability at 105% of nominal grid voltage

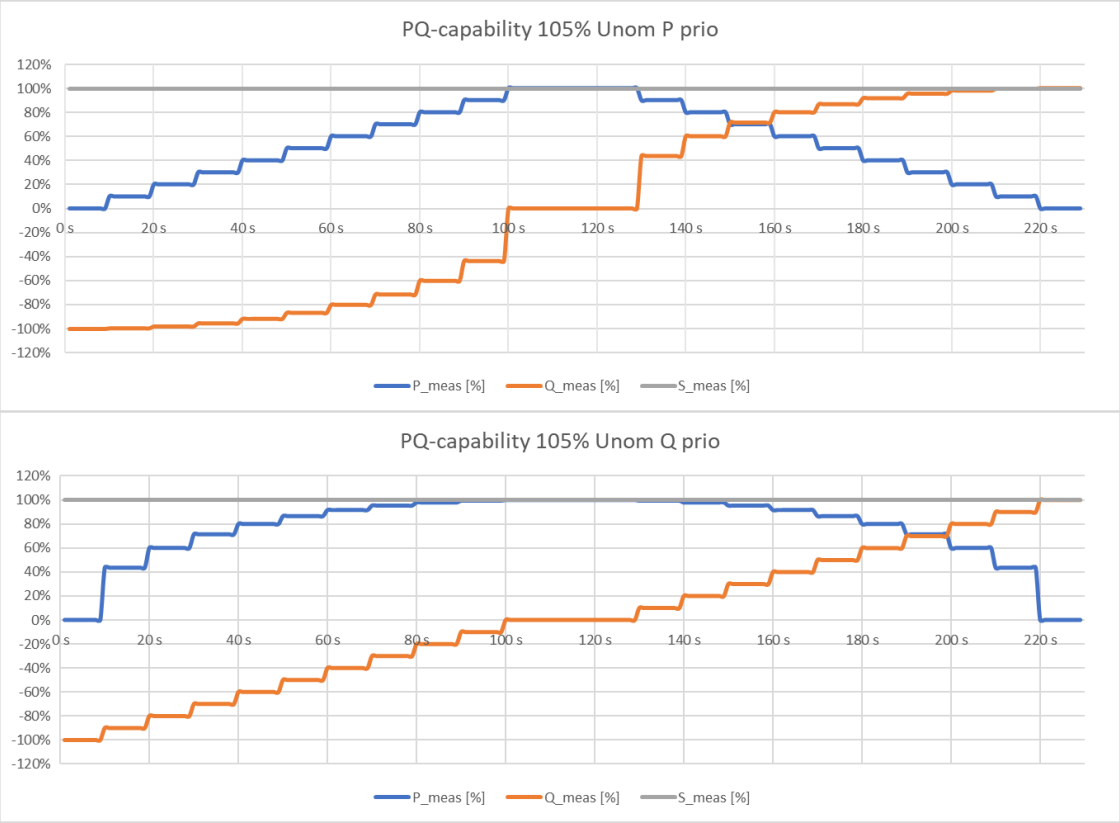


Figure 44: PQ-capability 105% Unom (P- and Q-prioritisation)

Table 10: PQ-capability at 105% Unom (P- and Q-prioritisation)

P Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-100%	100%
10%	-99%	100%
20%	-98%	100%
30%	-95%	100%
40%	-92%	100%
50%	-87%	100%
60%	-80%	100%
70%	-71%	100%
80%	-60%	100%
90%	-44%	100%
100%	0%	100%
90%	-44%	100%
80%	-60%	100%
70%	-71%	100%
60%	-80%	100%
50%	-87%	100%
40%	-92%	100%
30%	-95%	100%
20%	-98%	100%
10%	-99%	100%
0%	-100%	100%

Q Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-100%	100%
44%	-90%	100%
60%	-80%	100%
71%	-70%	100%
80%	-60%	100%
87%	-50%	100%
92%	-40%	100%
95%	-30%	100%
98%	-20%	100%
99%	-10%	100%
100%	0%	100%
99%	10%	100%
98%	20%	100%
95%	30%	100%
92%	40%	100%
87%	50%	100%
80%	60%	100%
71%	70%	100%
60%	80%	100%
44%	90%	100%
0%	100%	100%

2.5.6 PQ-Capability at 110% of nominal grid voltage

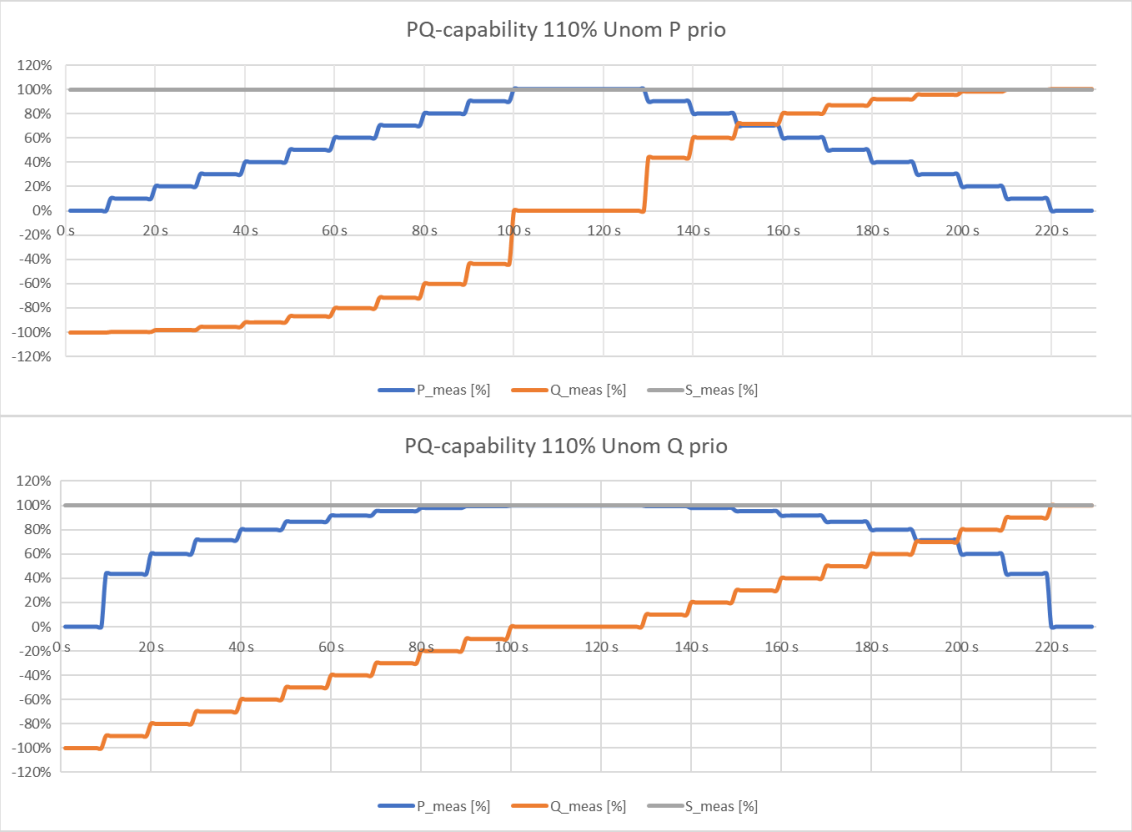


Figure 45: PQ-capability 110% Unom (P- and Q-prioritisation)

Table 11: PQ-capability at 110% Unom (P- and Q-prioritisation)

P Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-100%	100%
10%	-99%	100%
20%	-98%	100%
30%	-95%	100%
40%	-92%	100%
50%	-87%	100%
60%	-80%	100%
70%	-71%	100%
80%	-60%	100%
90%	-44%	100%
100%	0%	100%
90%	-44%	100%
80%	-60%	100%
70%	-71%	100%
60%	-80%	100%
50%	-87%	100%
40%	-92%	100%
30%	-95%	100%
20%	-98%	100%
10%	-99%	100%
0%	-100%	100%

Q Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-100%	100%
44%	-90%	100%
60%	-80%	100%
71%	-70%	100%
80%	-60%	100%
87%	-50%	100%
92%	-40%	100%
95%	-30%	100%
98%	-20%	100%
99%	-10%	100%
100%	0%	100%
99%	10%	100%
98%	20%	100%
95%	30%	100%
92%	40%	100%
87%	50%	100%
80%	60%	100%
71%	70%	100%
60%	80%	100%
44%	90%	100%
0%	100%	100%

2.5.7 PQ-Capability at 115% of nominal grid voltage

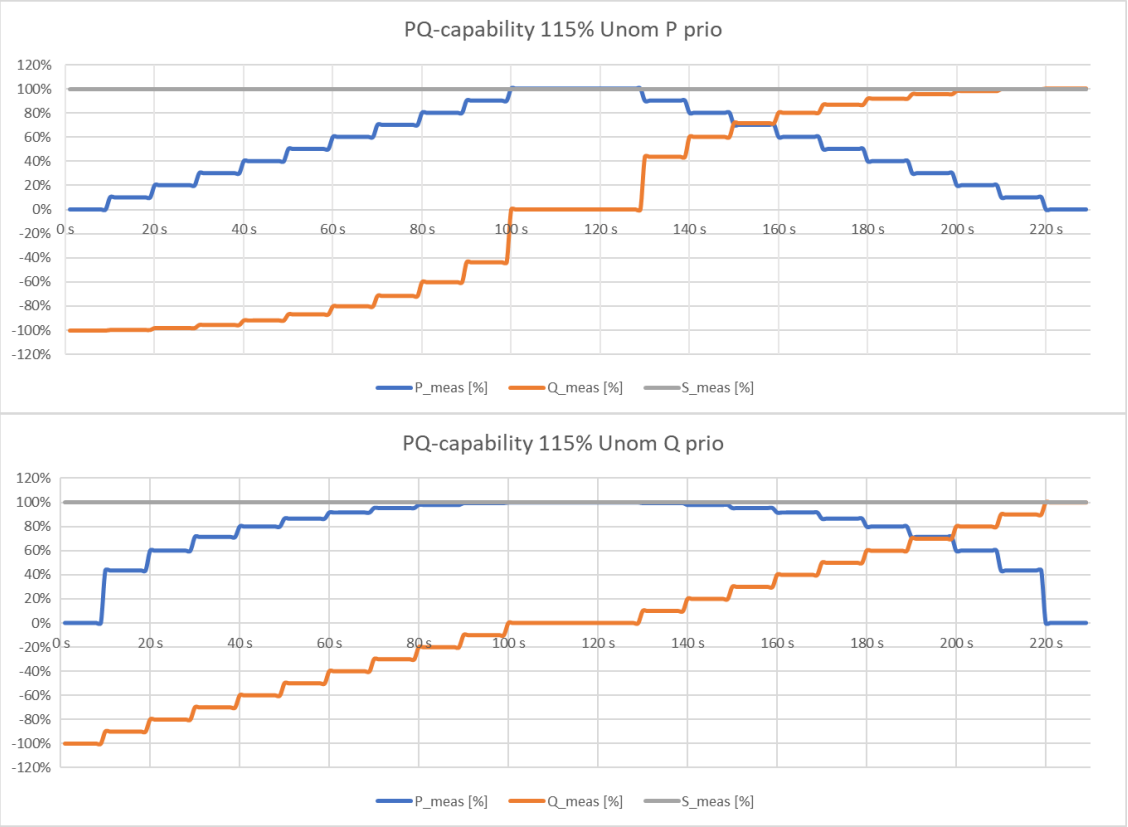


Figure 46: PQ-capability 115% Unom (P- and Q-prioritisation)

Table 12: PQ-capability at 115% Unom (P- and Q-prioritisation)

P Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-100%	100%
10%	-99%	100%
20%	-98%	100%
30%	-95%	100%
40%	-92%	100%
50%	-87%	100%
60%	-80%	100%
70%	-71%	100%
80%	-60%	100%
90%	-44%	100%
100%	0%	100%
90%	-44%	100%
80%	-60%	100%
70%	-71%	100%
60%	-80%	100%
50%	-87%	100%
40%	-92%	100%
30%	-95%	100%
20%	-98%	100%
10%	-99%	100%
0%	-100%	100%

Q Prio		
P_meas [%]	Q_meas [%]	S_meas [%]
0%	-100%	100%
44%	-90%	100%
60%	-80%	100%
71%	-70%	100%
80%	-60%	100%
87%	-50%	100%
92%	-40%	100%
95%	-30%	100%
98%	-20%	100%
99%	-10%	100%
100%	0%	100%
99%	10%	100%
98%	20%	100%
95%	30%	100%
92%	40%	100%
87%	50%	100%
80%	60%	100%
71%	70%	100%
60%	80%	100%
44%	90%	100%
0%	100%	100%

3 Inverter characteristic

The specific behaviour of the inverter is described in more detail in this chapter. The following points are addressed:

- Auxiliary Supply
- Measuring Accuracy
- Grid Side Inverter Impedance
- DC Common-Mode Voltage
- Short Circuit Current contribution
- Short Circuit Current withstand capability
- Losses during STATCOM operation

3.1 Auxiliary Supply

The APS has three separate inputs for the auxiliary supply. These different inputs are explained in the following chapters.

3.1.1 Control System

The control system is internal supplied with a 24 Vdc power supply. This device is feed from the interface of the 400/230 Vac control voltage supply.

The recommended external UPS should be ≥1000 VA (see operating instructions document).

The inrush current of the auxiliary input is shown in the diagrams below (Figure 47 and Figure 48).

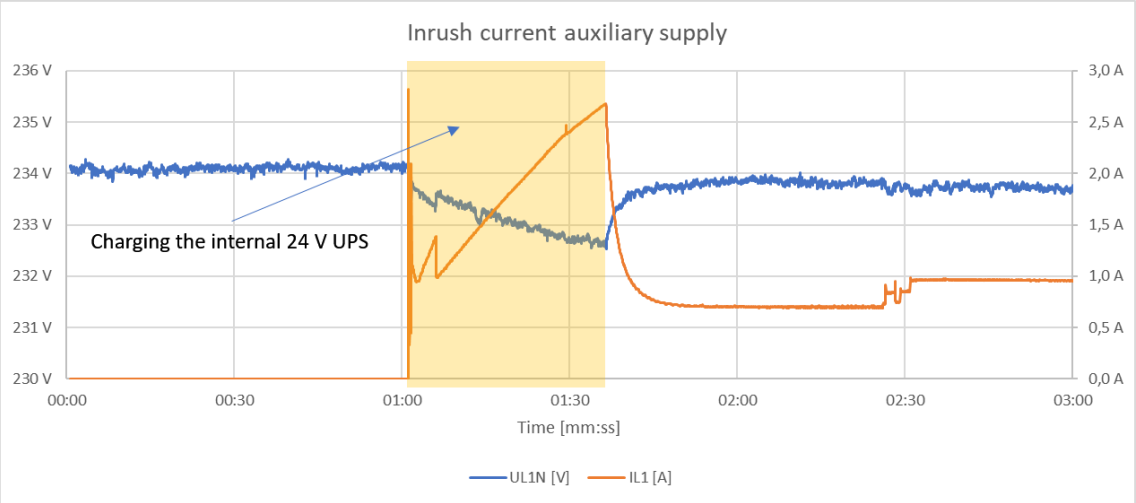


Figure 47: Inrush current auxiliary supply input (overview)

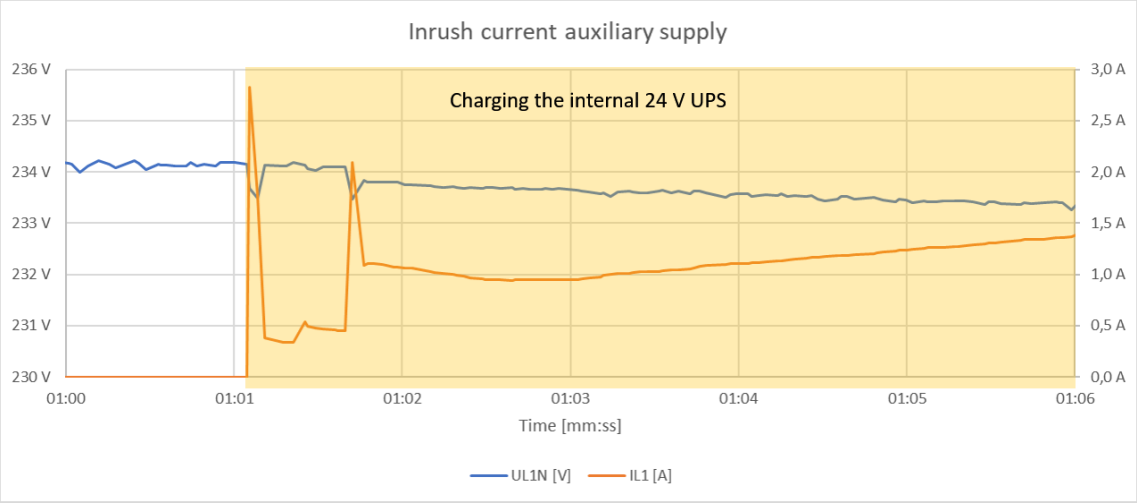


Figure 48: Inrush-current auxiliary supply input (detailed)

The power requirements of the control system are shown in Table 13.

Table 13: Auxiliary supply - control system

Component	Operating mode	S_{Peak}	P	Q	S
Control system	Startup	-	550 W		600 VA
Control system	Idle	-	350 W	-	390 VA
Control system	Full load*	-	480 W	-	550 VA

*Max. est. value

Detailed information about the active power demand in different situations are presented in the following chapters.

3.1.1.1 Standard Components

Table 14 presents the power demand of the APS in the 24 Vdc-Control part. This includes all parts like:

- APMC
- APU control cards and peripheries (IGBT driver and current sensors)
- Display
- Watchdog

These components have a different power demand depending on the operation state of the APS. The power demands marked for:

- Energy Saving (APUs Operating State 0)
- Standby (APUs Operating state up to AC connected)
- Idle (APUs operating with 0 kVA)
- Full load (APUs operating with full load)

Table 14: Auxiliary supply - basic control

Number of APUs	Energy saving	Standby	Idle	Full load
1	34.43 W	68.54 W	98.74 W	113.18 W
2	34.43 W	89.30 W	149.70 W	178.58 W
3	38.78 W	127.76 W	218.36 W	261.68 W
4	38.78 W	148.53 W	269.33 W	327.08 W

3.1.1.2 Optional Components

Table 15 presents the power demand of several other components of the APS. These are optional or only in temporary use.

Table 15: Auxiliary supply - optional components

Additional components	Value	Estimated usage in h/year
Lights, each	2.5 W	During service
Dehumidifier	120.0 W	-
CMB idle, each	22.8 W	8760 h
CMB full load, each	43.0 W	Depending on application
Display illuminated	1.5 W	200 h

3.1.2 Power Demand Cooling System

Table 16 presents the power demand of the different components of the APS cooling system. This table presents only the power demand of one cooling system. APS equipped with 3 or 4 APUs are using 2 of these cooling systems.

Table 16: Auxiliary supply - cooling system

Component	Operating mode	S_{Peak}	P	Q	S
Air-air heat exchanger	low speed	-	115 W	-32 var	119 VA
Air-air heat exchanger	high speed	-	750 W	6 var	750 VA
Water pump (50Hz Version)	on	-	637 W	843 var	1057 VA
Water pump (60Hz Version)	on	-	643 W	683 var	938 VA
Water-air heat exchanger (50Hz Version, 1500V-Serie)	on	-	963 W	823 var	1267 VA
Water-air heat exchanger (60Hz Version, 1500V-Serie)	on	-	942 W	858 var	1274 VA

There are no operating hours per year of the different components available. These are related to:

- Kind of using the APS
- Ambient conditions like temperature and wind
- Other

3.1.3 Power Demand Heating-System

Table 17 presents the power demand of the heating system of the APS. The heating is supplied from a central feeder. Due to this, the power demands have to be summed up.

Table 17: Auxiliary supply - heating

Component	Operating mode	S_{Peak}	P	Q	S
Heating control unit	on	-	400 W	- var	- VA
Heating power unit APU 1/2	on	-	1200 W	- var	- VA
Heating DC distribution APU 1/2	on	-	400 W	- var	- VA
Heating power unit APU 3/4	on	-	1200 W	- var	- VA
Heating DC distribution APU 3/4	on	-	400 W	- var	- VA

3.1.4 Option +-20% auxiliary supply

This option uses an auxiliary transformer with three output windings to compensate fluctuations up to +-20% of the grid voltage.

There are defined thresholds when to switch between the output windings to ensure an output voltage of +-10% of the nominal value.

The graph in Figure 49 shows the RMS of a simulated fluctuation of the primary voltage and the corresponding RMS of the secondary voltage when this option is active.

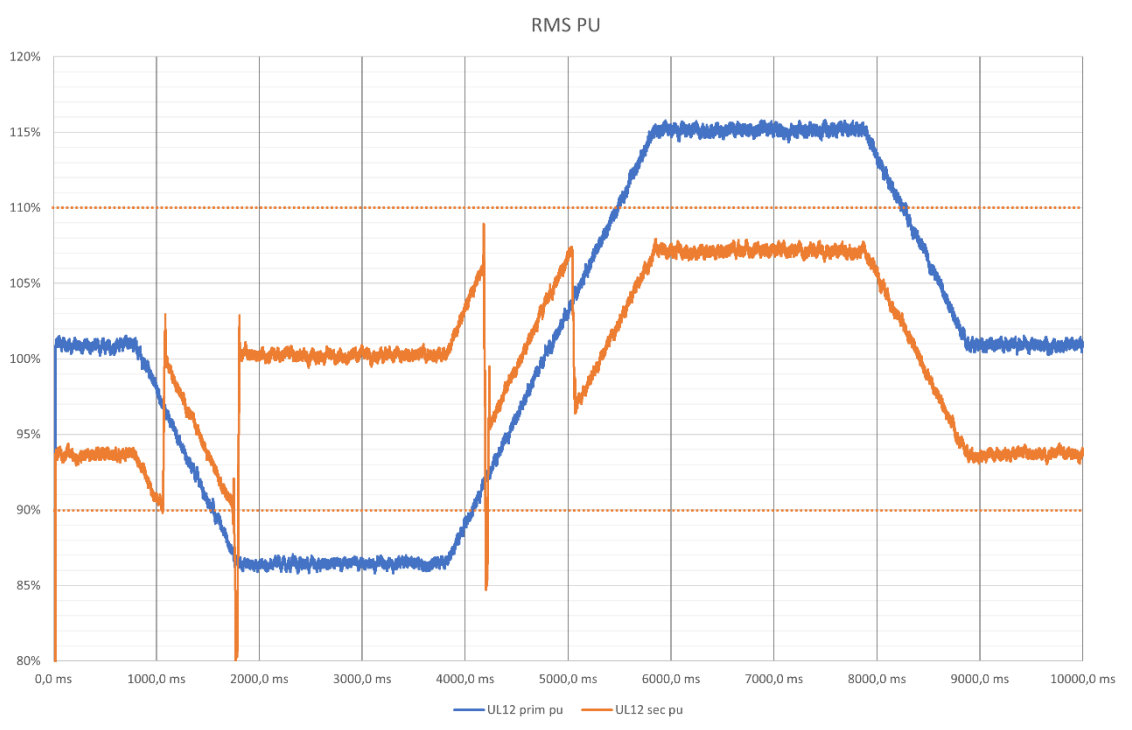


Figure 49: RMS of primary and secondary voltage

3.1.5 Buffer time (LVRT)

The APS has an internal buffer device (24 V-UPS) to withstand grid voltage dips and stays connected to the grid. This device will ensure the 24 V power supply using the internal auxiliary transformer and no external UPS.

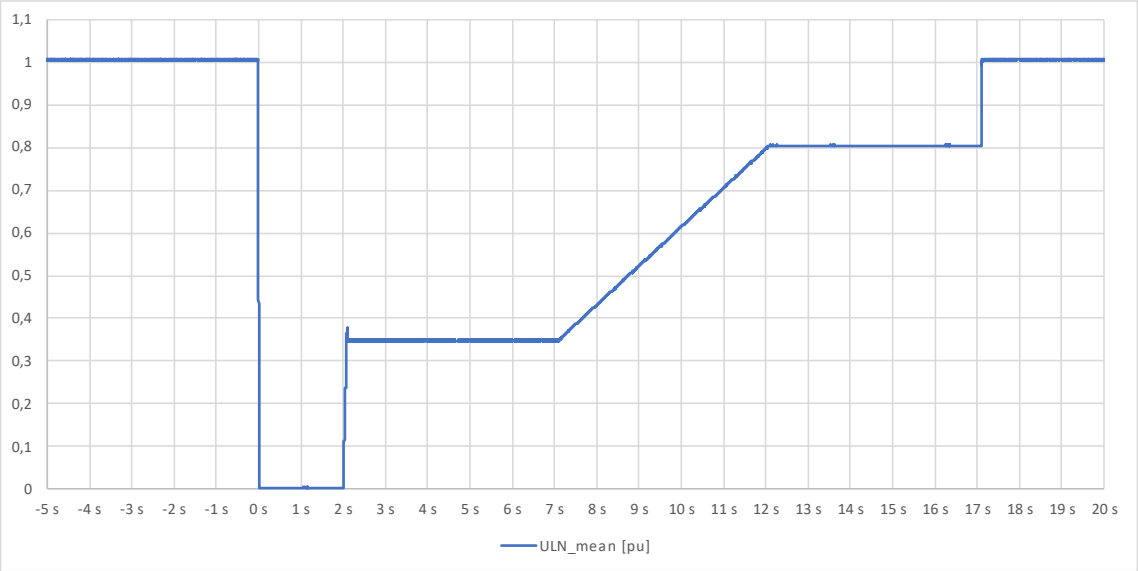


Figure 50: LVRT limitation curve

Figure 50 shows the LVRT limitation curve, which fulfil the VDE guideline. Below 35% of the nominal grid voltage the internal buffer device ensures the power supply for the PLC and the control cards. Only the water pump, the fans and the heating are not buffered.

Under full load the internal buffer device ensures the power supply for 2 s if the grid voltage is below 35 % of the nominal grid voltage.

3.2 Measuring Accuracy

3.2.1 APU Control Card

Table 18 shows the interfaces of the APU control card.

Table 18: Measuring accuracies of APU control card

Measurement	Measuring range	Measuring accuracy	Comments
AC voltage	- 1500 V_{AC} … + 1500 V_{AC}	± 1.0%	-
AC current	- 2000 A_{AC} … + 2000 A_{AC}	± 1.0%	-
frequency	40 … 70 Hz	+-10 mHz	relative
		+-50 mHz	absolute
DC voltage	- 25 V_{DC} … + 900 V_{DC}	± 1.0%	Pos. and neg. DC-link voltage
DC currents	- 1530 A_{DC} … + 1530 A_{DC}	± 1.0%	-
Temperature (ambient)	- 45 °C … + 125 °C	± 1.0% at + 5 °C … + 60 °C	-

3.2.2 Current Measurement Board

There are different current measurement boards (CMB), with different current ranges, available.

Table 19 shows the interfaces of the CMB.

Table 19: Measuring accuracies of Current measurement board

Measurement	Measuring range	Measuring accuracy	Comments
DC voltage	- 1800 V_{DC} … + 1800 V_{DC}	± 1.0%	All HW versions
DC current	- 300 A_{DC} … + 300 A_{DC}	± 0.4%	300A Board
DC current	- 420 A_{DC} … + 420 A_{DC}	± 0.4%	420A Board

3.3 Grid Side Inverter Impedance

Each APU has its own grid side harmonic filter. The single line diagram of each APU is shown in Figure 51 and Figure 52 depending on the maximum DC-link voltage.

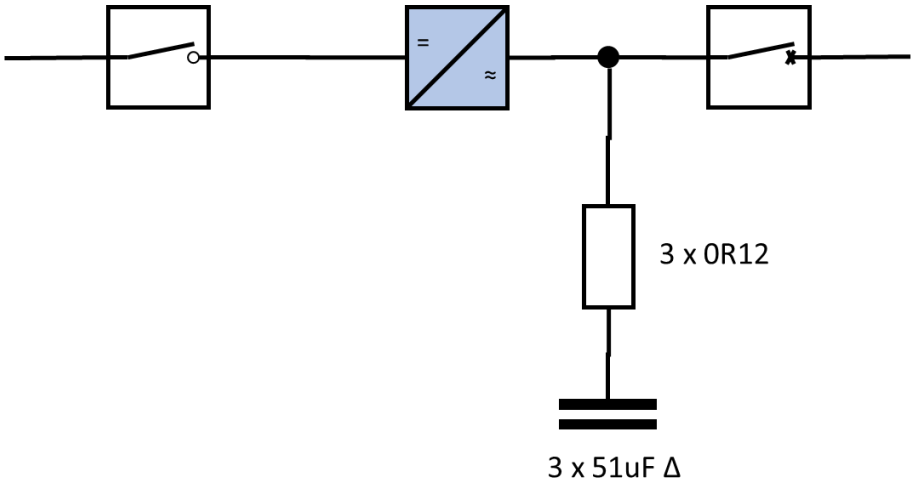


Figure 51: APU single line diagram grid side filter (1000 Vdc-Series)

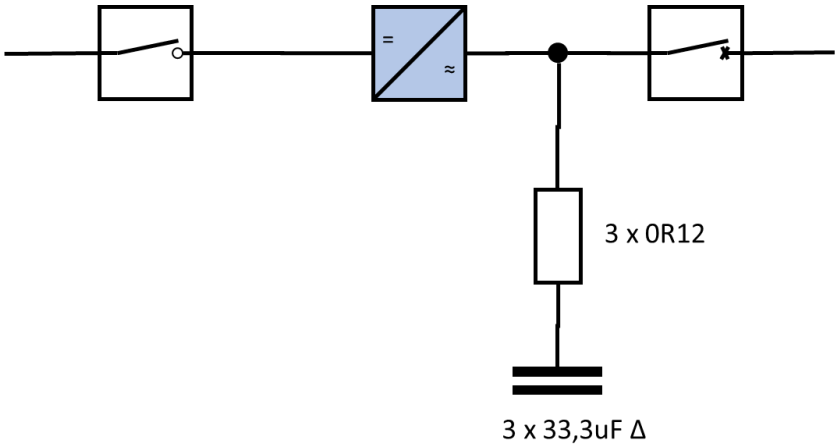


Figure 52: APU single line diagram grid side filter (1500 Vdc-Series)

The following diagrams, shown in Figure 53, Figure 54, Figure 55 and Figure 56, are only representing the grid side behaviour of one APU. If there are several APUs in an APS system installed, the total impedance has to be calculated depending on the operating state of all APUs.

3.3.1 APS Series 1000 Vdc

This chapter describes the grid side inverter impedance in different operation states. The operation states are categorised in three groups:

- Operating State <=200
- Operating State 500/600
- Operating State > 600

3.3.1.1 Operating State <=200 (“DC connected”)

If the APU is not connected to the AC grid, there is no grid side filter connected to the grid.

3.3.1.2 Operating State 500/600 (“AC connected”)

If the APU is coupled to the AC grid, the grid side filters are connected also. These filters have a characteristic shown in Figure 53 and Figure 54.

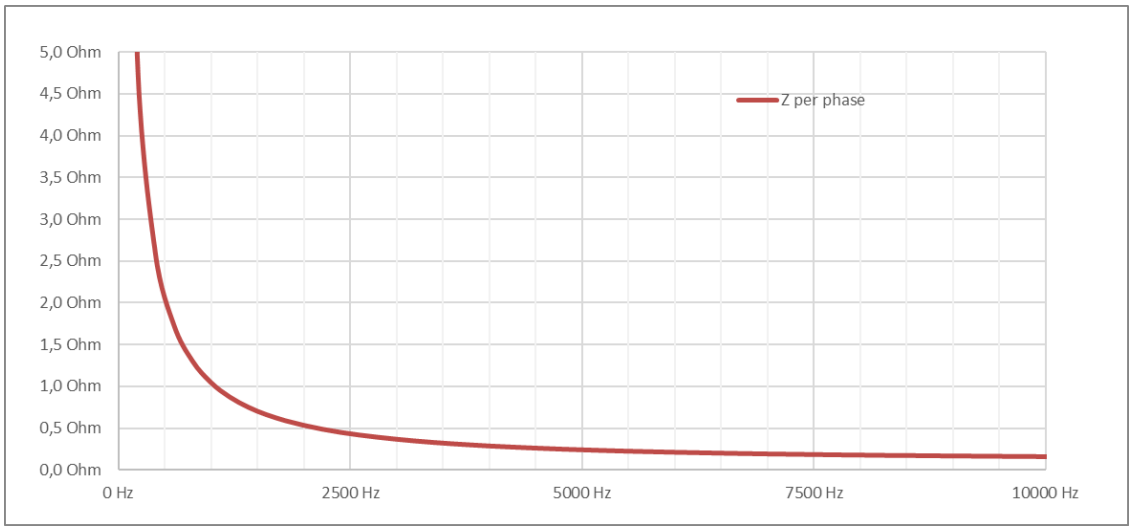


Figure 53: Grid side APU impedance 0…10 kHz (1000 Vdc-Series)

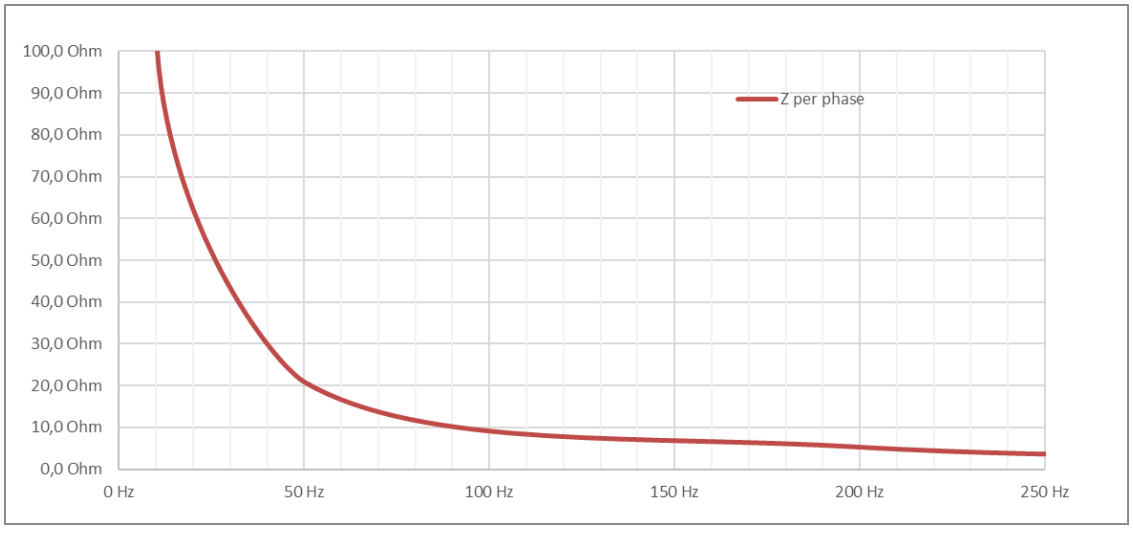


Figure 54: Grid side APU impedance 0…250 Hz (1000 Vdc-Series)

3.3.1.3 Operating State > 600 (Operating)

Operating the APU in a higher operating mode does not influence the grid side impedance of the APU. In normal grid feeding operation the APU has a behaviour like a current source to the AC grid and the same behaviour like in operating state 500/600.

3.3.2 APS Series 1500 Vdc

This chapter describes the grid side inverter impedance in different operation states. The operation states are categorised in three groups:

- Operating State <=200
- Operating State 500/600
- Operating State > 600

3.3.2.1 Operating State <=200 (“DC connected”)

If the APU is not connected to the AC grid, there is no grid side filter connected to the grid.

3.3.2.2 Operating State 500/600 (“AC connected”)

If the APU is coupled to the AC grid, the grid side filters are connected also. These filters have a characteristic shown in Figure 55 and Figure 56.

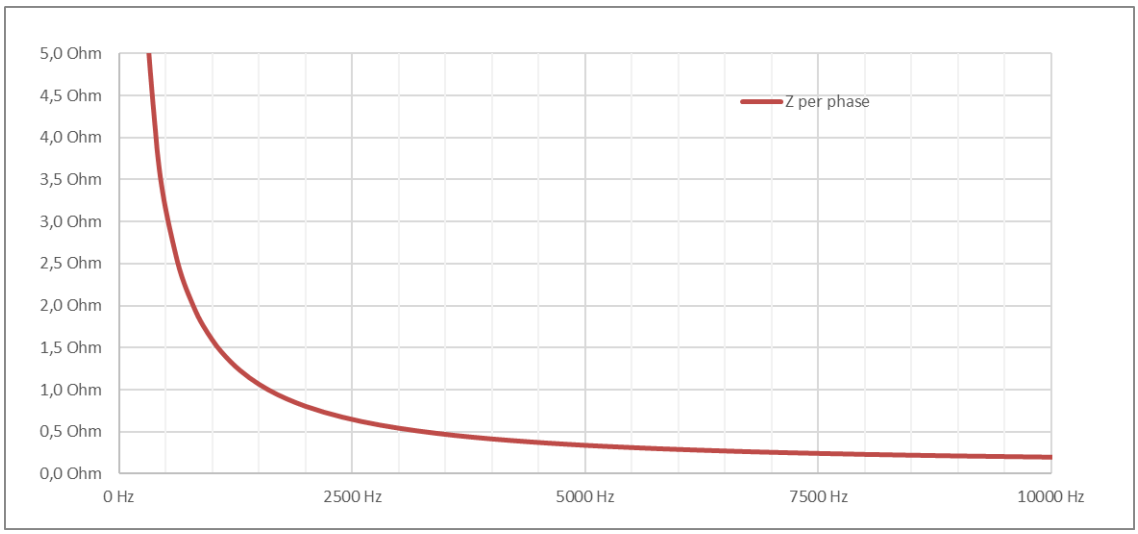


Figure 55: Grid side APU impedance 0…10 kHz (1500 Vdc-Series)

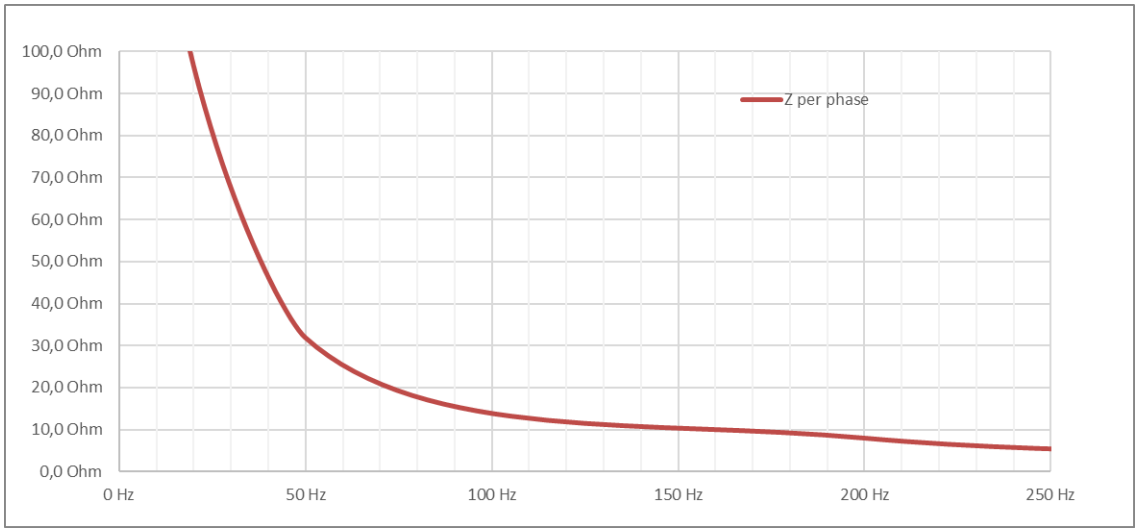


Figure 56: Grid side APU impedance 0…250 Hz (1500 Vdc-Series)

3.3.2.3 Operating State > 600 (Operating)

Operating the APU in a higher operating mode does not influence the grid side impedance of the APU. In normal grid feeding operation the APU has a behaviour like a current source to the AC grid and the same behaviour like in operating state 500/600.

3.3.3 Inverter impedance in case of grid faults

In some grid fault calculations, the impedance of the inverter has to be inserted. This inverter impedance is calculated by:

$$Z = \frac{U}{I}$$

In these calculations are the following parameters requested:

- Subtransient Z”(p.u.)
- Transient Z’(p.u.)
- Synchronous Z(p.u.)

To calculate the impedance the grid voltage has to be defined, at which fault condition the calculation has to done. This is often a voltage between 0…20% of the nominal grid voltage.

The current for these different parameters are different. The Ik’’ and Ik’ currents for the sub transient and transient impedance are shown in the datasheet of the inverter.

The current for the synchronous part is depending on the kind of grid fault. The voltage drop and the k-factor of the LVRT configuration defines the amplitude.

$$I = \left(1 - \frac{fault\ voltage}{nominal\ voltage}\right) \times k - factor \times nominal\ current$$

If the current value is more than nominal current, it is limited to that value. This value is called instantaneous controlled fault current magnitude.

This simple model is only valid for symmetric voltage drops. Due to this the negative sequence current is close to 0A. The negative sequence impedance is very high.

Due to the missing neutral connection of the inverter, the zero-sequence current is 0 A, the zero-sequence impedance is very high.

WSTECH strictly recommends using modern simulation tools for more information. The PSCAD model (or other provided models) directly calculates the current, that are fed into the grid in any different grid fault condition. Please use these currents for more complex grid fault studies.

In the following table an example for a symmetric grid fault in a 690Vgrid of 1 APU is shown.

Parameter	**0%**	**25%**	**50%**	**75%**	**100%**
	0 V rms	173 V rms	345 V rms	518 V rms	690 V rms
Sub transient Z"(p.u.)	0,000 Ω	0,094 Ω	0,188 Ω	0,282 Ω	0,376 Ω
Transient Z'(p.u.)	0,000 Ω	0,094 Ω	0,188 Ω	0,282 Ω	0,376 Ω
Synchronous Z(p.u.)	0,000 Ω	0,095 Ω	0,190 Ω	0,285 Ω	0,379 Ω
Neg.-seq. Z2(p.u.)	9999 Ω	9999 Ω	9999 Ω	9999 Ω	9999 Ω
Zero-seq. Z0(p.u.)	9999 Ω	9999 Ω	9999 Ω	9999 Ω	9999 Ω
Instantaneous Controlled Fault Current Magnitude	1050 A rms	1050 A rms	1050 A rms	1050 A rms	1050 A rms

3.4 DC Common-Mode Voltage

3.4.1 Equipment

The following equipment was used for the measurement:

- APS-000064
- 400 kWh battery from NEC with a voltage of 887 V_{DC}
- Tektronix Scope DPO 5054B
- Tektronix Probe THDP0100
- Test field Halle 2

3.4.2 DC-coupled APUs

This chapter describes the measurement setup and results for DC-coupled APUs.

3.4.2.1 Measurement Setup

The common mode voltage is measured between the positive DC-Bus voltage to earth.

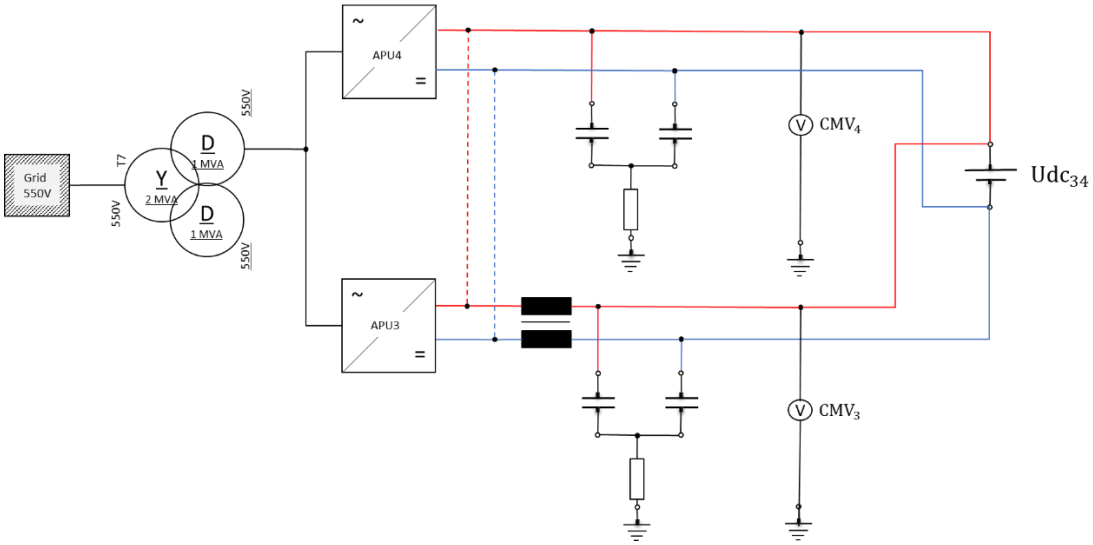


Figure 57: Measurement setup with coupled APUs

3.4.2.2 Measurements

The measurements were performed in the operation mode “CC-CV” and “Standby”. The results are presented in the next sections.

3.4.2.2.1 APU-Pair in Operating Mode CC-CV with 50 kW Charging

Figure 58 and Figure 59 are showing the voltage of an operating system. In both figures we can see an DC-voltage of ~ 443,5 V. The total DC-voltage was 887 Vdc.



Figure 58: +Udc to PE 20 ms/div at an operating APU-pair

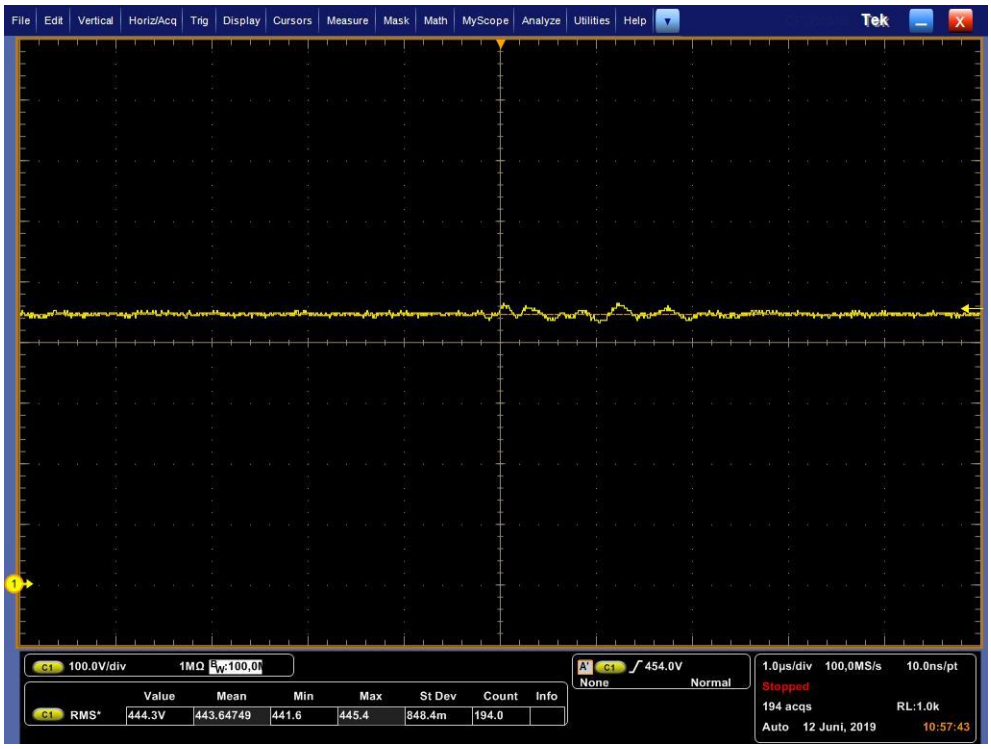


Figure 59: +Udc to PE 1 µs/div at an operating APU-pair

3.4.2.2.2 APU in Operating Mode Standby

Figure 60 shows a reference measurement. During that time the APS was in standby. No influence of the inverter to the DC-voltage is present.

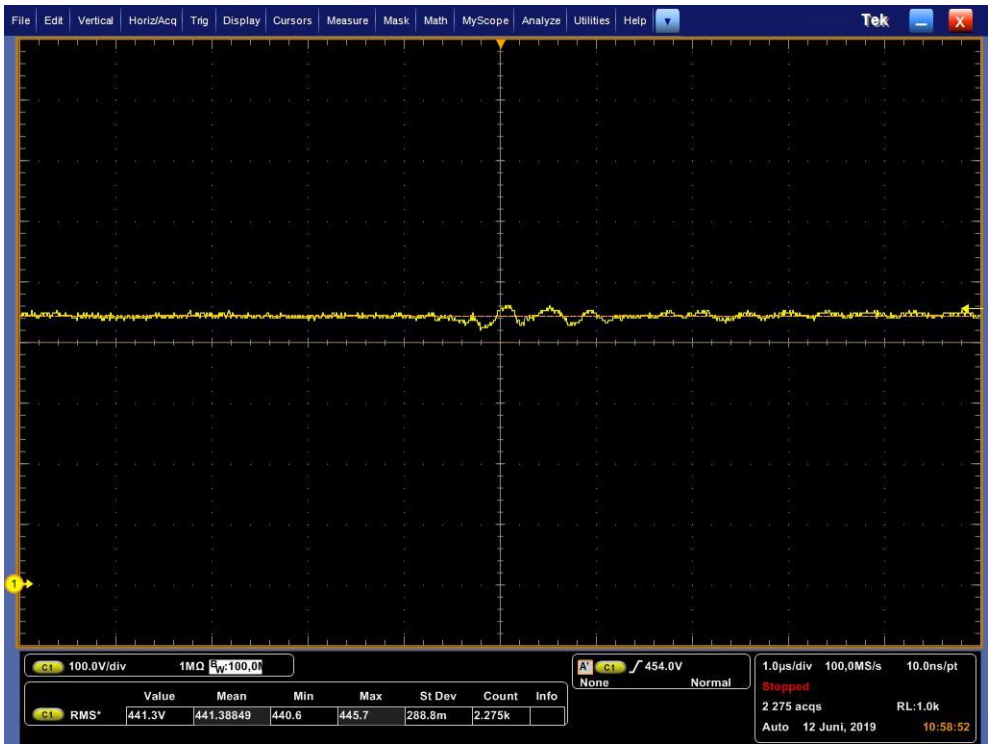


Figure 60: +Udc to PE 1 µs/div reference measurement

3.4.2.3 Result

There is no common-mode voltage measured. The voltage is very stable with no spikes. The DC-link voltage is symmetric to the earth potential.

The spikes on the measurement signal in Figure 59 are measurement error. This is documented with the reference measurement of Figure 60. During the test other inverters have been operating in the test field. May they create this effect. The other inverters are not connected to this measurement setup directly.

3.4.3 DC separated APUs

This chapter describes the measurement setup and results for DC separated APUs.

3.4.3.1 Measurement Setup

The common-mode voltage (CMV) will be measured between the positive DC-Bus voltage to earth.

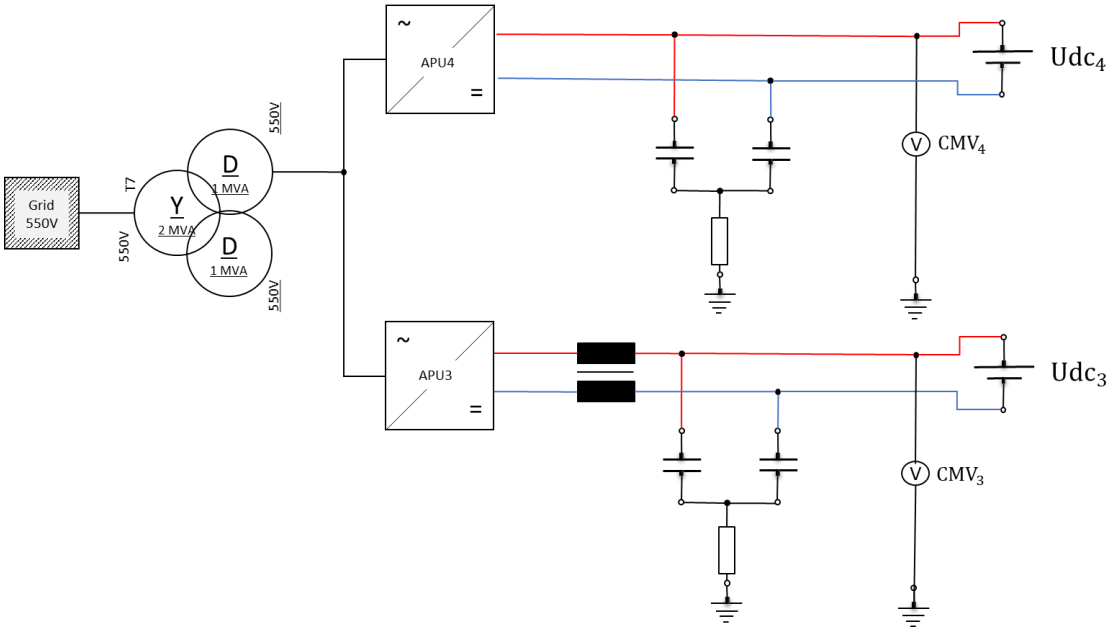


Figure 61: Measurement setup with separated APUs

3.4.3.2 Measurements

The measurements were performed in the operation mode “CC-CV” with 0% active power and 50% active power.

3.4.3.2.1 APU3/4 in Operating Mode CC-CV 0% Active Power

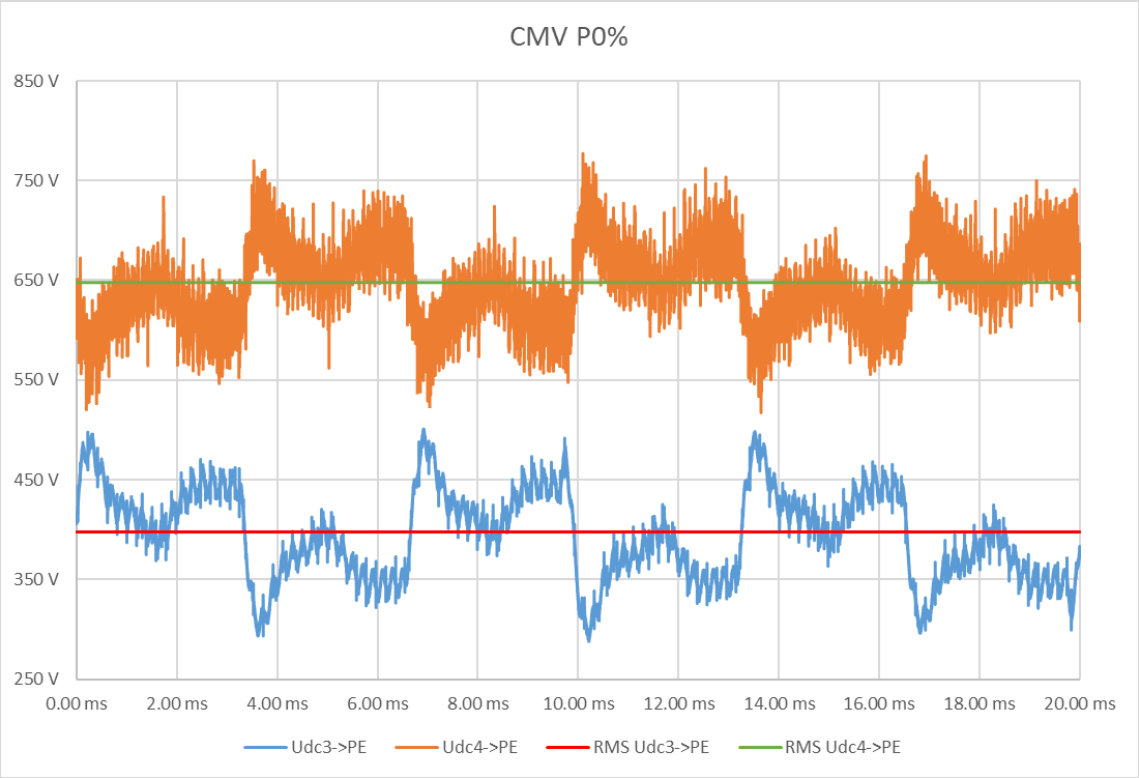


Figure 62: CMV 0% active power

Figure 62 shows the common mode voltage of a separated operating system.

In Table 20 are the parameters of the test listed.

Table 20: Test parameters CMV 0% active power

Component	Description	Value
$\mathbf{Udc_3}$	DC-Voltage APU3	800 Vdc
$\mathbf{Udc_4}$	DC-Voltage APU4	1300 Vdc
$\mathbf{P_3}$	Active power APU3	0 kW
$\mathbf{P_4}$	Active power APU4	0 kW
$\mathbf{CMV_{3RMS}}$	RMS DC-Voltage against PE APU3	398 Vdc
$\mathbf{CMV_{3peak-peak}}$	CMV Peak to Peak APU3	207 Vdc
$\mathbf{CMV_{4RMS}}$	RMS DC-Voltage against PE APU4	648 Vdc
$\mathbf{CMV_{4peak-peak}}$	CMV Peak to Peak APU4	251 Vdc

3.4.3.2.2 APU3/4 in Operating Mode CC-CV 50% Active Power

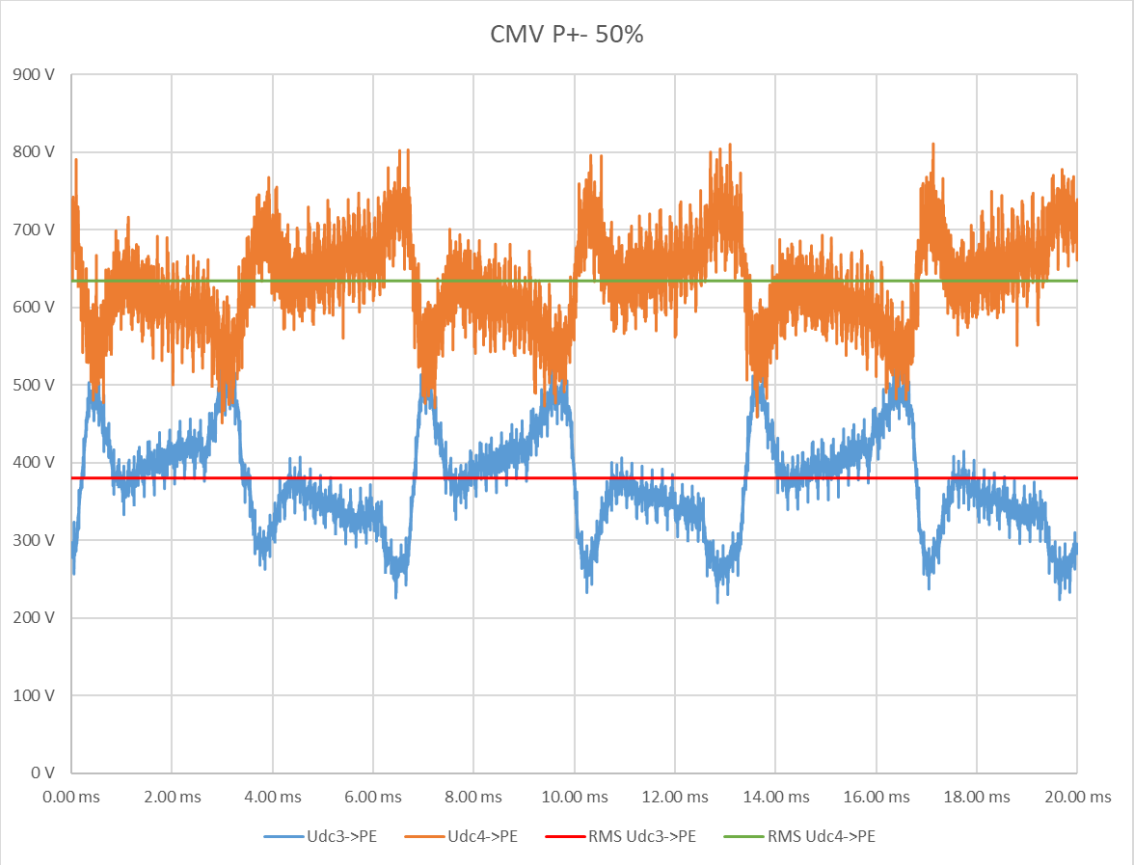


Figure 63: CMV 50% active power

Figure 63 shows the common-mode voltage of a separated operating system.

In Table 21 are the parameters of the test listed.

Table 21: Test parameters CMV 50% active power

Component	Description	Value
Udc_3	DC-Voltage APU3	760 Vdc
Udc_4	DC-Voltage APU4	1268 Vdc
P_3	Active power APU3	500 kW
P_4	Active power APU4	-500 kW
CMV_{3RMS}	RMS DC-Voltage against PE APU3	380 Vdc
$CMV_{3peak-peak}$	CMV Peak to Peak APU3	310 Vdc
CMV_{4RMS}	RMS DC-Voltage against PE APU4	634 Vdc
$CMV_{4peak-peak}$	CMV Peak to Peak APU4	324 Vdc

3.4.3.3 Result

The CMV is depending on the DC voltage- and power difference between the AC coupled systems. With higher DC voltage- or power difference the CMV increases.

3.5 Short Circuit Current contribution

In the following sections the short circuit current contribution is described for the AC side and the DC side.

3.5.1 AC Short Circuit

If a short circuit occurs between the AC terminals of the APS or between the AC terminals of the transformer connected to the APS, each power stack of the APS feeds following currents into the short circuit.

Table 22: Short circuit currents (nominal)

Power stacks	I_P	I_k''	I_k'
1000 V	1674 Â	1137 A	1137 A
1500 V	1544 Â	1050 A	1050 A

The peak current I_P is about 4% higher than the amplitude of the nominal currents due to the internal harmonic filters, which feed the charged energy into the short circuit as well. The values Ik'' and Ik' are the nominal short-circuit currents in the steady state.

These values have a maximum error of 1% of the values listed in Table 22.

Table 23 shows the maximum values considering the errors.

Table 23: Short circuit currents (maximum values)

Power stacks	I_P	I_k''	I_k'
1000 V	1690 Â	1148 A	1148 A
1500 V	1560 Â	1060 A	1060 A

These short circuit current values are the worst-case values for 3-phase and phase-to-phase short circuits. In case of other short circuits, the currents will be less than the above listed values.

3.5.2 DC short circuit test

To know the real short circuit current in case of a DC-short circuit at the DC-main terminals of the APS, a short circuit test has been done.

3.5.2.1 Test Setup

For this test one APU of an APS was used. The DC-disconnector was connected, the DC-circuit was charged up from AC-side and the AC-circuit breaker closed after charging up the DC-circuit.

At the DC-main terminals a DC-disconnector was connected to realize the short-circuit (Figure 64). The points of measuring are shown in Figure 64 and in detail in Figure 65.

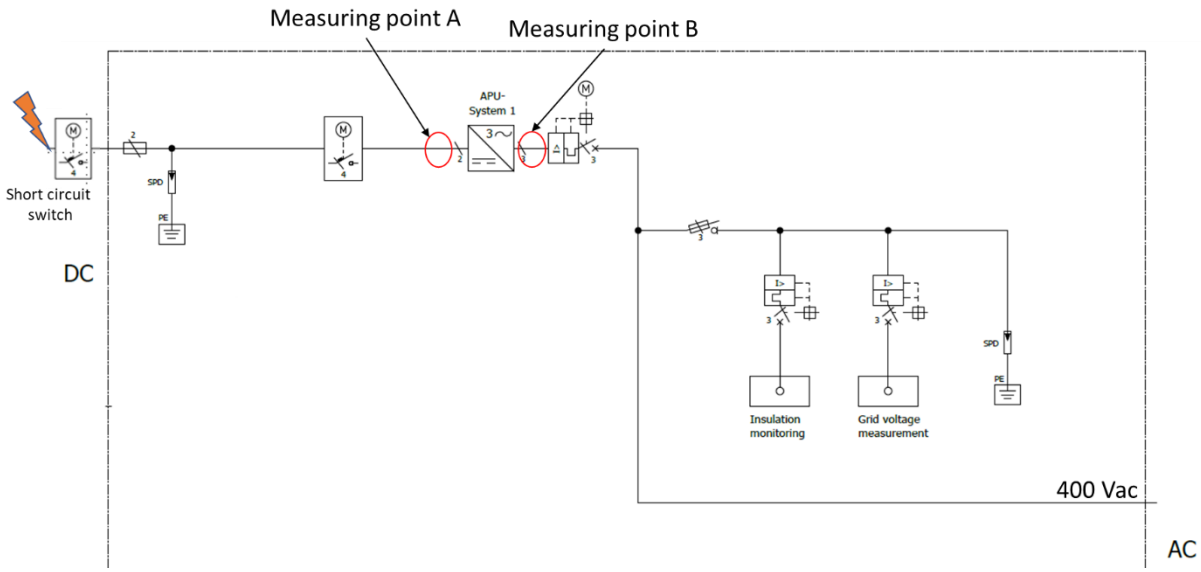


Figure 64: Schematic test setup

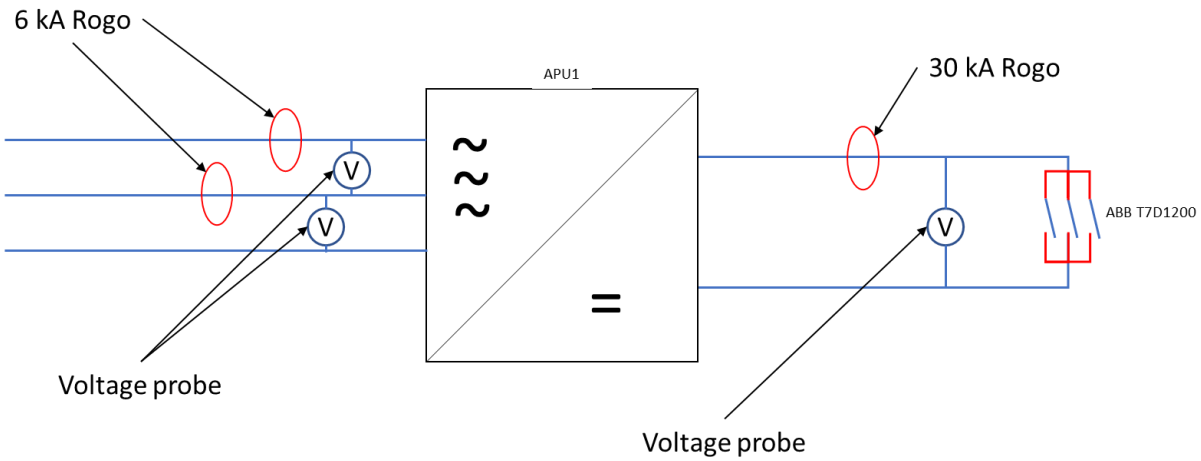


Figure 65: Used measurement equipment

3.5.2.2 Measurement results

The short circuit current runs through the AC-circuit breaker, the main inductance, the diodes in the power stack, the DC-disconnector and the DC-fuses.

The DC-fuses are four 500 A 1500 V fuses in parallel connection.

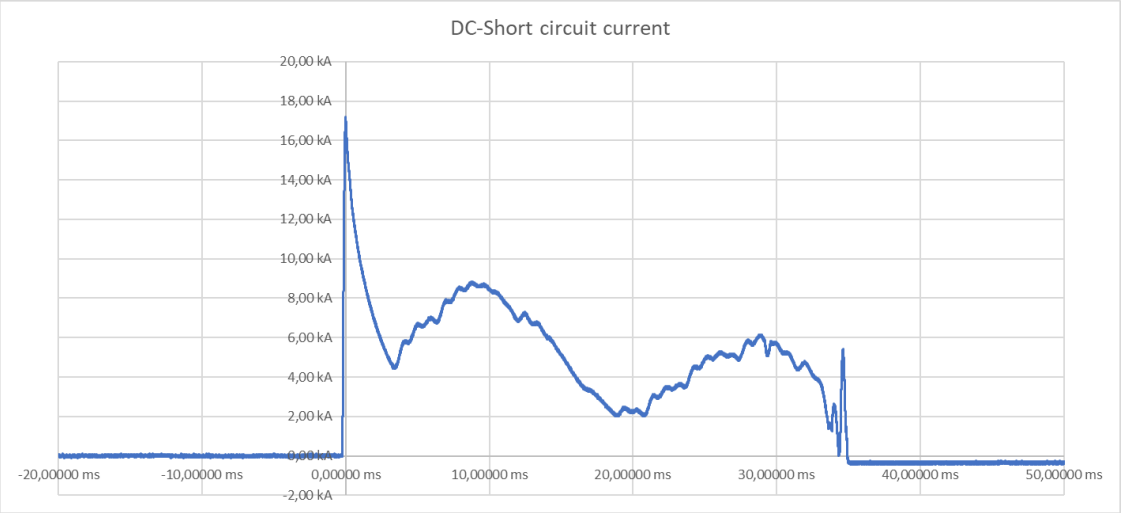


Figure 66: short circuit current DC

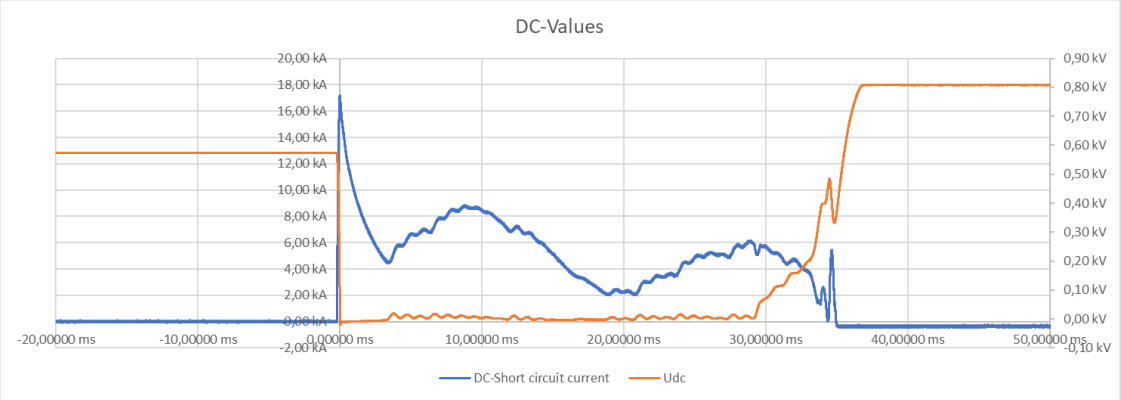


Figure 67: DC-link values

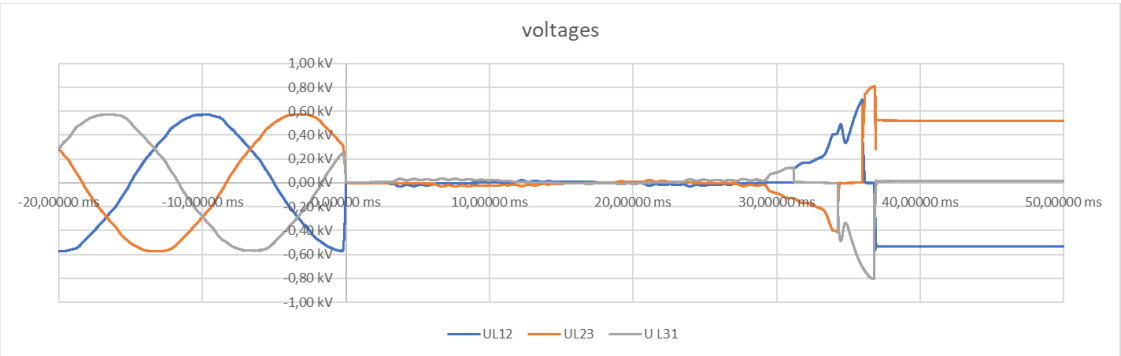


Figure 68: AC-grid voltage behind circuit breaker

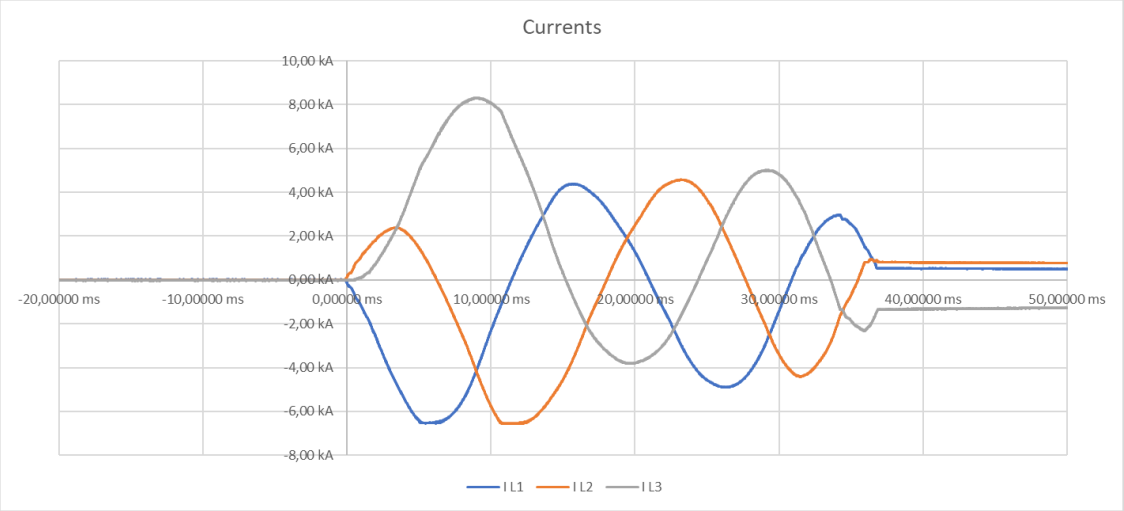


Figure 69: AC short circuit currents

3.5.2.3 Pictures

Figure 70 shows the tripped AC-circuit breaker. This can be seen by the small grey button on the lower left side of the breaker. A close-up is shown in Figure 71.



Figure 70: tripped AC breaker

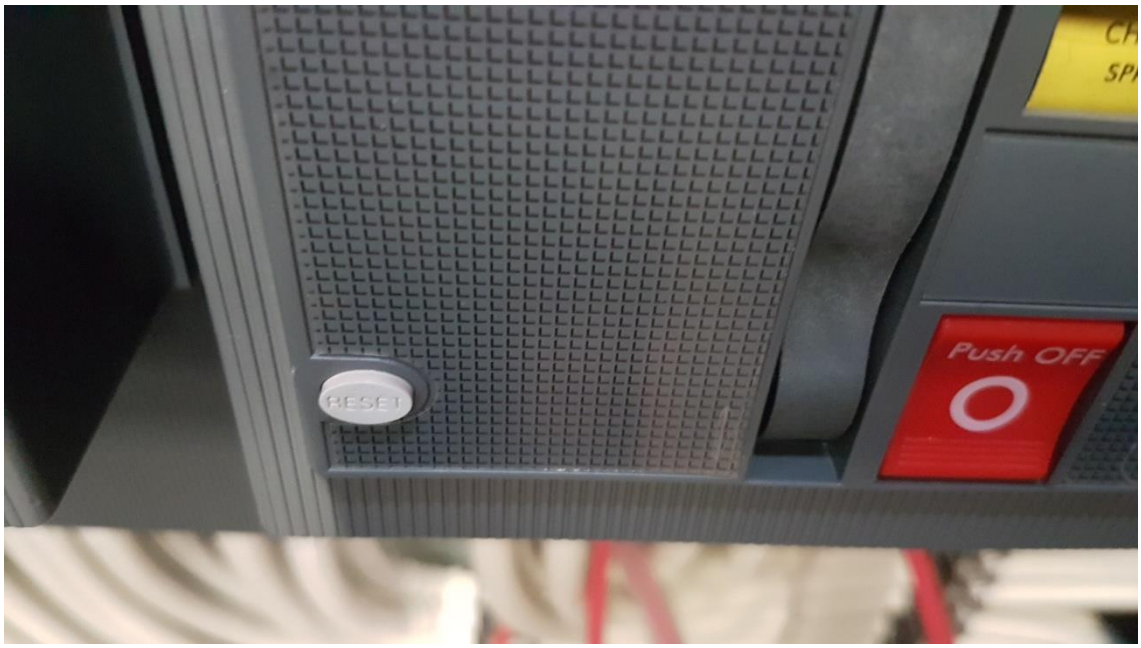


Figure 71: tripped AC breaker - detail

3.5.2.4 Test results

The mayor parameters that have been measured in this test are shown in Table 24.

Table 24: Test results

Parameter	Value	Comment
Max. DC-short circuit current	17 kA @ 400 Vac	See Figure 66
Tripping time AC-Breaker	Disconnection starts at 36 ms	See Figure 68
Tripping time DC-Fuses	Disconnection starts at 29 ms	See Figure 67

The first spike of the dc-current is the discharge current of the internal dc-link capacitors. Thereafter the current is relatively stable until timestamp 30 ms. Then the DC-fuses starts are already disconnecting before the ac-breaker opens the connects. The AC-Breaker was tripping itself and cleared the short circuit.

3.6 Short circuit withstand capability

3.6.1 AC-interface

The maximum short circuit withstand capability is marked in the datasheet.

3.6.2 DC-interface

There are 3 different level defined for the DC short circuit withstand capability.

- 6 kA
- 30 kA
- 140 kA up to 0,5 ms

The dc-disconnector has an operating range of up to 6 kA short circuit current. The breaker does normally open after 30 ms on an overcurrent above maximum dc current

The standard fuses are able to protect up to 30 kA short circuit currents. Please use the datasheet of the fuses for more information. (Article number 500 A, 1500 Vdc fuse: 500492)

If there is an external protection used, the inverter is able to handle currents up to 140 kA up to 0.5 ms in maximum. The derating of this current and the duration time is shown in the following figure.

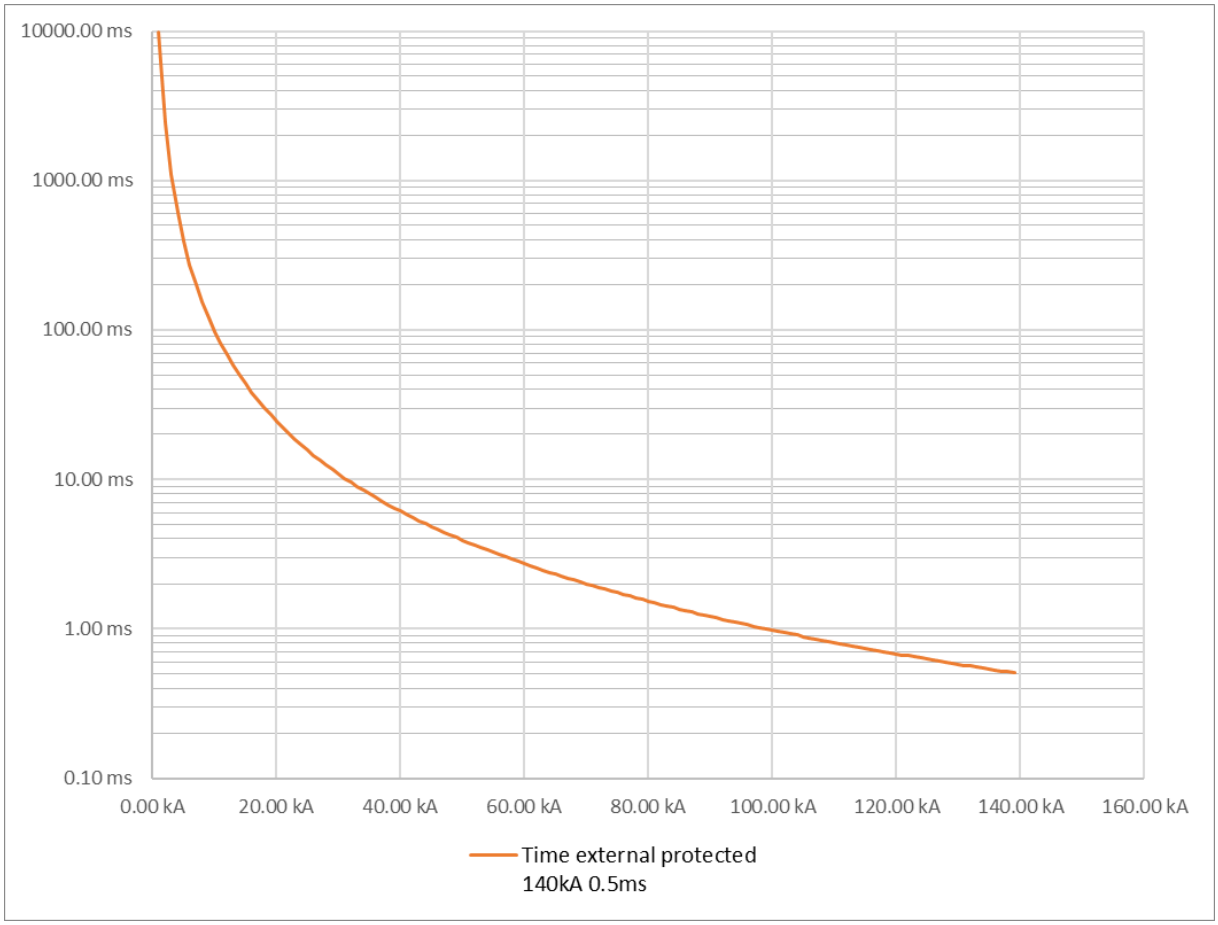


Figure 72: dc short circuit withstand capability with external protection

3.7 Losses during STATCOM operation

Since no DC source is connected to the APS in STATCOM mode, all losses are taken from the grid (active power supply). The losses of the inverter are similar in operations points of active and reactive power. Due to this the losses of the inverter can be calculated like:

$$P = \sim \left(S - \frac{Q}{\eta}\right)$$

In the validation process we measured the losses and calculated the “efficiency” of the system with

$$\eta = \sim \left(\frac{Q}{S - P}\right)$$

The results are shown in the following figure.

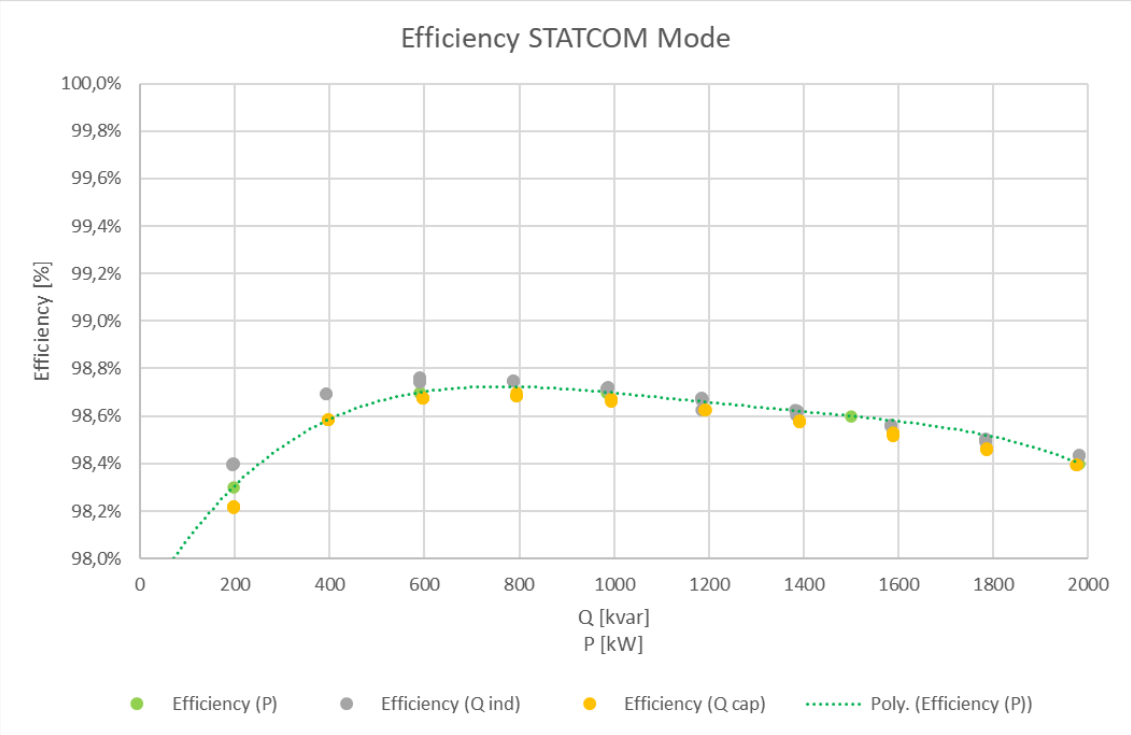


Figure 73: STATCOM Mode Efficiency

The green line in Figure 73 shows the efficiency of active power transmission. The grey and yellow points showing the “efficiency at ind/cap reactive power.

The measurements above show, that the STATCOM mode efficiency is comparable to the active power efficiency.

4 Operational characteristics

In this section several diagrams are showing the response and behaviour during different operation points:

- start-up
- grid parallel operation
- grid forming operation

4.1 Start Sequencies

For the start sequence, the cases DC-coupled and DC-separated are a little different. The start procedure is described in the following sections.

4.1.1 DC-coupled

A start sequence consists of the following mayor steps:

- Energy Saving
- Sleep
- Off
- DC-Connect
- AC-Connect
- Operation

The transitions from the individual steps are shown below.

The control command for the operating mode of the coupled APUs has to be send.

Both APUs are controlled in the same way (operating mode and power)

The internal state machine is taking care on a sequential operation of the breakers, to avoid unwanted effects.

In the following figure a typical sequence of the connection of the ac and dc breakers is shown. Furthermore, the dc-link voltages of both units are shown. The pre-charge will be operated also in a sequence.

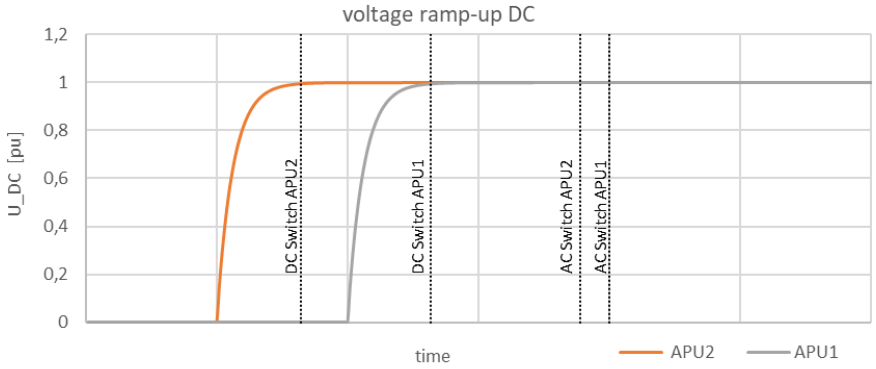


Figure 74: typical connection sequence of ac/dc breakers

4.1.1.1 Energy Saving (0) => Sleep (20) => Off (50)

In these steps the APU controller and the driver getting switched on. The controller is booting and the system will be brought to a defined switched off operating point.

4.1.1.2 OFF (50) => DC-Connect (200)

When the inverter has to perform the DC-connection the following steps will be done:

- Init DC connect (100)
- Wait DC connect (130)
- Pre-charge DC (160)
- DC-connect (200)

These are described in more detail in the following chapters.

4.1.1.2.1 Init DC connect and Wait DC connect

Init DC connect and Wait DC connect are wait states to prevent synchronous switching.

4.1.1.2.2 Pre-charge DC

During the pre-charge the inverter does several steps automatically, after the operating mode is set. These are:

1. Closing the pre-charge circuit
2. DC link gets charged
3. Activate monitoring
    - Duration (time constant)
    - Increase of voltage
4. Open the pre-charge circuit

Pre-charge will be OK, after the time constant and no further du/dt inside the dc-link is detected. If pre-charge fails, the inverter will stop with an error.

The DC-pre-charge uses DC-power. Each APU has its own DC-pre-charge circuit. The pre-charge function consists out of:

- Charge resistor (94 Ω)
- DC link (4.68 mF)
- Pre-charge monitoring

In Figure 75 a simple circuit diagram of the pre-charge circuit is shown. It is used to run the simulation.

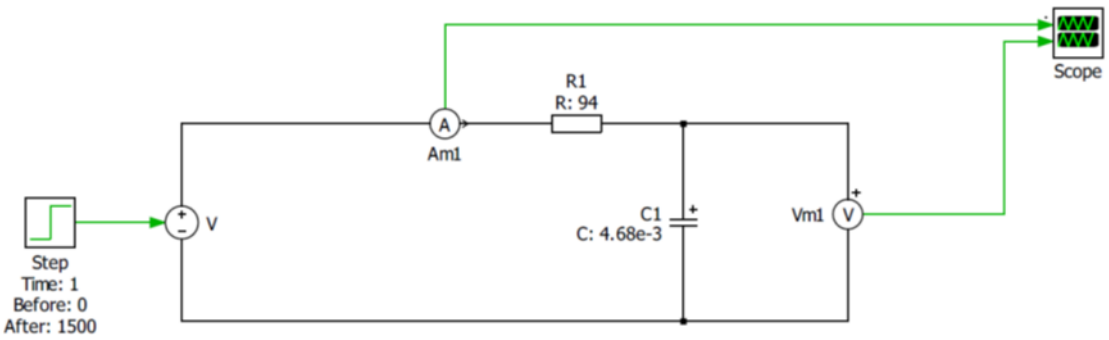


Figure 75: Circuit diagram (simple)

The pre-charge of the APS takes around 2 s. The peak current during this is around 16 A (I = U/R = 1500 V/96 Ohm) (see Figure 77 and Figure 77: DC-pre-charge – currentFigure 77).

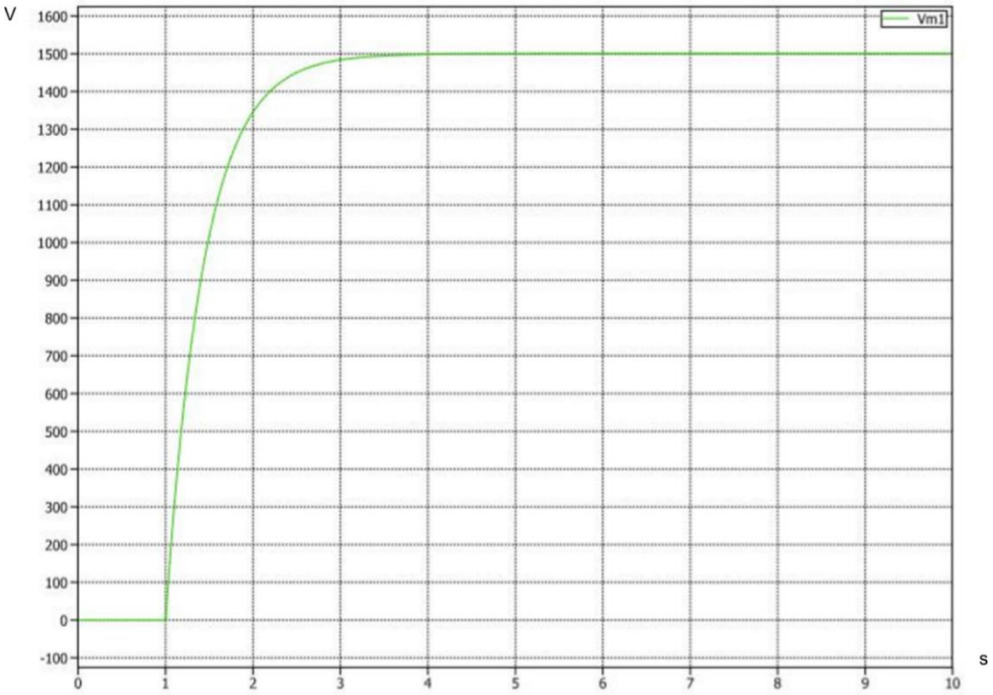


Figure 76: DC-pre-charge – voltage

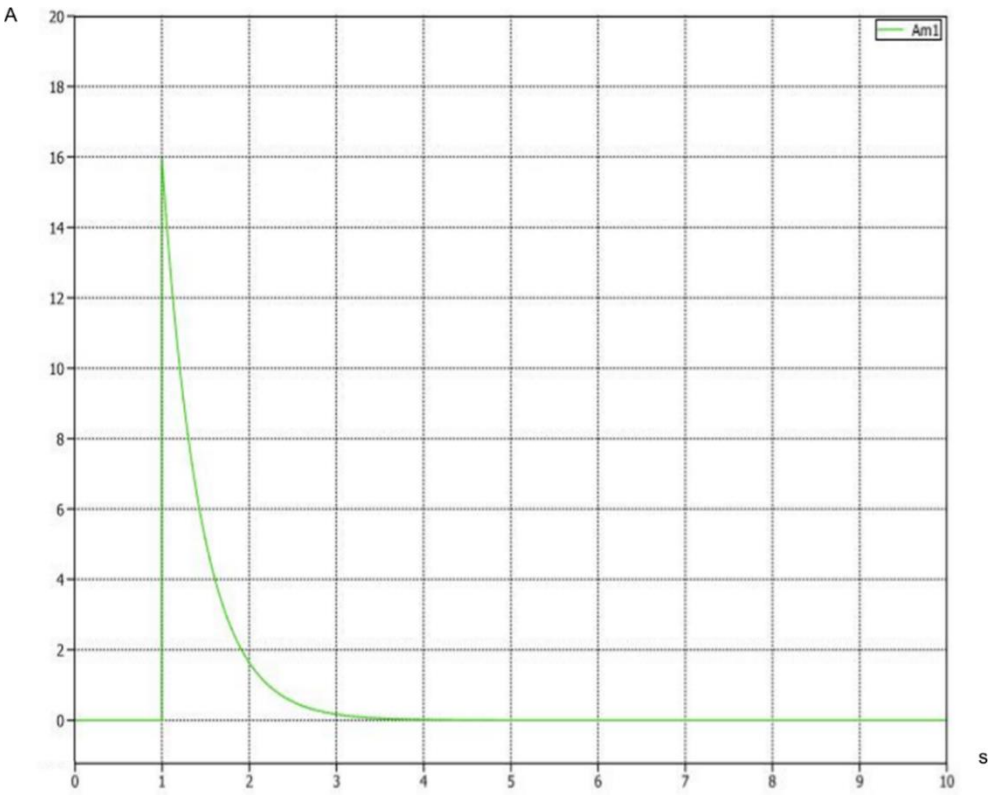


Figure 77: DC-pre-charge – current

4.1.1.2.3 DC connect

After opening the pre-charge circuit, the main DC contact will close. In that moment the DC grid is directly connected with the internal DC-link of the inverter.

4.1.1.3 DC-Connect (200) => AC-Connect (500)

When the inverter has to perform the AC-connection the following steps will be done:

- Init AC connect (330)
- Wait AC connect (360)
- Pre-charge AC (optional) (400)
- AC connect (500)

These are descripted more in detail in the following chapters.

4.1.1.3.1 Init AC connect and Wait AC connect

Init AC connect and wait AC connect are wait states to prevent synchronous switching.

4.1.1.3.2 Pre-charge AC (optional feature for STATCOM)

This function is used to charge the dc-link from the AC grid. Normally the dc link is already charged by the dc-interface. Only in STATCOM operating mode the DC link will charged by the AC-grid.

During the charging the dc-link will be monitored in a similar way compared to the dc-pre-charge function (see 4.1.1.2.2).

The ac-pre-charge will also be done via pre-charge resistors. The impact on the grid is very limited due to this.

4.1.1.3.3 AC connect

The AC circuit breaker will close.

4.1.1.4 AC-Connect (500) => Operation (xxxx)

After reaching the AC Connect state, it is possible to switch to a regular operating mode.

During operation the PWM pulse pattern is controlling the IGBTs. The control-loops are active and running.

The individual operating modes are explained in more detail in chapter 4.2 and 4.3.

4.1.2 DC-Separated (ES only)

A start sequence consists of the following steps:

- Energy Saving
- Sleep
- Off
- DC-Connect
- AC-Connect
- Operation

The transitions from the individual steps are shown below.

The control command for the operating mode of each independent APUs has to be send.

Each APU can be controlled totally independent from the other.

If both APUs are set in the same moment to a new operating mode, the internal state machine is taking care on a sequential operation of the breakers, to avoid unwanted effects.

4.1.2.1 Energy Saving (0) => Sleep (20) => Off (50)

This function is the same as in a DC-coupled system.

Please note, if one APU of the pair is set to operating mode OFF, the other will follow directly. This is caused by the usage of a common control card.

4.1.2.2 OFF (50) => DC-Connect (200)

This function is the same as in a DC-coupled system.

4.1.2.3 DC-Connect (200) => AC-Connect (500)

This function is the same as in a DC-coupled system.

4.1.2.4 AC-Connect (500) => Operation (xxxx)

This function is the same as in a DC-coupled system.

4.2 Grid parallel operation

In this section several diagrams are showing the response and behaviour during different operation points:

- start-up
- changing setpoints
- response to grid faults

4.2.1 Start-up to operation

This chapter describes the start sequence of the following operation modes:

- MPPT
- CC-CV

4.2.1.1 Start sequence MPPT

The MPPT tracker will start automatically with a voltage setpoint of ~85% of the actual dc-voltage (Udc).

The load will be monitored during start up. If there is to less power the inverter will stop the PWM and will try it again in a moment.

If the PWM is active the MPPT is searching with a continuous tracker for the best dc-voltage to get the highest power from the PV-field. A cyclic tracker is activated regularly to scan the PV-field in a wide voltage range to avoid an operation in a local maximum.

4.2.1.2 Start sequence CC-CV

The cc-cv control is working below the normal power control function. It can be seen as a cp–cc-cv control (constant power/current/voltage).

When the inverter is starting, it will try to reach the active power setpoint.

If the dc-link current is reaching the controlled limit, it will be continuing the operation with the max controlled current.

If the dc-link voltage is reaching the controlled limit, it will be continuing the operation at the voltage limit (min or max).

If the operating point of the system is outside the limits of the dc-grid, it will try to reach the dc limits due to an active power flow (into the grid or out of the grid). In that case the power setpoint is not anymore relevant and will be ignored.

Example:

The inverter is controlled like:

Control parameter	**Value**
Pset	10 kW
Idc max	100 A
Idc min	-100 A
Udc max	1200 Vdc
Udc min	1000 Vdc

In the moment of the start, the dc-link voltage is 950 Vdc.

The inverter will try to get the dc-link voltage in the range of 1000…1200 Vdc. In this case it has to be brought energy to the dc-link. The inverter will feed power into the dc-link. The power value can be up to max. inverter power. The control parameter Pset will be ignored, because the inverter was already out of the correct operating range.

To avoid this behaviour the dc-link voltage has to be between the Udc min/max parameters during start up.

4.2.2 Response to Change of Set Point

The response of the APS due to a change of set points is an important information to create fast over all control loops. In the following chapters are several step changes in load documented.

The complete set-up of the following tests is:

- PLC operating as external control system
- Communication interval 10 ms
- Communication via ModbusTCP (all control and status parameters)
- PLC provides analogue outputs with signals of:
  - Set point Pac/Qac
  - Response Pac/Qac
- APS provides digital output with an impulse caused by a change of a received set point (only active power set point change)
- DC current is measured with a current clamp to measure the active power jump.
- AC current is measured with a current clamp to measure the reactive power jump.

All signals are measured with an oscilloscope. An overview is marked in Figure 78.

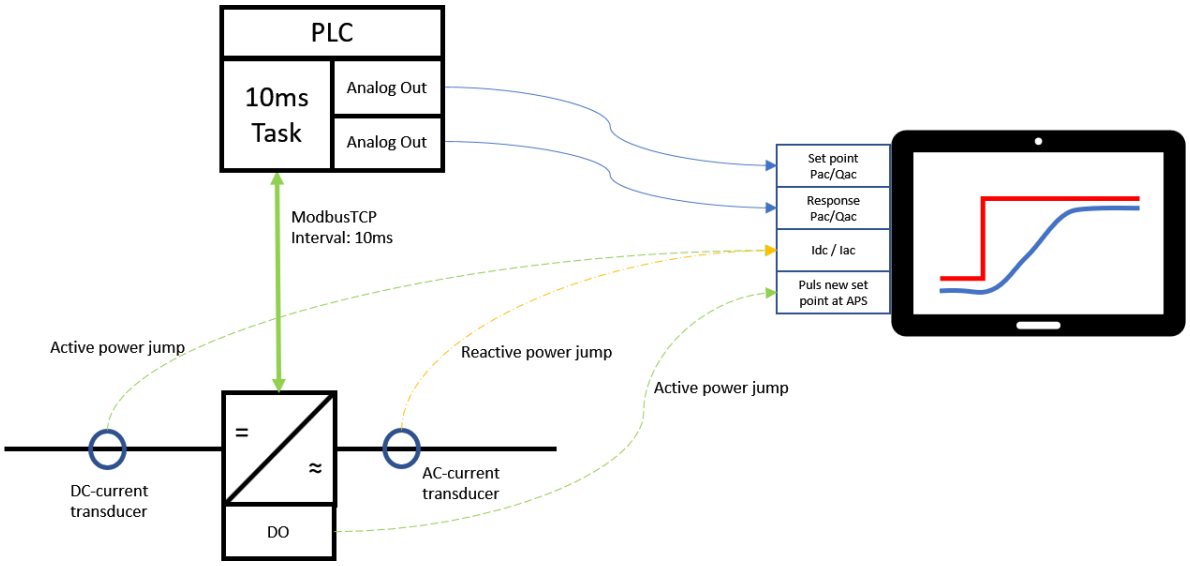


Figure 78: Test set-up response measurements

All tests are done with an APS650-ES-1-330-1 in CC-CV-Mode. The APU is not operating in any kind of DC-limitation. The DC voltage and DC current limits are not involved in that responses. The usage of these limitation can reduce the response speed of the system.

The signal presented in the following chapter are defined in Table 25.

Table 25: Response signal description

Signal Name	Description
Set point Pac/Qac	Analog output of the external overall controller The Pac/Qac set point is the requested set point to the APU
Response Pac/Qac	Analog output of the external overall controller The response of Pac/Qac is the status parameter of the APU transmitted to the overall controller. The steps in the signal are caused by the communication interval of 10 ms.
Idc total (only active power jump)	Directly measured DC current signal of the APU, represents the active power response of the APU to the grid.
Iac L1 total (only reactive power jump)	Measured the AC-current of one phase represent the reactive power response.
Puls new set point at APS (only active power jump)	APU test-signal is creating an impulse of 10 ms caused by a change of a set point

4.2.2.1 Load Jump 0 kW to 500 kW

Figure 79 shows the response to a load jump from 0 kW to 500 kW.

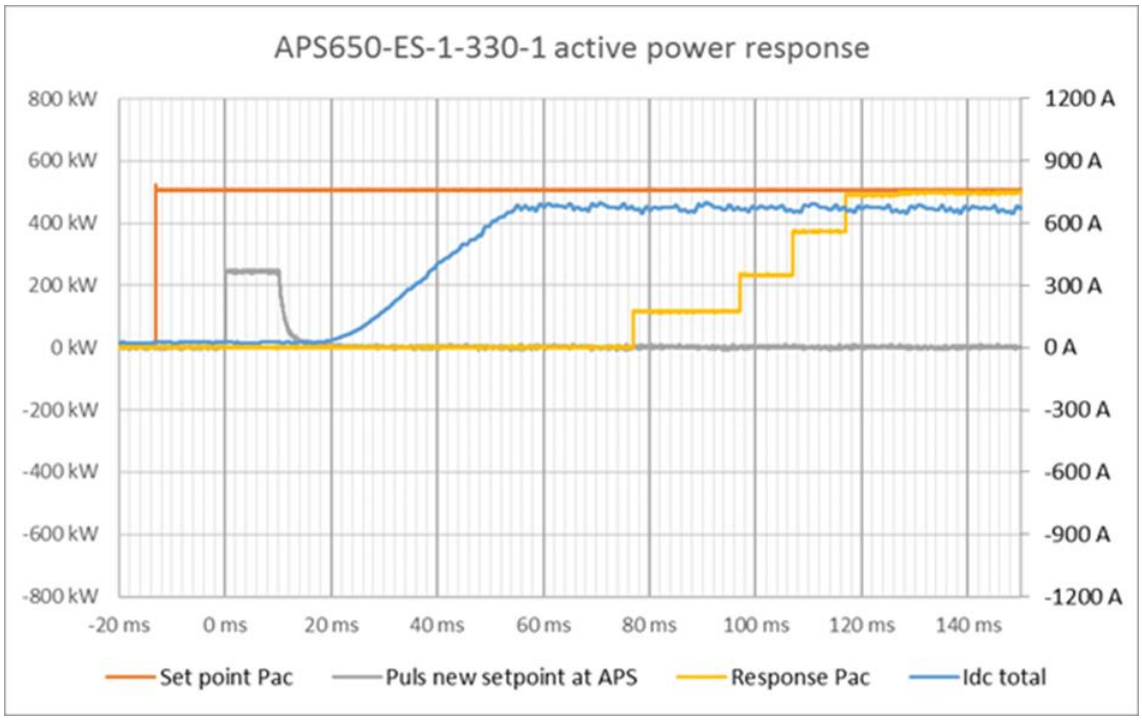


Figure 79: Response time - change of set point (0 kW to 500 kW)

4.2.2.2 Load Jump -650 kW to 650 kW

Figure 80 shows the response to a load jump from -650 kW (minimum power) to 650 kW (maximum power).

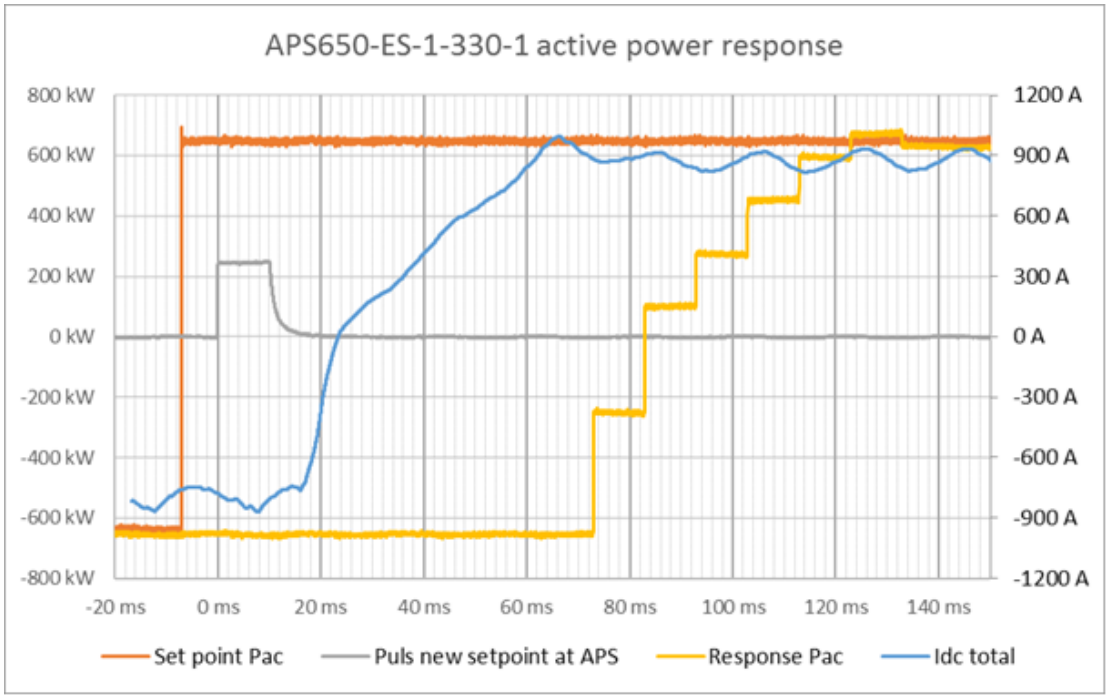


Figure 80: Response time - change of set point (-650 kW to 650 kW)

4.2.2.3 Reactive Power Jump 0%-100% (Qnom)

Figure 81 shows the response to a reactive power jump from 0%-100% (Qnom).

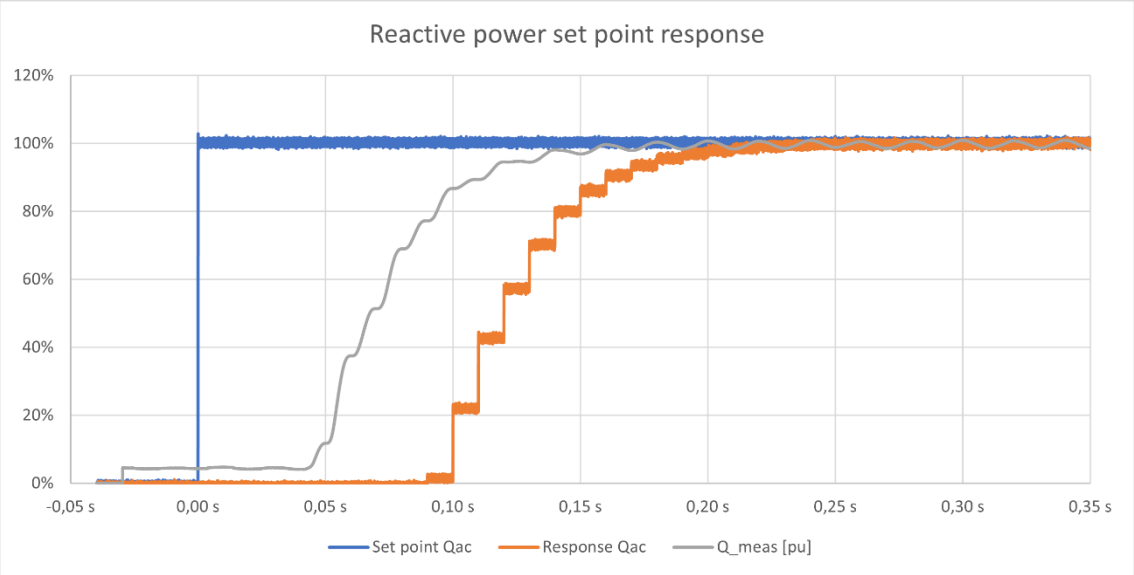


Figure 81: Response time – change reactive power set point (0% to 100%)

4.2.2.4 Reactive Power Jump -100%-100% (Qnom)

Figure 82 shows the response to a reactive power jump from -100%-100% (Qnom).

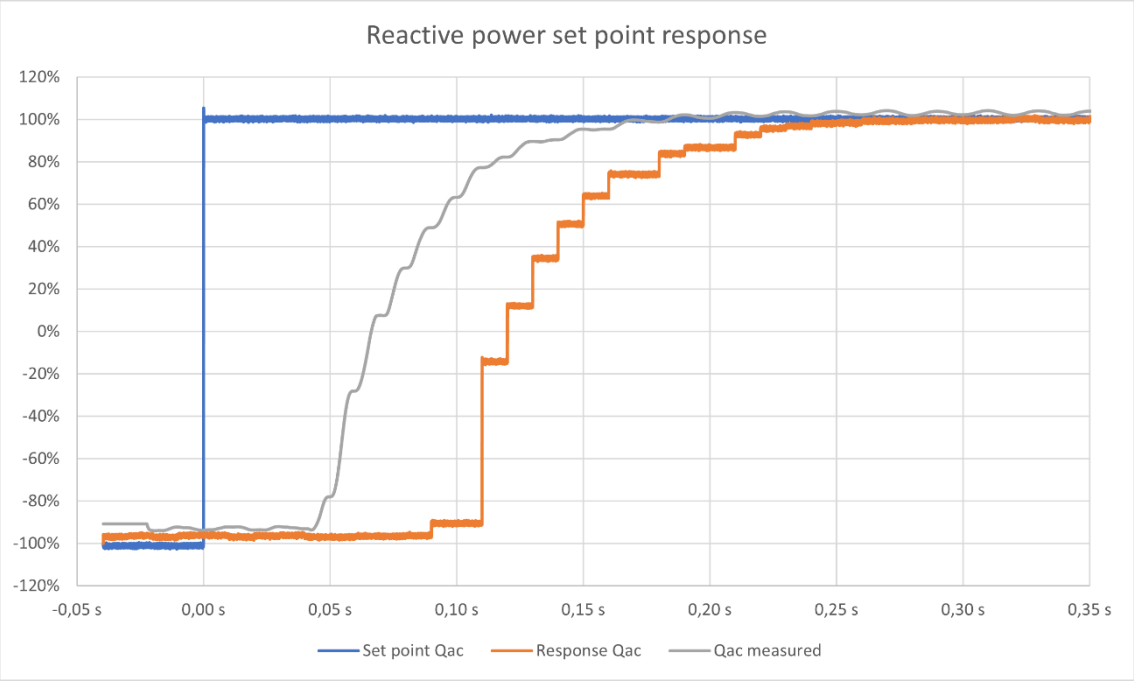


Figure 82: Response time – change reactive power set point (-100% to 100%)

4.2.3 Response to Grid Faults

The voltage and current behaviour of the APS caused by different grid faults is shown in this chapter. All tests are done with a voltage level in the DC-link of 750 Vdc. The results with these error messages can be vary on phases or sequence. These error messages are the result of that measurement.

4.2.3.1 Response with 100% Load

The following tests have been done with 100% load of the inverter.

4.2.3.1.1 Grid Disconnection (High impedance grid loss)

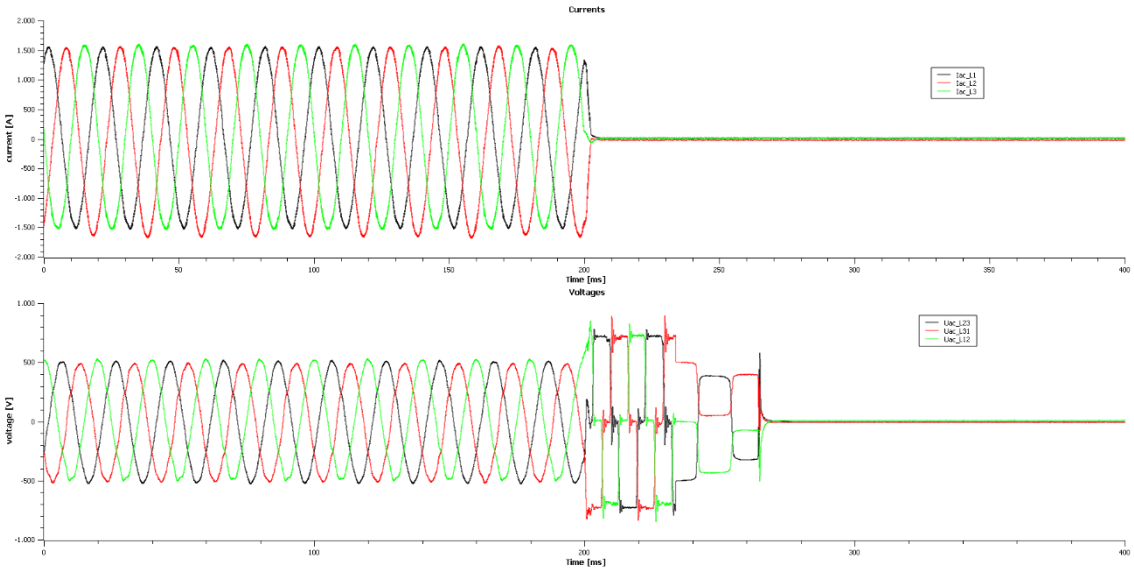


Figure 83: Grid disconnection 100% (measurement 17)

Result:

The APS stops due to an APU error message “96 - Overvoltage Stage 4”.

4.2.3.1.2 Short Circuit Phase to Earth

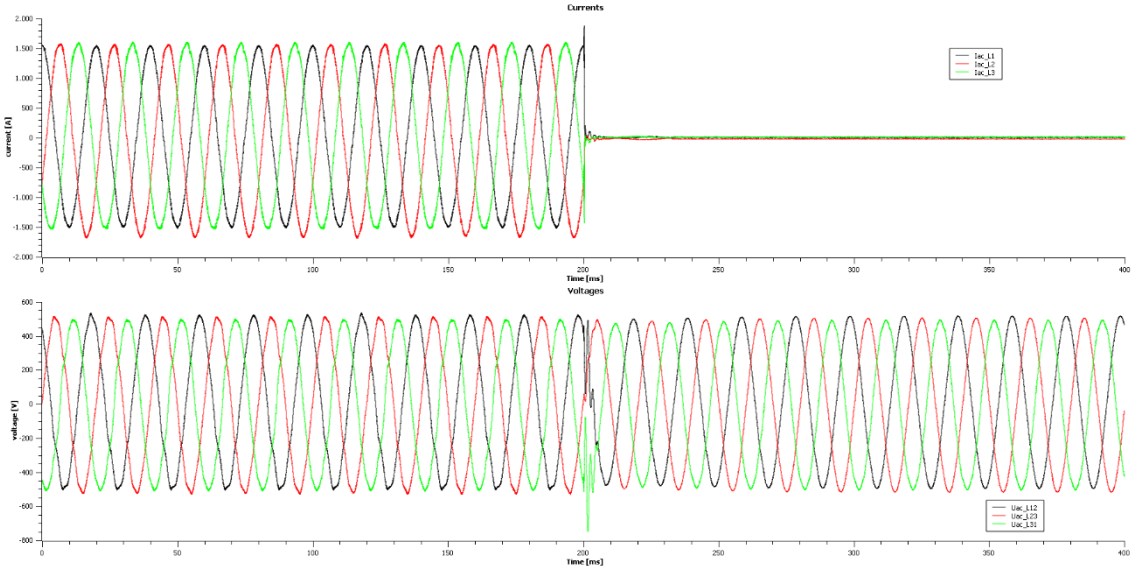


Figure 84: Short circuit L1 to PE (measurement 18)

Result:

The APS stops due to an APU error message “40 - over current Idc HW”.

4.2.3.1.3 Short circuit between two Phases

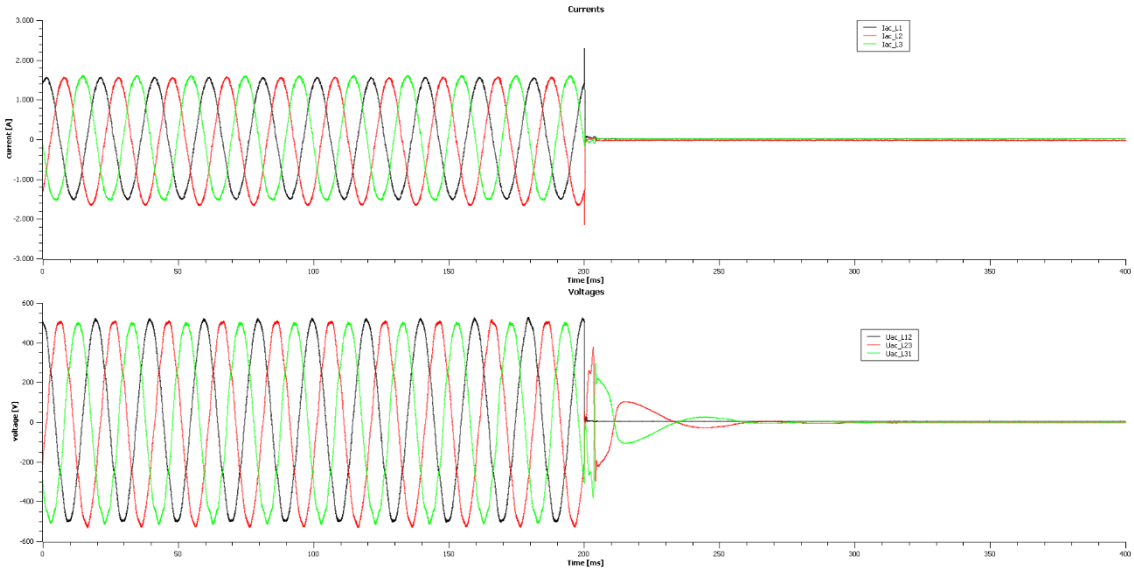


Figure 85: Short circuit L1 to L2 (measurement 19)

Result:

The APS stops due to an APU error message “41 - Over current L1 HW”.

4.2.3.1.4 Short Circuit between all Phases

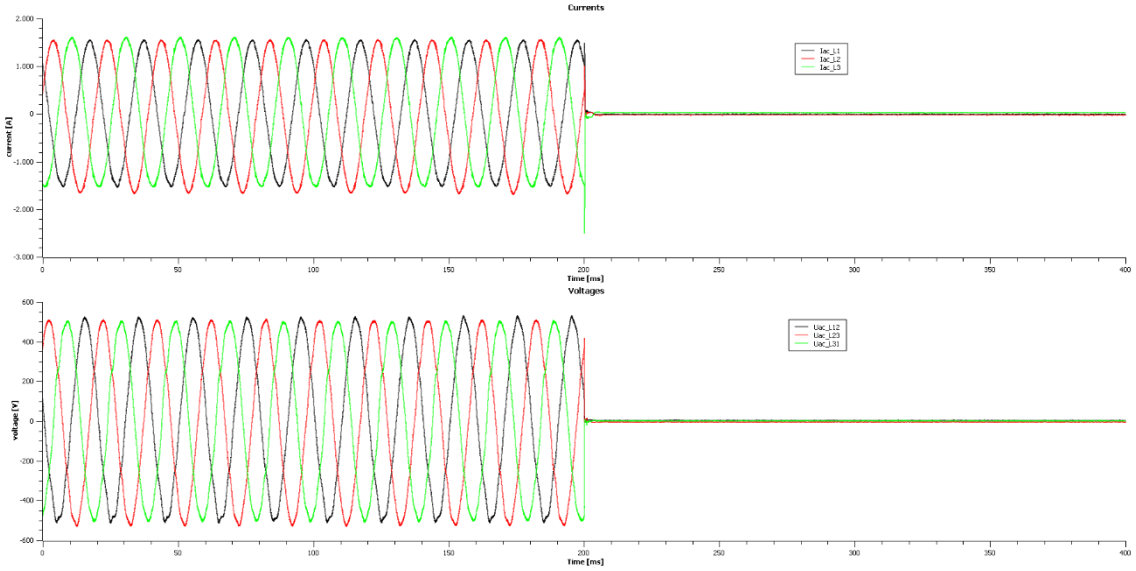


Figure 86: Short circuit between all phases (measurement 20)

Result:

The APS stops due to an APU error message “43 - Over current L3 HW”.

4.2.3.2 Response with 50% Load

The following tests have been done with 50% load of the inverter.

4.2.3.2.1 Grid Disconnection (High impedance grid loss)

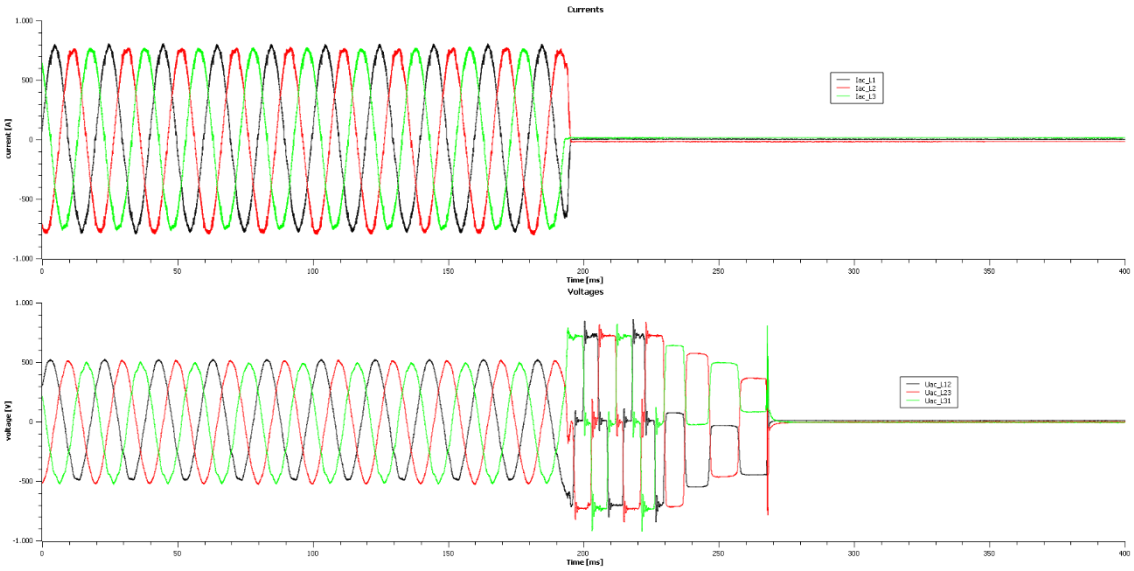


Figure 87: Grid disconnection (measurement 24)

Result:

The APS stops due to an APU error message “96 - Overvoltage Stage 4”.

4.2.3.2.2 Short Circuit Phase to Earth

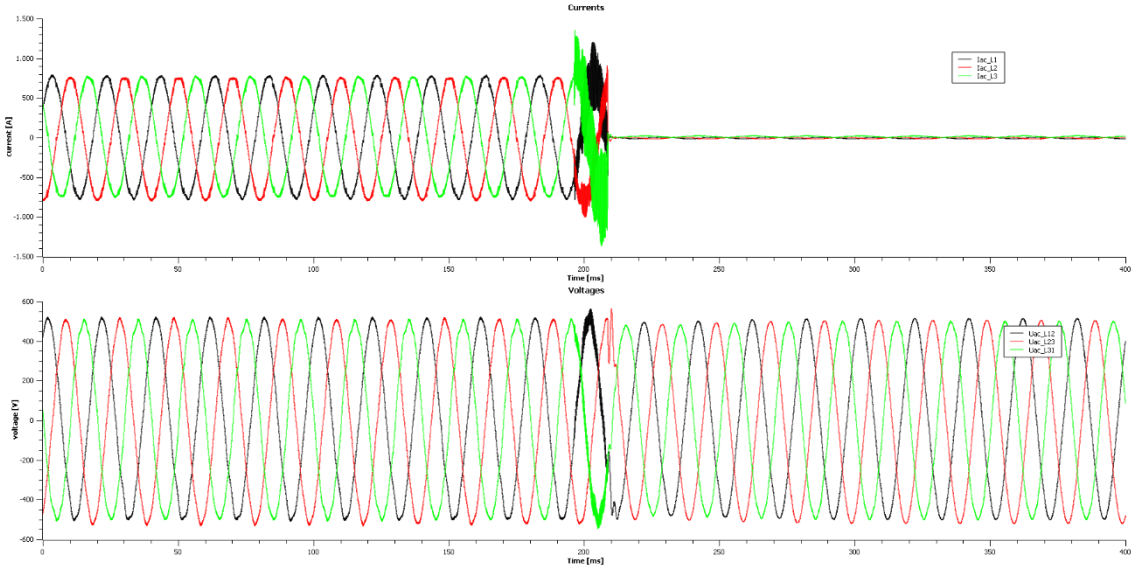


Figure 88: Short circuit L1 to PE (measurement 23)

Result:

The APS stops due to an APU error message “75 - current sum error”.

4.2.3.2.3 Short Circuit between two Phases

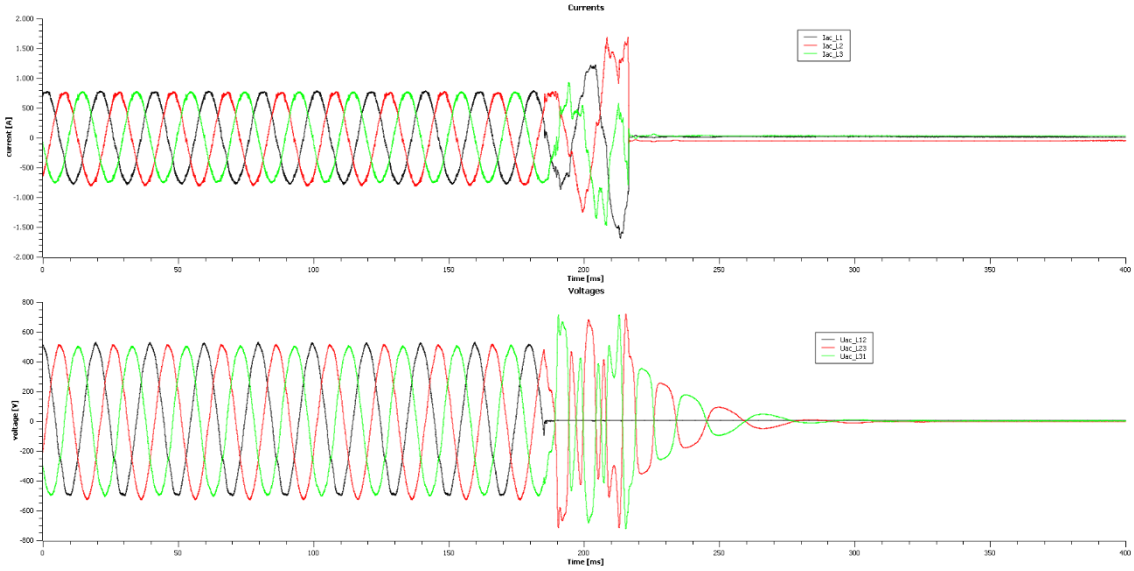


Figure 89: Short circuit L1 to L2 (measurement 22)

Result:

The APS stops due to an APU error message “13 - Over current Iac L2 SW”.

4.2.3.2.4 Short Circuit between all Phases

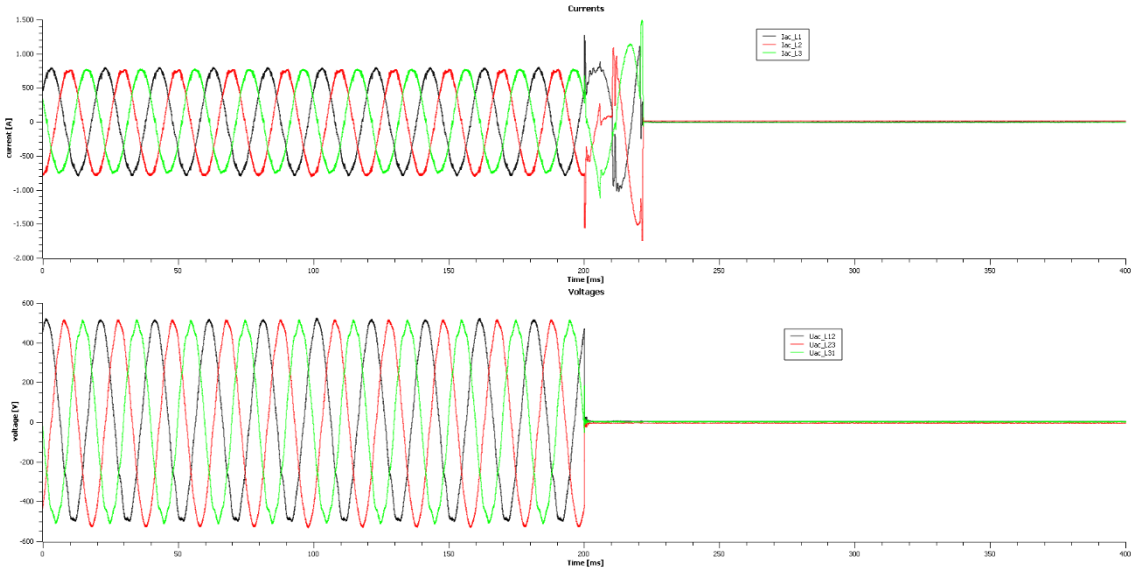


Figure 90: Short circuit between all phases (measurement 21)

Result:

The APS stops due to an APU error message “42 - Over current Iac L2 HW”.

4.3 Grid forming (island grid)

In this section several diagrams are showing the response and behaviour during different operation points:

- start-up
- changing setpoints
- load jumps

4.3.1 Start-up to operation

During start-up of the island grid operating mode, there are different cases on the grid status that can be identified:

- Grid available, even when it is instable or out of normal conditions.
- No grid available

The inverter response during start-up scenarios are explained in the following chapters.

4.3.1.1 Island grid mode in a non-existing grid (“Black start”)

The start-up sequence for a black start is shown in Figure 91. The ramp is configured with 20 %/s

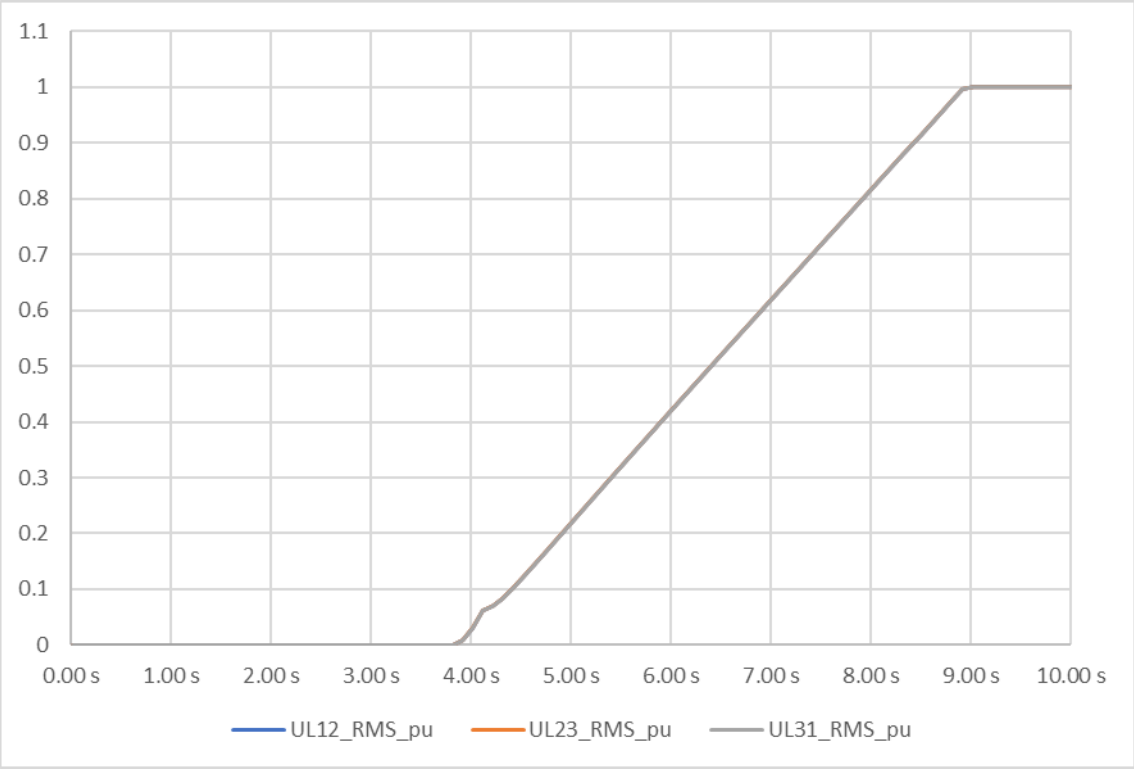


Figure 91: Start up "black start"

4.3.1.2 Island grid mode parallel to an existing grid

The inverter can be directly started in island grid mode when an existing grid is available. The switch on condition, that are used for grid feeding operation, are not active in the moment. The inverter can even start parallel to a grid, that is out of that range.

The inverter will synchronize with the phase of the grid automatically and will start the operation. The droop curves P(f) and Q(U) will define the output power of the inverter. To avoid high inrush power oscillations the frequency and voltage setpoint of the inverter should be set to the present values of the grid. In that case the power output is limited.

In the following figure a connection process is shown. This behaviour can be different in each connection event.

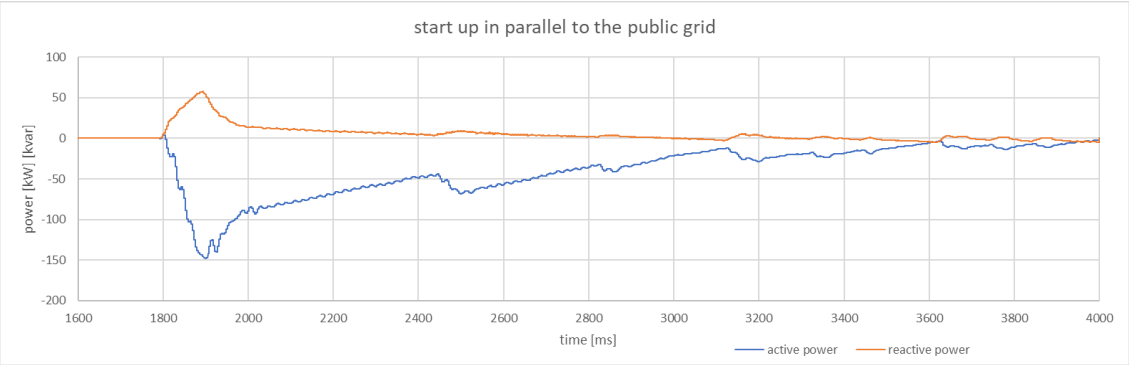


Figure 92: Start up in parallel to public grid

4.3.2 Response to Change of Set Point

This section is in progress. Further information is available in the *Function and Interface Description*.

4.3.2.1 Voltage set point

The voltage set point can be used to increase or decrease the island grid voltage in case of over or under voltage. A change of this parameter is just a move of the Q(U) function on the X-axes.

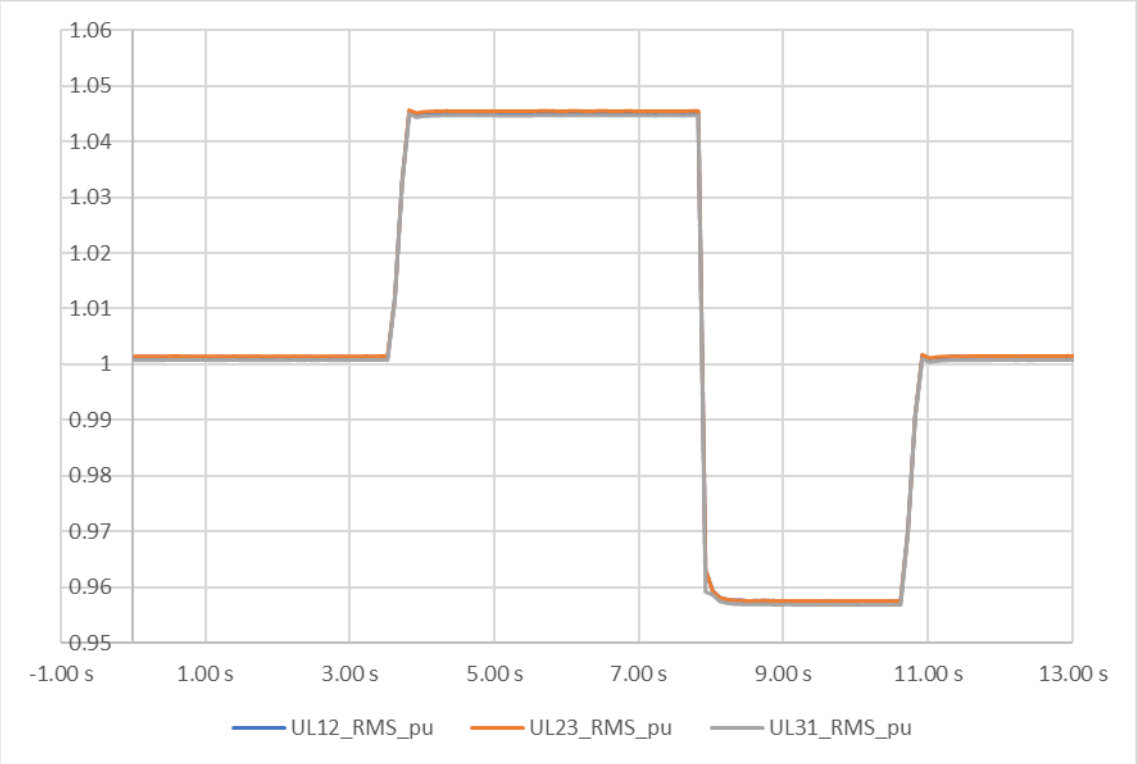


Figure 93: Voltage set point response (island grid)

4.3.2.2 Reactive power setpoint

A change on the reactive power set point is just a move of the Q(U) function in the Y-axes.

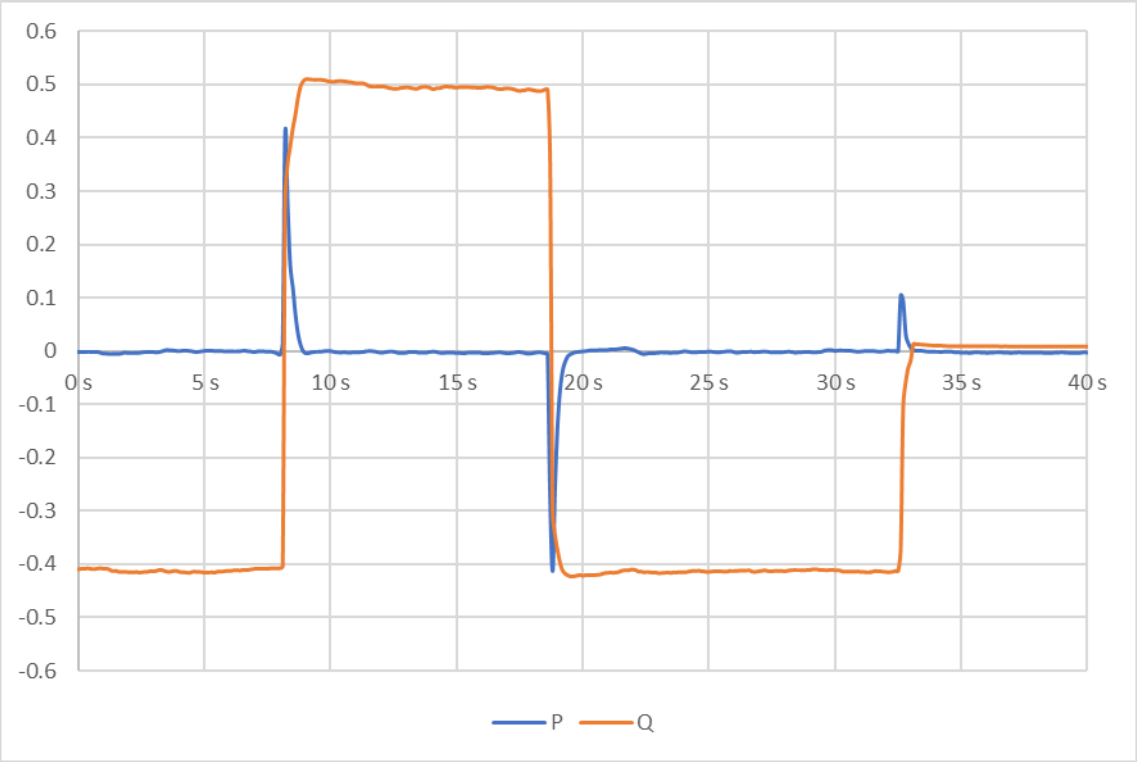


Figure 94: Reactive power set point response (public grid)

4.3.2.3 Frequency set point

The frequency set point is just a move of the P(f) function in the X-axes. This parameter can be used to stabilize the island grid frequency or to transfer active power from one inverter to another. The next diagram shows a set point change of 49Hz / 51Hz.

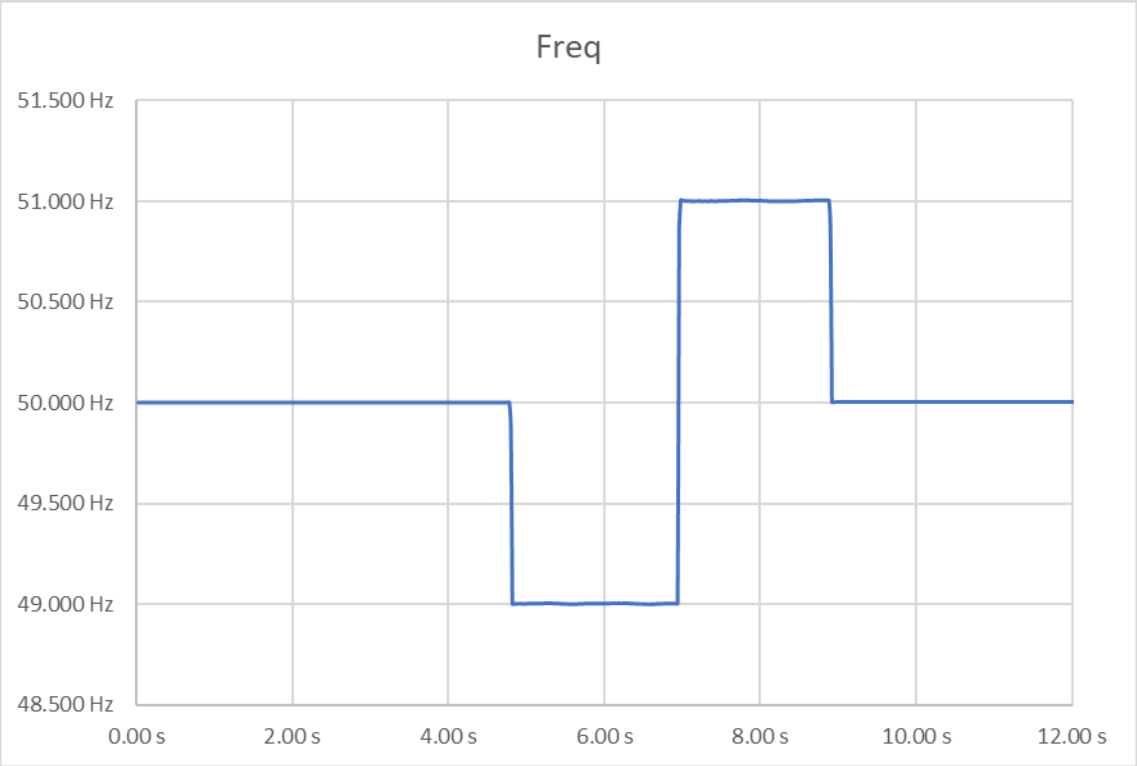


Figure 95: Frequency set point response (island grid)

4.3.2.4 Active power set point

During operation in parallel to the public grid or a frequency-controlled island grid, the response of an active power set point is shown in Figure 96. The diagram shows a set point change of +-500 kW. A change of this parameter is just a move of the P(f) function in the Y-axes.

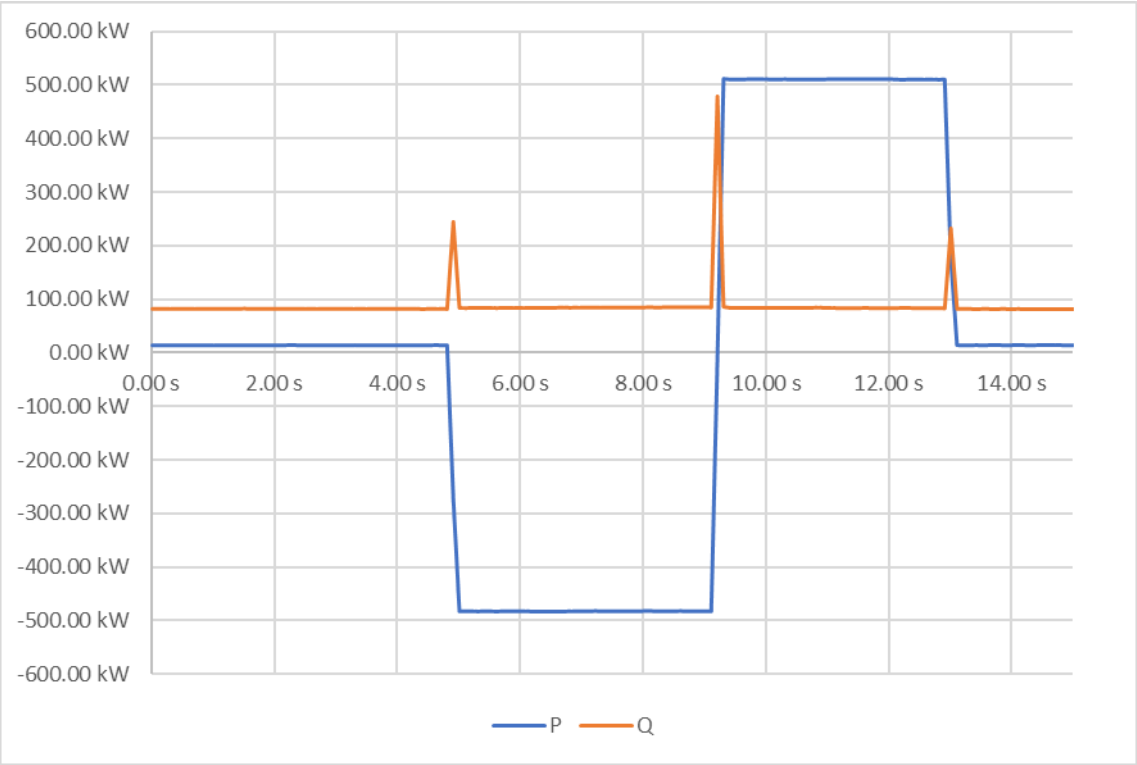


Figure 96: active power set point response (public grid)

4.3.2.5 CCCV behaviour

If more than one energy source is active in the island grid, it is possible to transfer energy between the APS and the other sources.

This is possible by using the control parameter “F_island” or “P_set”. Using the control parameter “P_set”, the CCCV-behaviour with the parameter “Vdc_min”, “Vdc_max”, “Idc_min” and “Idc_max” is available.

Is the island grid frequency equal to the parameter “F_island” the active power is 0 kW. If the other energy sources control the island grid frequency, the APS can charge the battery by setting the control parameter “P_set” to a negative value.

If now the DC-voltage reaches the value of the parameter “Vdc_max” the active power will increase to limit the DC-voltage at its set point. Same is, if the DC-current reaches the value of the parameter “Idc_min” the active power will be limited to control the DC-current.

This behaviour only limits the active power set point (P_set). If the APS feeding the island grid with active power due to a deviation of the grid frequency and the frequency set point, the CCCV-behaviour can’t limit the active power.

4.3.3 Response due to Change of load

The following sections contain the latest measurements regarding island grid behaviour. Schematic load changes are shown in Figure 97, Figure 98 and Figure 99.

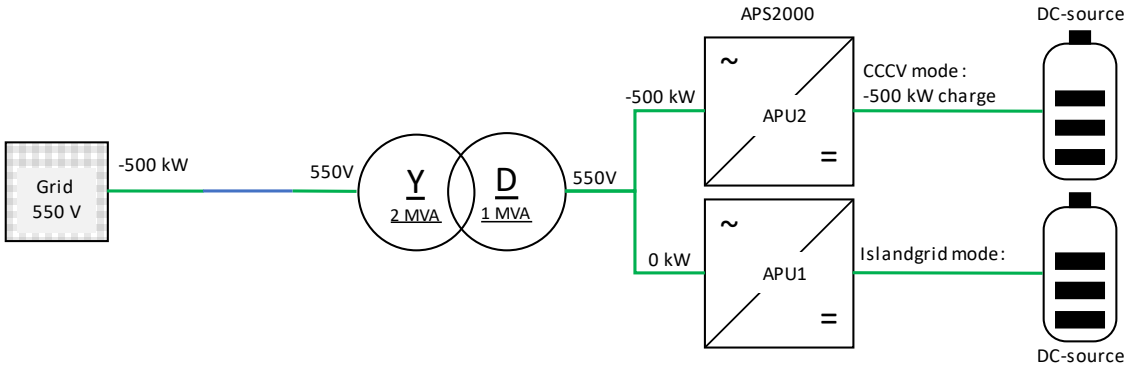


Figure 97: Running APS in parallel to public grid

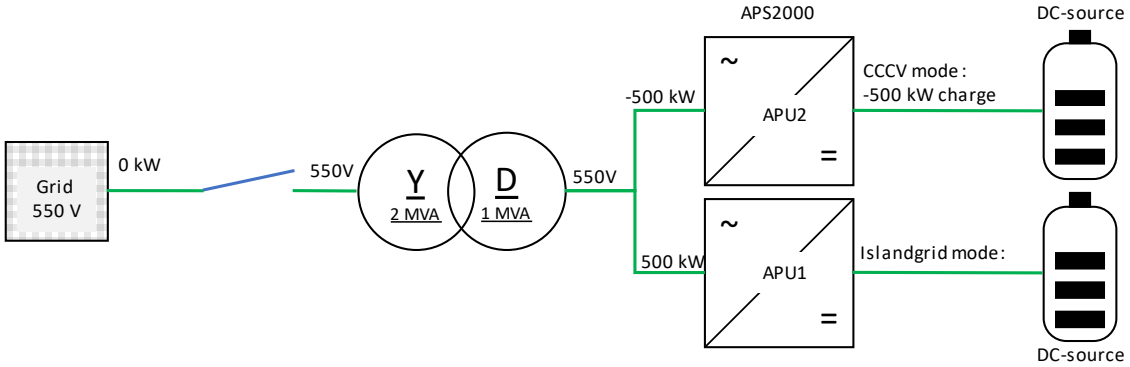


Figure 98: Running APS in island grid with load

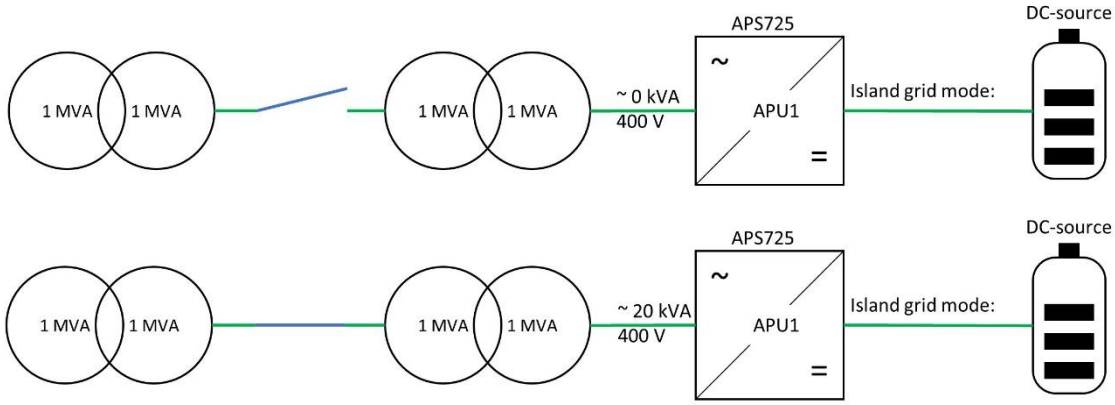


Figure 99: APS in island grid mode with and without load (2 MVA transformer)

4.3.4 Connection of a 2 MVA transformer

For the load jump test a coupled APU pair is running in island grid mode without an external load. As load a 2 MVA transformer is connected to the running APS. The following figures shows the transformer connection three times.

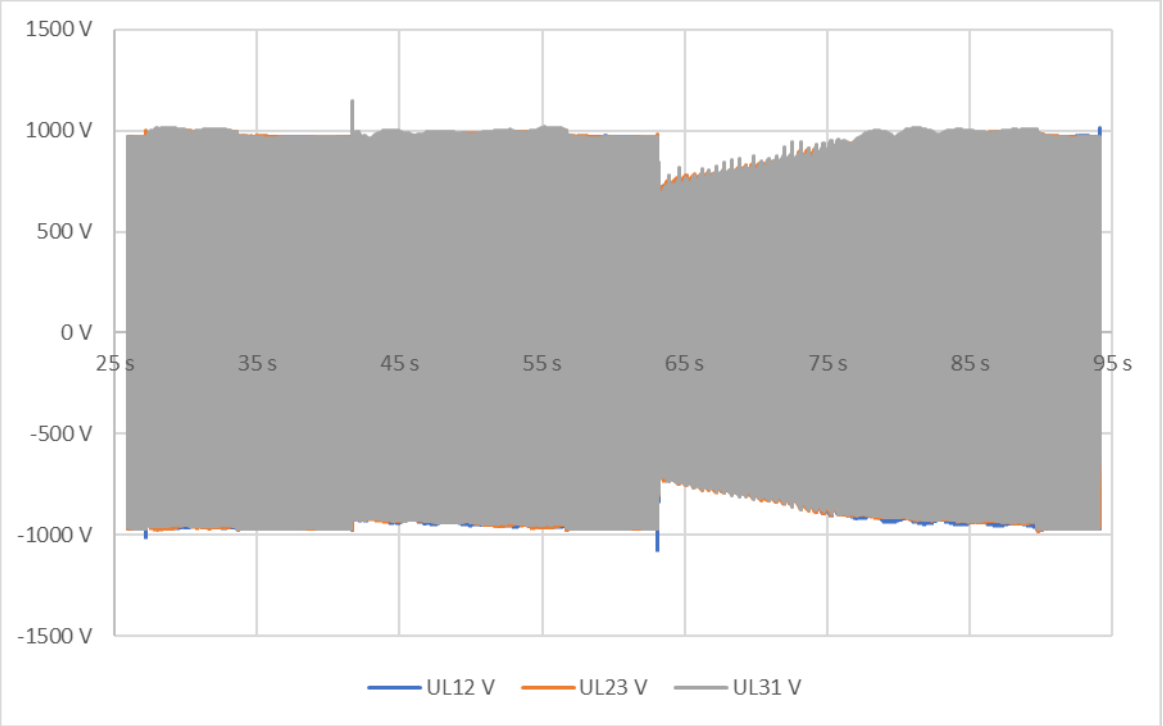


Figure 100: Sample values of voltages

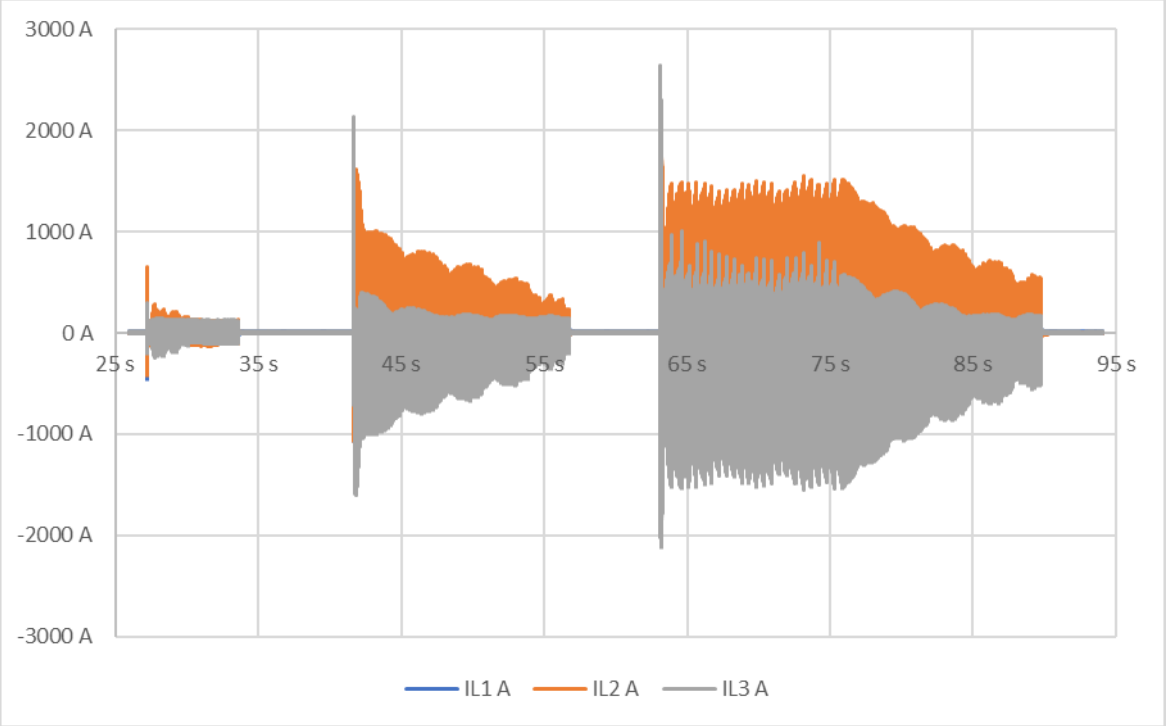


Figure 101: Sample values of currents

4.3.5 Response due to droop curves

The internal drop curves:

- P(f)
- Q(U)

Can be configured. These curves define the ratio between the power output of the inverter and the grid voltage parameters (frequency and amplitude).

These curves can be seen from both views:

- If there is a grid frequency brought to the inverter, the inverter will response with a related active power output.
- If there is a load jump (caused by a demand of the grid), the inverter will response with a related frequency deviation.

In the following chapters this behaviour will be documented for some specific situations. In any case a PSCAD simulation with the settings, behaviour and configuration of your specific project are strictly recommended.

4.3.5.1 Response on P(f) drop curve

The inverter response of the drop curve P(f) can be caused by:

- Load jump in the grid with a single inverter as grid forming device.
- Frequency ramp in the island grid
- Combination of both effects

In the following chapter some responses are shown, to get a feeling on the inverter behaviour caused by these cases. Please note that this is only an example, the behaviour in your island grid can be different. A simulation with PSCAD is strictly recommended to get proper results.

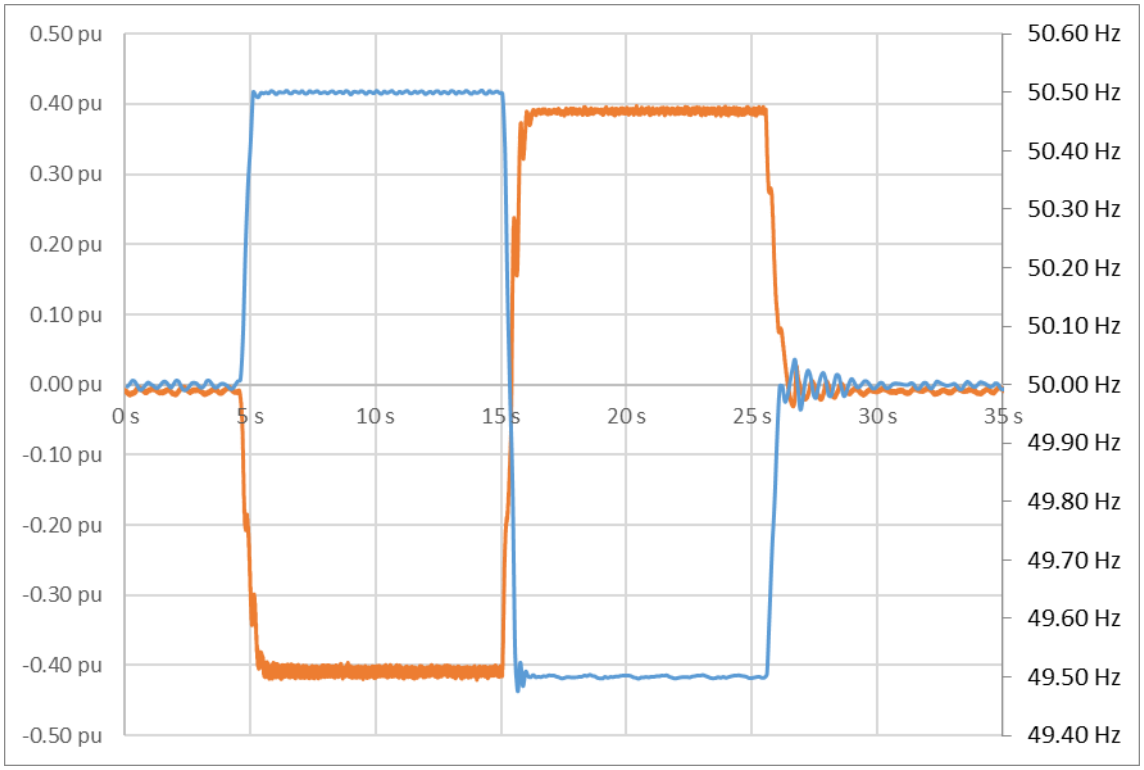


Table 26: Test 10 Configuration Island P(f)

UC Parameter	Parameter	Value
UC0070	Island Mode Voltage Limit	10 [%]
UC0060	Island Mode Frequency Limit	1.0 [Hz]
UC0050	Island Mode Reactive Power Limit	1000 [kvar]
UC0040	Island Mode Active Power Limit	1000 [kW]
UC1490	Island Mode P Controller KP	1.0
UC1500	Island Mode P Controller KI	0.0
UC1510	Island Mode Q Controller KP	0.0
UC1520	Island Mode Q Controller KI	0.0001

4.3.5.2 Response on Q(U) drop curve

The inverter feeds the reactive power depending on the configured Q(U) curve. The offset in the reactive power injection is depending on the grid.

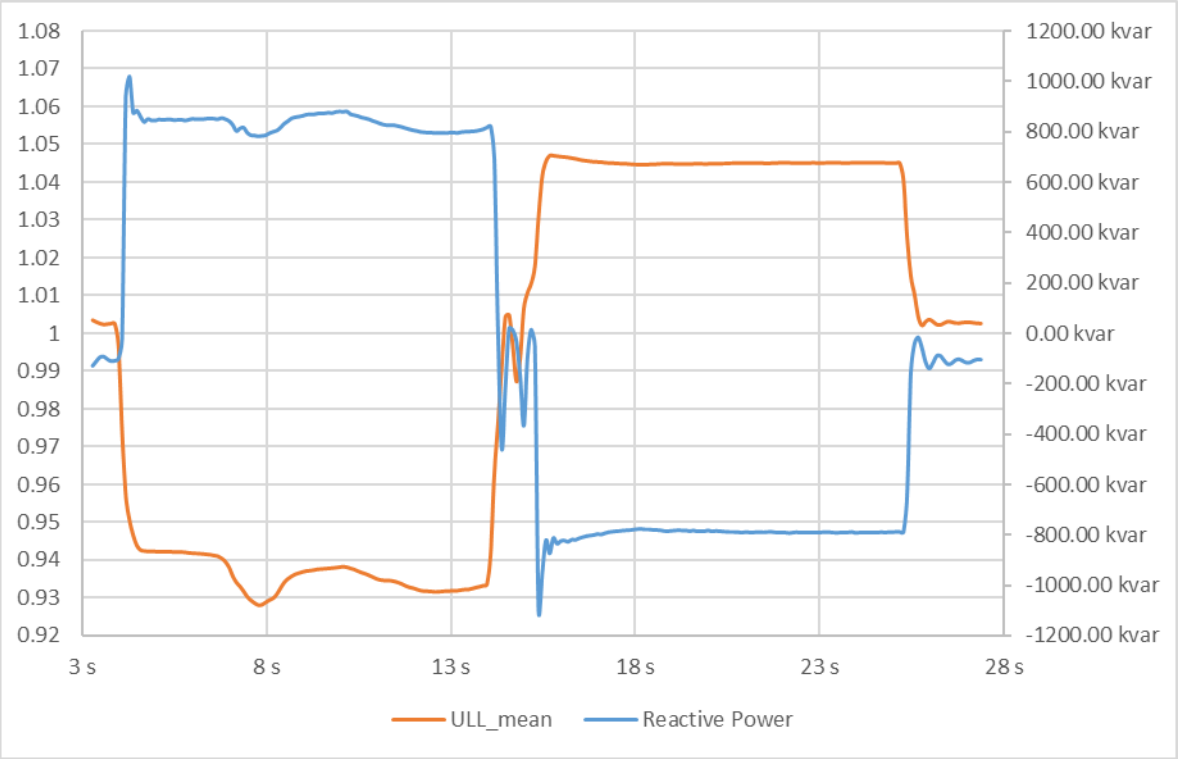


Table 27: Test 10 Configuration Island Q(U)

UC Parameter	Parameter	Value
UC0070	Island Mode Voltage Limit	10 [%]
UC0060	Island Mode Frequency Limit	1.0 [Hz]
UC0050	Island Mode Reactive Power Limit	1000 [kvar]
UC0040	Island Mode Active Power Limit	1000 [kW]
UC1490	Island Mode P Controller KP	1.0
UC1500	Island Mode P Controller KI	0.0
UC1510	Island Mode Q Controller KP	0.0
UC1520	Island Mode Q Controller KI	0.003

The response of the system can be simulated with the PSCAD model.

5 Revision History

Version 01

- First release.

Version 02

- Add PQ diagrams for APS3200-xx-4-440-5.

Version 03

- Add PQ diagrams for APS5000-xx-4-690-5.
- Add chapter “Heating”.

Version 04

- Add power demand of the heating.
- Add chapter “APS Configuration”.

Version 05

- Add PQ diagrams for APS2400-xx-2-660-5.

Version 06

- Change DC-link voltage in simulation overview table of APS2400-xx-2-660-5.

Version 07

- Revise chapter “PQ Diagrams”.
- Revise chapter “Cooling System”.

Version 08

- Add chapter “Short Circuit Currents”.
- Revise chapter “Water-Air Heat Exchanger”.

Version 09

- Revise chapter “Illustrations”.
- Revise chapter “Calculation of Active Power and Reactive Power for 2-4 APUs”.
- Revise chapter “Cooling System”.

Version 10

- Add chapter “Measuring Accuracy”.
- Add chapter “Revision History”.

Version 110

- Delete chapter “Cooling System” without chapter “Power Demand Cooling System”;
- Add chapter “APS0800-xx-1-440-5 (60 Hz).
- Add frequency in every headline of chapter PQ diagrams.
- Add frequency in every figure designation of chapter PQ diagrams.

Version 140

- Add chapter “DC Common-Mode Voltage”.
- Change PQ-diagrams of 1500 Vdc APS-Series.

Version 150

- Add PQ-diagram for 660 V grid voltage.

Version 160

- Add PQ-diagram for 690 V grid voltage at 50 Hz.

Version 170

- Add Udcmin diagrams for Udcmin (cos(Phi)) function.

Version 180

- Add PQ diagrams for APS1090-xx-1-600-5 (60 Hz) and APS1140-xx-1-630-5 (60 Hz).

Version 190

- Add application Island Grid to chapter “Applications”.
- Add options for HW130 to chapter “Options”.
- Add options for HW130 to chapter “APU Options and Features”.
- Add chapter DC-Pre-charge.
- Update values for UPS in chapter 12.

Version 200

- Change designation of APS group in chapter 2.6.
- Change designation of APS group in chapter 2.12
- Change designation of APS group in chapter 2.14
- Change designation of APS group in chapter 2.16
- Change designation of APS group in chapter 2.18
- Add chapter “STATCOM Mode Efficiency”.
- Add chapter “Island Grid Behaviour”.

Version 210

- Add abstracts from Function and Interface Description to chapter Island Grid Behaviour”.

Version 220

- Add PQ diagrams for APS1000-xx-1-550-5 (60 Hz), add 500 A Fuses, 1000 Vdc fuses removed.

Version 230

- Add common-mode voltage in chapter 3.4.3

Version 240

- Rework PQ-diagrams (chapter 2 PQ Diagrams)
- Rework U_{DCmin}(cos(phi)) diagrams in chapter 2

Version 250

- Add symmetrical 100 kW load jump in chapter (reworked in V351)
- Add asymmetrical 100 kW load jump in chapter (reworked in V351)

Version 260

- Add U_{DCmin}(cos(phi)) APS1200-xx-1-660-5 (50 Hz) in chapter 2

Version 270

- Add chapter 2.1 Pac(Udc)

Version 280

- Add diagrams of the inrush-current in chapter 3.1.1 Control System.

Version 290

- New Chapters 4.2.2.3,4.2.2.4 “reactive set points” added.
- New Chapter 2.5

Version 300

- Add PQ diagrams for APS0600-xx-1-330-5
- Add PQ diagrams for APS0650-xx-1-360-5

Version 310

- Delete diagrams in chapter 2.2 PQ Diagrams
- Change Chapter 4 Udcmin Diagrams to chapter 2.2.2
- Add U_{DCmin}(cos(phi)) APS600-xx-1-330-5 and U_{DCmin}(cos(phi)) APS1250-xx-1-690-5 in chapter 2.2.2
- Rework chapter 2.1. Include all APS types in the diagrams.

Version 320

- Change chapter 4.3.1.1
- Chapter 3.1.2 60 Hz values added.
- Chapter 3.1.4 added.
- Chapter 3.2.2 new current measurement range added.

Version 330

- Restructuring of the chapters
- Chapter 2.4.2 added.
- Chapter 3.6 added.
- Chapter 4.1 added.
- Chapter 4.2.1.1 and 4.2.1.2 added.
- Chapter 4.3.1 reworked.

Version 340

- Chapter 3.3.3 added.
- Chapter 2.5 Inverters bigger than 690 Vac are removed from list.
- Chapter 2.5 Note added.

Version 350

- Rework chapter 4.3 / Change measurement results
- Chapter 2.3 added.

Version 360

- Add chapter 3.1.5
- Rework chapter 4.3

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